MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3890 REAGENTS – SMBS & PEROXIDE STORAGE AND MIXING STATION
DESIGN CALCULATION OF FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The Purpose of this document is to analysis & design of **3890 - REAGENTS - SMBS & PEROXIDE** structure foundation. The dimensions and reinforcement are to be calculated. The report describes the methodology and the design parameters used in the design calculation. The analysis & design is carried out in accordance with project specifications , codes & standards.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah- Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD Pro. RCC column provided below the steel structure shed. The columns are RCC cast-in-situ. Columns rest on isolated founfation, suitably designed to transfer design load on soil.
Stability check against Overturning & Sliding are done in the excel sheet. Base Pressure check has been done in the excel sheet. For strength calculation of column Bending Moment & Shear Force values are directly taken from STAAD & designed in the standard excel sheet.

4. DESIGN REQUIREMENTS

4.1 Reference Codes & Books

Reference is made in this standard as indicated in STRUCTURAL ENGINEERING DESIGN CRITERIA Doc No. : MD-216-1000-EG-ST-CRT-1001-A03 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

4.2 Materials Property

A. Density

4.2.A.(i) Density of Dry Soil : 1.9 T/m^3
4.2.A.(ii) Density of Concrete : 2.5 T/m^3
4.2.A.(iii) Density of Water : 1.0 T/m^3

B. Concrete Grade

4.2.B.(i) Foundation and Structural Members: 35 Mpa Strength of 28 days
4.2.B.(ii) Blinding Concrete below foundation bases: 15 Mpa Strength of 28 days

C. Reinforcement Grade

4.2.C.(i) Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
4.2.C.(ii) Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

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D. Grout Grade

4.2.D.(i) Grout shall be 50 mm cementitious of Strength 40 MPa.

4.3 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.4 Concrete Clear Cover

Concrete Exposure	Member	MinimumCove (mm)
Exposed to Soil	Foundation	50
	Column	50

5. LOAD

5.1 Primary Loads

3890 - REAGENTS - SMBS & PEROXIDE Shed the primary load values are taken from the steel structure load data drawing **Drg. No. : MD-216-3890-EG-ST-GAD-2365**

1 Dead (D)
2 Collateral (C)
3 Live (L)
4 Wind_Left1 (W_L1)
5 Wind_Right1 (W_R1)
6 Wind_Left2 (W_L2)
7 Wind_Right2 (W_R2)
8 Wind_Long1 (W_LG1)
9 Wind_Long2 (W_LG2)
10 Seismic_L (E_L)
11 Seismic_R (E_R)
12 Temperatur_(+)25C (T_(+)25)
13 Temperatur_(-)25C (T_(-)25)

5.2 Load Combinations for Stress Calculation

L/C	LOAD COMBINATION
101	D+C+T_(+)25
102	D+C+T_(-)25
103	D+C+LR
104	D+C+0.75*T_(+)25+0.75*LR
105	D+C+0.75*T_(-)25+0.75*LR
106	D+C+W_L1
107	D+C+W_R1
108	D+C+W_L2
109	D+C+W_R2
110	D+C+W_LG1
111	D+C+W_LG2
112	D+C+0.7*E_L
113	D+C+0.7*E_R
114	D+C+0.75*W_L1+0.75*LR
115	D+C+0.75*W_R1+0.75*LR
116	D+C+0.75*W_L2+0.75*LR
117	D+C+0.75*W_R2+0.75*LR
118	D+C+0.75*W_LG1+0.75*LR

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119	D+C+0.75*W_LG2+0.75*LR
120	D+C+0.75*(0.7*E_L)+0.75*LR
121	D+C+0.75*(0.7*E_R)+0.75*LR
122	0.6*(D+C)+W_L1
123	0.6*(D+C)+W_R1
124	0.6*(D+C)+W_L2
125	0.6*(D+C)+W_R2
126	0.6*(D+C)+W_LG1
127	0.6*(D+C)+W_LG2
128	0.6*(D+C)+0.7*E_L
129	0.6*(D+C)+0.7*E_R

5.3 Load Combinations for Strength Calculation

L/C	LOAD COMBINATION
201	1.4*(D+C)
202	1.4*(D+C+T_(+)25)+0.5*LR
203	1.4*(D+C+T_(-)25)+0.5*LR
204	1.2*(D+C)+1.6*LR+0.8*W_L1
205	1.2*(D+C)+1.6*LR+0.8*W_R1
206	1.2*(D+C)+1.6*LR+0.8*W_L2
207	1.2*(D+C)+1.6*LR+0.8*W_R2
208	1.2*(D+C)+1.6*LR+0.8*W_LG1
209	1.2*(D+C)+1.6*LR+0.8*W_LG2
210	1.2*(D+C)+1.6*W_L1+0.5*LR
211	1.2*(D+C)+1.6*W_R1+0.5*LR
212	1.2*(D+C)+1.6*W_L2+0.5*LR
213	1.2*(D+C)+1.6*W_R2+0.5*LR
214	1.2*(D+C)+1.6*W_LG1+0.5*LR
215	1.2*(D+C)+1.6*W_LG2+0.5*LR
216	1.2*(D+C)+E_L
217	1.2*(D+C)+E_R
218	0.9*(D+C)+1.6*W_L1
219	0.9*(D+C)+1.6*W_R1
220	0.9*(D+C)+1.6*W_L2
221	0.9*(D+C)+1.6*W_R2
222	0.9*(D+C)+1.6*W_LG1
223	0.9*(D+C)+1.6*W_LG2
224	0.9*(D+C)+E_L
225	0.9*(D+C)+E_R

6. SAFETY FACTOR AGAINST STABILITY

6.1 Overturning : For foundation design a minimum factor of safety against overturning is consider 2.0
6.2 Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

7. ALLOWABLE SOIL BEARING CAPACITY

The net safe bearing capacity of soil for foundations design as obtained from page 39 of Doc. No. MD-216-0000-EM-GM-RPT-0003 is 670 KN/m^2 at 1.2m depth below FGL
The neareast Bore Hole of **3890 - REAGENTS - SMBS & PEROXIDE** is **BH-04**. The Net Safe Bearing Capacity of soil as per approved L&T soil test report (Table-6 of Doc. No. MD-216-3000-EG-CV-RPT-1008-A02) is as mention below..

(i) 2m X 2m Net SBC at 1.5 below FGL is 590 KN/SQM
(ii) 3m X 3m Net SBC at 1.5 below FGL is 450 KN/SQM
(iii) 4m X 4m Net SBC at 1.5 below FGL is 375 KN/SQM
(iv) 5m X 5m Net SBC at 1.5 below FGL is 325 KN/SQM

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8.0 DESIGN OF FOUNDATION

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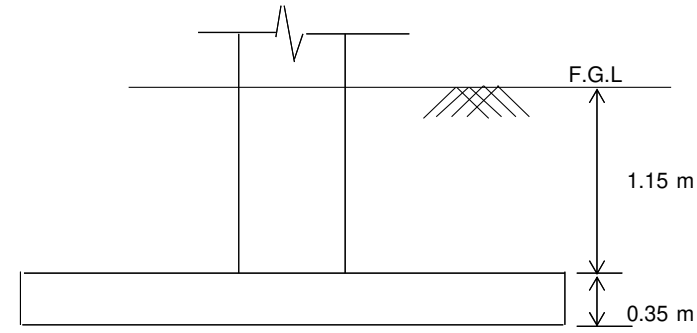
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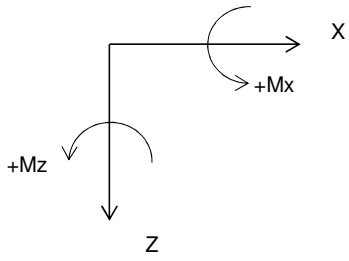
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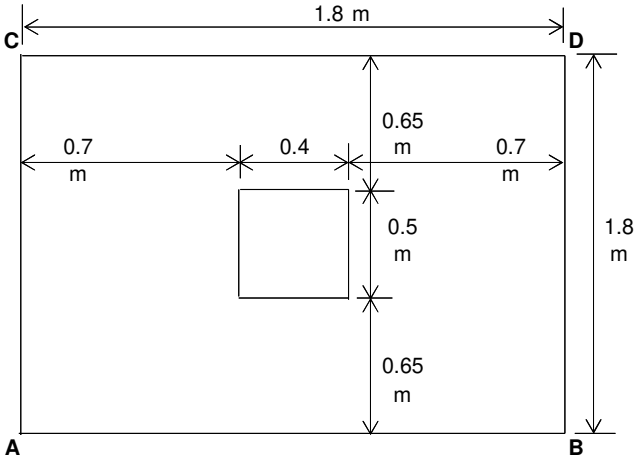
8.1 DESIGN OF FOUNDATION MKD. F1 :

(NODE NO. : 1, 8, 18, 25)

Net Safe Bearing Capacity of Soil =	59	Mton/m^2	*(Ref. Nearest Bore Hole Mkd. BH-04)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	59 + 1.5 X 1.9 =	61.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	1.8	m
Width of Foundation (Lz) =	1.8	m
Thickness of Base Raft =	0.35	m
Length of Column (lx) =	0.4	m
Width of Column (lz) =	0.5	m

Area , A = Lx*Lz
= 3.24 m^2
Zxx = Lx*Lz^2/6
= 0.972 m^3
Zzz = Lz*Lx^2/6
= 0.972 m^3

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WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	1.8X1.8X0.35X2.5 =	2.835
Column =	0.4X0.5X1.15X2.5 =	0.575
Soil =	((1.8X1.8)-(0.4X0.5))X1.15X1.9 =	6.642
	Total =	10.052

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK :

(Staad Group : F1)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	1	101 D+C+T	**0**	1.18	-0.009	-0.022	0	0
Min Fx	1	101 D+C+T	**0**	1.18	-0.009	-0.022	0	0
Max Fy	1	108 D+C+V	0	**1.377**	0.17	0.399	0	0
Min Fy	1	126 0.6*(D	0	**0.32**	-0.061	-0.143	0	0
Max Fz	1	108 D+C+V	0	1.377	**0.17**	0.399	0	0
Min Fz	18	106 D+C+V	0	1.186	**-0.124**	-0.291	0	0
Max Mx	1	108 D+C+V	0	1.377	0.17	**0.399**	0	0
Min Mx	18	106 D+C+V	0	1.186	-0.124	**-0.291**	0	0
Max My	1	101 D+C+T	0	1.18	-0.009	-0.022	**0**	0
Min My	1	101 D+C+T	0	1.18	-0.009	-0.022	**0**	0
Max Mz	1	101 D+C+T	0	1.18	-0.009	-0.022	0	**0**
Min Mz	1	101 D+C+T	0	1.18	-0.009	-0.022	0	**0**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
101	11.232	0.009	0.5	5.616	624	1.5	O.K.
101	11.232	0.009	0.5	5.616	624	1.5	O.K.
108	11.429	0.17	0.5	5.715	33.62	1.5	O.K.
126	10.372	0.061	0.5	5.186	85.02	1.5	O.K.
108	11.429	0.17	0.5	5.715	33.62	1.5	O.K.
106	11.238	0.124	0.5	5.619	45.31	1.5	O.K.
108	11.429	0.17	0.5	5.715	33.62	1.5	O.K.
106	11.238	0.124	0.5	5.619	45.31	1.5	O.K.
101	11.232	0.009	0.5	5.616	624	1.5	O.K.
101	11.232	0.009	0.5	5.616	624	1.5	O.K.
101	11.232	0.009	0.5	5.616	624	1.5	O.K.
101	11.232	0.009	0.5	5.616	624	1.5	O.K.

OVERTURNING CHECK:

L/C : 126 0.6*(D+C)+W_LG1

Max Upword Force =	0	Mton		
Self Wt. of Foundation =	10.052	Mton		
Overturning Moment =	0 X 1.8/2 + 0.143 =		0.143	Mton-m
Resisting Moment =	10.052 X 0.9 X 1.8/2 =		8.142	Mton-m
F.O.S =	56.94	>.	2	O.K.

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BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A	Mx/Zxx	Mz/Zzz	BASE PRESSURE AT POINT				Remarks
	(Mton/m^2)	(Mton/m^2)	(Mton/m^2)	A	B	C	D	
101	3.467	0.023	0	3.49	3.49	3.444	3.444	O.K.
101	3.467	0.023	0	3.49	3.49	3.444	3.444	O.K.
108	3.527	0.41	0	3.937	3.937	3.117	3.117	O.K.
126	3.201	0.147	0	3.348	3.348	3.054	3.054	O.K.
108	3.527	0.41	0	3.937	3.937	3.117	3.117	O.K.
106	3.469	0.299	0	3.768	3.768	3.17	3.17	O.K.
108	3.527	0.41	0	3.937	3.937	3.117	3.117	O.K.
106	3.469	0.299	0	3.768	3.768	3.17	3.17	O.K.
101	3.467	0.023	0	3.49	3.49	3.444	3.444	O.K.
101	3.467	0.023	0	3.49	3.49	3.444	3.444	O.K.
101	3.467	0.023	0	3.49	3.49	3.444	3.444	O.K.
101	3.467	0.023	0	3.49	3.49	3.444	3.444	O.K.
			MAX =	3.937	3.937	3.444	3.444	
			MIN =	3.348	3.348	3.054	3.054	

SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

(Staad Group : F1)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	1	201 1.4*(D	**0**	1.674	0.002	0.004	0	0
Min Fx	1	201 1.4*(D	**0**	1.674	0.002	0.004	0	0
Max Fy	1	212 1.2*(D	0	**1.753**	0.267	0.628	0	0
Min Fy	1	222 0.9*(D	0	**0.441**	-0.097	-0.228	0	0
Max Fz	1	220 0.9*(D	0	1.366	**0.271**	0.636	0	0
Min Fz	18	210 1.2*(D	0	1.441	**-0.201**	-0.473	0	0
Max Mx	1	220 0.9*(D	0	1.366	0.271	**0.636**	0	0
Min Mx	18	210 1.2*(D	0	1.441	-0.201	**-0.473**	0	0
Max My	1	201 1.4*(D	0	1.674	0.002	0.004	**0**	0
Min My	1	201 1.4*(D	0	1.674	0.002	0.004	**0**	0
Max Mz	1	201 1.4*(D	0	1.674	0.002	0.004	0	**0**
Min Mz	1	201 1.4*(D	0	1.674	0.002	0.004	0	**0**

STRESS CALCULATION FOR STRENGTH DESIGN :

L/C	(Wf+Fy)/A	Mx/Zx	Mz/Zz	BASE PRESSURE AT POINT			
	(Mton/m^2)	(Mton/m^2)	(Mton/m^2)	A	B	C	D
201	3.619	0.004	0	3.623	3.623	3.615	3.615
201	3.619	0.004	0	3.623	3.623	3.615	3.615
212	3.644	0.646	0	4.29	4.29	2.998	2.998
222	3.239	0.235	0	3.474	3.474	3.004	3.004
220	3.524	0.654	0	4.178	4.178	2.87	2.87
210	3.547	0.487	0	4.034	4.034	3.06	3.06
220	3.524	0.654	0	4.178	4.178	2.87	2.87
210	3.547	0.487	0	4.034	4.034	3.06	3.06
201	3.619	0.004	0	3.623	3.623	3.615	3.615
201	3.619	0.004	0	3.623	3.623	3.615	3.615
201	3.619	0.004	0	3.623	3.623	3.615	3.615
201	3.619	0.004	0	3.623	3.623	3.615	3.615

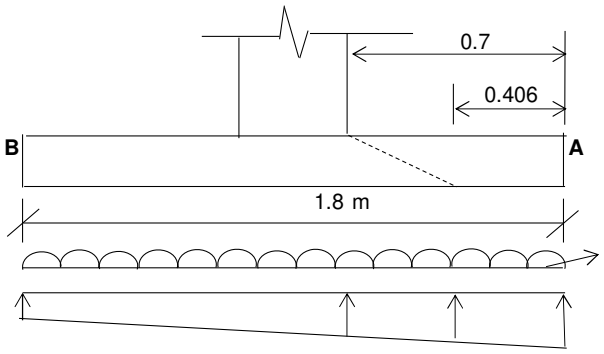
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PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point B (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
201	3.623	3.623	3.62	3.62	0.138	0.229
201	3.623	3.623	3.62	3.62	0.138	0.229
212	4.29	4.29	4.29	4.29	0.301	0.499
222	3.474	3.474	3.47	3.47	0.102	0.169
220	4.178	4.178	4.18	4.18	0.274	0.454
210	4.034	4.034	4.03	4.03	0.239	0.396
220	4.178	4.178	4.18	4.18	0.274	0.454
210	4.034	4.034	4.03	4.03	0.239	0.396
201	3.623	3.623	3.62	3.62	0.138	0.229
201	3.623	3.623	3.62	3.62	0.138	0.229
201	3.623	3.623	3.62	3.62	0.138	0.229
201	3.623	3.623	3.62	3.62	0.138	0.229
				MAX =	0.301	0.499

ALONG X - DIRECTION :

Mu = 0.301 Mton-m

Φ*Kn = Mu/bd^2 deff. = 294 mm

Mu/bd^2 = 0.301X10^7/(1000X294^2) = 0.035 N/mm^2 < 6.9 N/mm^2

Pt = 0.0001

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X294/100 = 5.29 cm^2

Provided **12** mm @ **150** C/C Provide Ast = 7.54 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0026

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ONE-WAY SHEAR CHECK

Vu = 0.499 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X0.499X10^4X294/0.301/10^7)X1000X294/10^4

= 28.463 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X294/10^4

= 29.129 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X294/10^4

= 50.44 Mton

HENCE, Vc = 28.463 Mton

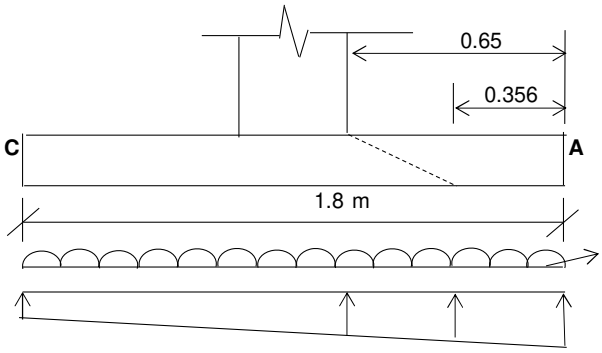
Φt*Vc = 0.75X28.463 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 21.347 Mton **> 0.499Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(28.463X10^4+0.66X35^0.5X1000X294)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 88.231 Mton **> Vu= 0.499Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point C (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
201	3.623	3.615	3.62	3.62	0.119	0.201
201	3.623	3.615	3.62	3.62	0.119	0.201
212	4.29	2.998	3.82	4.03	0.326	0.484
222	3.474	3.004	3.3	3.38	0.112	0.164
220	4.178	2.87	3.71	3.92	0.302	0.444
210	4.034	3.06	3.68	3.84	0.256	0.381
220	4.178	2.87	3.71	3.92	0.302	0.444
210	4.034	3.06	3.68	3.84	0.256	0.381
201	3.623	3.615	3.62	3.62	0.119	0.201
201	3.623	3.615	3.62	3.62	0.119	0.201
201	3.623	3.615	3.62	3.62	0.119	0.201
201	3.623	3.615	3.62	3.62	0.119	0.201
				MAX =	0.326	0.484

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ALONG Z - DIRECTION :

Mu = 0.326 T-m

Φ*Kn = Mu/bd^2 deff. = 294 mm

Mu/bd^2 = 0.326X10^7/(1000X294^2) = 0.038 N/mm^2 < 6.9 N/mm^2

Pt = 0.0001

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X294/100 = 5.29 cm^2

Provided **12** mm @ **150** C/C Provide Ast = 7.54 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0026

ONE-WAY SHEAR CHECK

Vu = 0.484 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X0.484X10^4X294/0.326/10^7)X1000X294/10^4

= 28.396 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X294/10^4

= 29.129 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X294/10^4

= 50.44 Mton

HENCE, Vc = 28.396 Mton

Φt*Vc = 0.75X28.396 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 21.297 Mton **> 0.484Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(28.396X10^4+0.66X35^0.5X1000X294)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 88.226 Mton **> Vu= 0.484Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 3.24 m^2

Perimeter (bo) = 2 X ((0.4 + 0.294) + (0.5 + 0.294)) = 2.976 m

β = 1.25

αs = 20 (Corner Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 3.937 $Mton/m^2$

Vu = ((3.24 - (0.4X0.5))X3.937

= 11.968 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X2.976X10^3X294/10^4

= 170.816 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1.25)X1X35^0.5X2.976X10^3X294/10^4

= 228.79 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+20X294/(2.976X10^3))X1X35^0.5X(2.976X10^3)X294/10^4

= 170.812 Mton

HENCE, Vc = 170.812 Mton

Φt*Vn = Φt*Vc = 0.75X170.812 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

128.109 Mton **> 11.968Mton , Hence O.K.**

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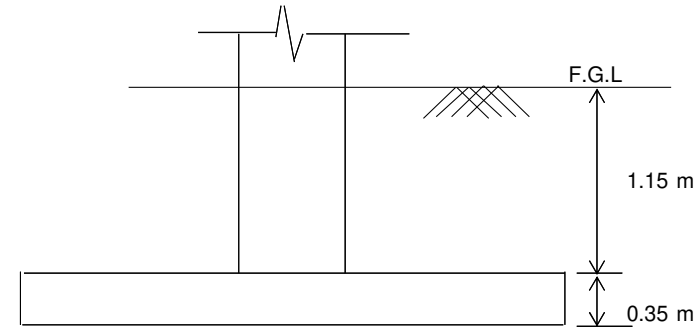
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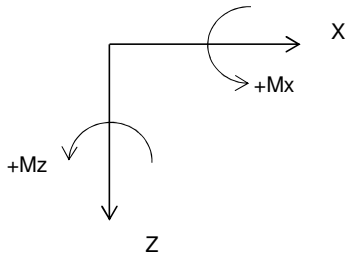
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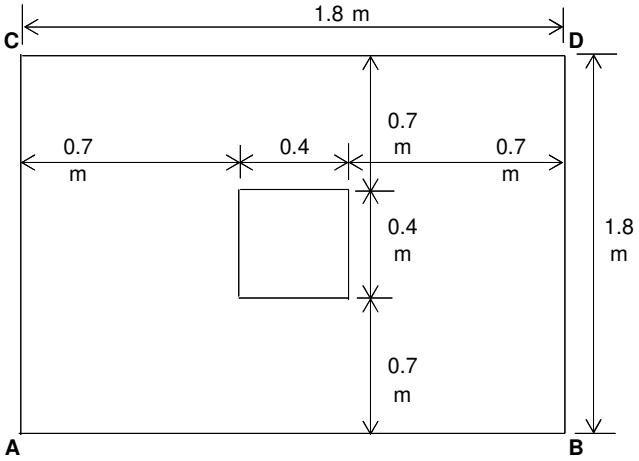
8.2 DESIGN OF FOUNDATION MKD. F2 :

(NODE NO. : 9 11 12 14 15 17)

Net Safe Bearing Capacity of Soil =	59	Mton/m^2	*(Ref. Nearest Bore Hole Mkd. BH-04)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	59 + 1.5 X 1.9 =	61.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	1.8	m
Width of Foundation (Lz) =	1.8	m
Thickness of Base Raft =	0.35	m
Length of Column (lx) =	0.4	m
Width of Column (lz) =	0.4	m

Area , A = Lx*Lz
= 3.24 m^2
Zxx = Lx*Lz^2/6
= 0.972 m^3
Zzz = Lz*Lx^2/6
= 0.972 m^3

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WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	1.8X1.8X0.35X2.5 =	2.835
Column =	0.4X0.4X1.15X2.5 =	0.46
Soil =	((1.8X1.8)-(0.4X0.4))X1.15X1.9 =	6.73
	Total =	10.025

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK :

(Staad Group : F2)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	12	111 D+C+V	**0.167**	0.808	-0.003	-0.007	0	-0.393
Min Fx	12	106 D+C+V	**-0.174**	0.737	-0.003	-0.007	0	0.408
Max Fy	9	103 D+C+L	0	**1.067**	0.002	0.005	0	0
Min Fy	15	123 0.6*(D	-0.169	**0.311**	0	0	0	0.398
Max Fz	9	101 D+C+T	0	0.951	**0.002**	0.005	0	0
Min Fz	12	101 D+C+T	0	0.949	**-0.003**	-0.007	0	0
Max Mx	9	101 D+C+T	0	0.951	0.002	**0.005**	0	0
Min Mx	12	101 D+C+T	0	0.949	-0.003	**-0.007**	0	0
Max My	9	101 D+C+T	0	0.951	0.002	0.005	**0**	0
Min My	9	101 D+C+T	0	0.951	0.002	0.005	**0**	0
Max Mz	12	106 D+C+V	-0.174	0.737	-0.003	-0.007	0	**0.408**
Min Mz	12	111 D+C+V	0.167	0.808	-0.003	-0.007	0	**-0.393**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
111	10.833	0.167	0.5	5.417	32.44	1.5	O.K.
106	10.762	0.174	0.5	5.381	30.93	1.5	O.K.
103	11.092	0.002	0.5	5.546	2773	1.5	O.K.
123	10.336	0.169	0.5	5.168	30.58	1.5	O.K.
101	10.976	0.002	0.5	5.488	2744	1.5	O.K.
101	10.974	0.003	0.5	5.487	1829	1.5	O.K.
101	10.976	0.002	0.5	5.488	2744	1.5	O.K.
101	10.974	0.003	0.5	5.487	1829	1.5	O.K.
101	10.976	0.002	0.5	5.488	2744	1.5	O.K.
101	10.976	0.002	0.5	5.488	2744	1.5	O.K.
106	10.762	0.174	0.5	5.381	30.93	1.5	O.K.
111	10.833	0.167	0.5	5.417	32.44	1.5	O.K.

OVERTURNING CHECK:

L/C : 123 0.6*(D+C)+W_R1

Max Upword Force = 0 Mton

Self Wt. of Foundation = 10.025 Mton

Overturning Moment = 0 X 1.8/2 + 0 = 0 Mton-m

Resisting Moment = 10.025 X 0.9 X 1.8/2 = 8.12 Mton-m

F.O.S = >. 2 **O.K.**

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BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A (Mton/m^2)	Mx/Zxx (Mton/m^2)	Mz/Zzz (Mton/m^2)	BASE PRESSURE AT POINT A	B	C	D	Remarks
111	3.344	0.007	0.404	3.755	2.947	3.741	2.933	O.K.
106	3.322	0.007	0.42	3.749	2.909	3.735	2.895	O.K.
103	3.423	0.005	0	3.428	3.428	3.418	3.418	O.K.
123	3.19	0	0.409	3.599	2.781	3.599	2.781	O.K.
101	3.388	0.005	0	3.393	3.393	3.383	3.383	O.K.
101	3.387	0.007	0	3.394	3.394	3.38	3.38	O.K.
101	3.388	0.005	0	3.393	3.393	3.383	3.383	O.K.
101	3.387	0.007	0	3.394	3.394	3.38	3.38	O.K.
101	3.388	0.005	0	3.393	3.393	3.383	3.383	O.K.
101	3.388	0.005	0	3.393	3.393	3.383	3.383	O.K.
106	3.322	0.007	0.42	3.749	2.909	3.735	2.895	O.K.
111	3.344	0.007	0.404	3.755	2.947	3.741	2.933	O.K.
			MAX =	3.755	3.428	3.741	3.418	
			MIN =	3.393	2.781	3.38	2.781	

SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

(Staad Group : F2)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	12	215 1.2*(D	**0.268**	0.959	-0.004	-0.009	0	-0.63
Min Fx	12	210 1.2*(D	**-0.278**	0.846	-0.004	-0.009	0	0.653
Max Fy	9	203 1.4*(D	0	**1.43**	0.003	0.007	0	0
Min Fy	15	219 0.9*(D	-0.271	**0.442**	0	0	0	0.637
Max Fz	9	201 1.4*(D	0	1.357	**0.003**	0.007	0	0
Min Fz	12	201 1.4*(D	0	1.336	**-0.004**	-0.01	0	0
Max Mx	9	201 1.4*(D	0	1.357	0.003	**0.007**	0	0
Min Mx	12	201 1.4*(D	0	1.336	-0.004	**-0.01**	0	0
Max My	9	201 1.4*(D	0	1.357	0.003	0.007	**0**	0
Min My	9	201 1.4*(D	0	1.357	0.003	0.007	**0**	0
Max Mz	12	210 1.2*(D	-0.278	0.846	-0.004	-0.009	0	**0.653**
Min Mz	12	215 1.2*(D	0.268	0.959	-0.004	-0.009	0	**-0.63**

STRESS CALCULATION FOR STRENGTH DESIGN :

L/C	(Wf+Fy)/A (Mton/m^2)	Mx/Zx (Mton/m^2)	Mz/Zz (Mton/m^2)	BASE PRESSURE AT POINT A	B	C	D
215	3.39	0.009	0.648	4.047	2.751	4.029	2.733
210	3.355	0.009	0.672	4.036	2.692	4.018	2.674
203	3.535	0.007	0	3.542	3.542	3.528	3.528
219	3.231	0	0.655	3.886	2.576	3.886	2.576
201	3.513	0.007	0	3.52	3.52	3.506	3.506
201	3.506	0.01	0	3.516	3.516	3.496	3.496
201	3.513	0.007	0	3.52	3.52	3.506	3.506
201	3.506	0.01	0	3.516	3.516	3.496	3.496
201	3.513	0.007	0	3.52	3.52	3.506	3.506
201	3.513	0.007	0	3.52	3.52	3.506	3.506
210	3.355	0.009	0.672	4.036	2.692	4.018	2.674
215	3.39	0.009	0.648	4.047	2.751	4.029	2.733

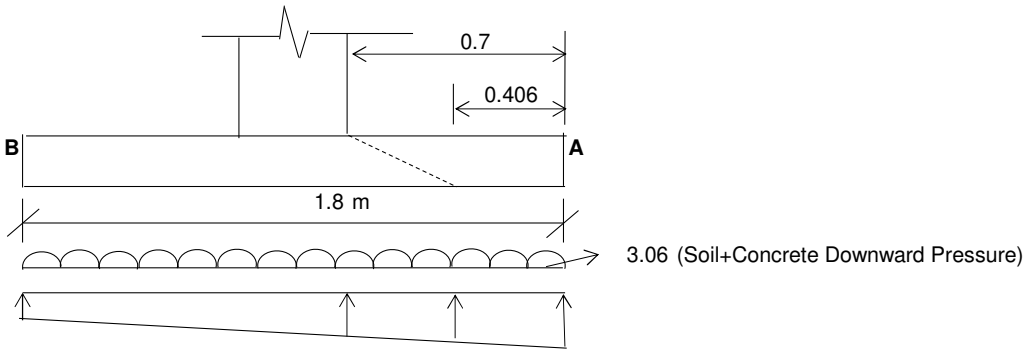
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PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point B (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
215	4.047	2.751	3.54	3.75	0.325	0.461
210	4.036	2.692	3.51	3.73	0.325	0.458
203	3.542	3.542	3.54	3.54	0.118	0.196
219	3.886	2.576	3.38	3.59	0.285	0.395
201	3.52	3.52	3.52	3.52	0.113	0.187
201	3.516	3.516	3.52	3.52	0.111	0.184
201	3.52	3.52	3.52	3.52	0.113	0.187
201	3.516	3.516	3.52	3.52	0.111	0.184
201	3.52	3.52	3.52	3.52	0.113	0.187
201	3.52	3.52	3.52	3.52	0.113	0.187
210	4.036	2.692	3.51	3.73	0.325	0.458
215	4.047	2.751	3.54	3.75	0.325	0.461
				MAX =	0.325	0.461

ALONG X - DIRECTION :

Mu = 0.325 Mton-m

Φ*Kn = Mu/bd^2 deff. = 294 mm

Mu/bd^2 = 0.325X10^7/(1000X294^2) = 0.038 N/mm^2 < 6.9 N/mm^2

Pt = 0.0001

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X294/100 = 5.29 cm^2

Provided **12** mm @ **150** C/C Provide Ast = 7.54 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0026

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ONE-WAY SHEAR CHECK

Vu = 0.461 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X0.461X10^4X294/0.325/10^7)X1000X294/10^4

= 28.371 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X294/10^4

= 29.129 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X294/10^4

= 50.44 Mton

HENCE, Vc = 28.371 Mton

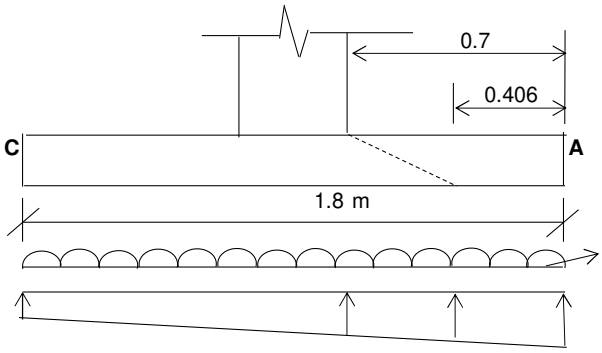
Φt*Vc = 0.75X28.371 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 21.278 Mton **> 0.461Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(28.371X10^4+0.66X35^0.5X1000X294)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 88.225 Mton **> Vu= 0.461Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point C (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
215	4.047	4.029	4.04	4.04	0.243	0.402
210	4.036	4.018	4.03	4.03	0.24	0.397
203	3.542	3.528	3.54	3.54	0.118	0.196
219	3.886	3.886	3.89	3.89	0.202	0.335
201	3.52	3.506	3.51	3.52	0.114	0.187
201	3.516	3.496	3.51	3.51	0.113	0.186
201	3.52	3.506	3.51	3.52	0.114	0.187
201	3.516	3.496	3.51	3.51	0.113	0.186
201	3.52	3.506	3.51	3.52	0.114	0.187
201	3.52	3.506	3.51	3.52	0.114	0.187
210	4.036	4.018	4.03	4.03	0.24	0.397
215	4.047	4.029	4.04	4.04	0.243	0.402
				MAX =	0.243	0.402

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ALONG Z - DIRECTION :

Mu = 0.243 T-m

Φ*Kn = Mu/bd^2 deff. = 294 mm

Mu/bd^2 = 0.243X10^7/(1000X294^2) = 0.028 N/mm^2 < 6.9 N/mm^2

Pt = 0.0001

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X294/100 = 5.29 cm^2

Provided **12** mm @ **150** C/C Provide Ast = 7.54 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0026

ONE-WAY SHEAR CHECK

Vu = 0.402 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X0.402X10^4X294/0.243/10^7)X1000X294/10^4

= 28.461 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X294/10^4

= 29.129 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X294/10^4

= 50.44 Mton

HENCE, Vc = 28.461 Mton

Φt*Vc = 0.75X28.461 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 21.346 Mton **> 0.402Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(28.461X10^4+0.66X35^0.5X1000X294)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 88.231 Mton **> Vu= 0.402Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 3.24 m^2

Perimeter (bo) = 2 X ((0.4 + 0.294) + (0.4 + 0.294)) = 2.776 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 3.755 Mton/m^2

Vu = ((3.24 - (0.4X0.4))X3.755

= 11.565 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X2.776X10^3X294/10^4

= 159.336 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X2.776X10^3X294/10^4

= 246.247 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X294/(2.776X10^3))X1X35^0.5X(2.776X10^3)X294/10^4

= 207.48 Mton

HENCE, Vc = 159.336 Mton

Φt*Vn = Φt*Vc = 0.75X159.336 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

119.502 Mton **> 11.565Mton , Hence O.K.**

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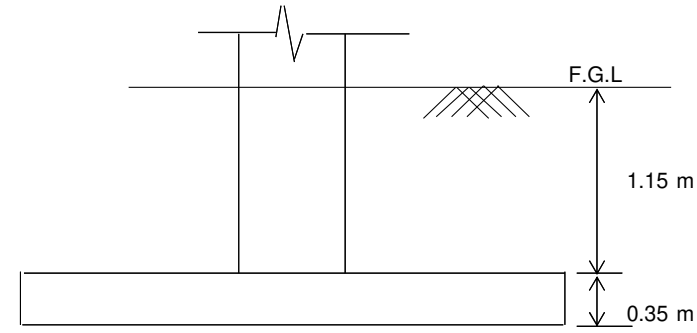
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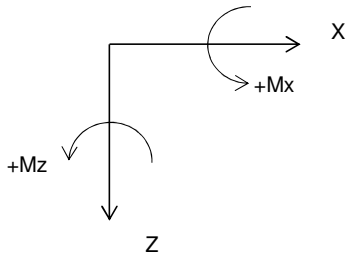
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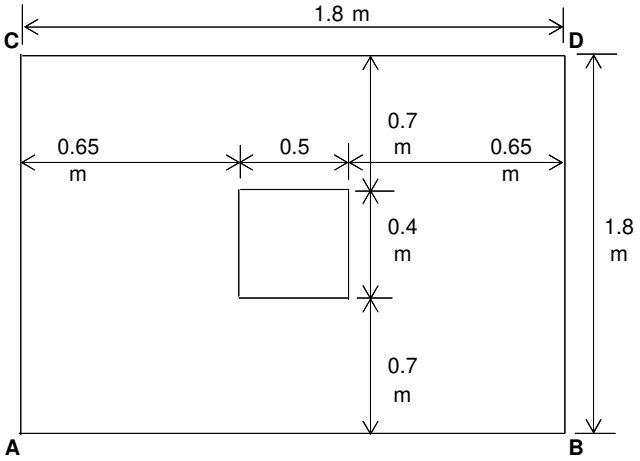
8.3 DESIGN OF FOUNDATION MKD. F3 :

(NODE NO. : 2 To 7 19 To 24)

Net Safe Bearing Capacity of Soil =	59 Mton/m^2	*(Ref. Nearest Bore Hole Mkd. BH-04)*
Depth of Foundation from FGL =	1.5 m	
Gross Allowable Bearing Capacity of Soil =	59 + 1.5 X 1.9 = 61.85 Mton/m^2	
Compressive Strength of Concrete , fc' =	35 N/mm^2	
Yield Strength of Main Reinf. , fy =	420 N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420 N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9	
Strength Reduction Factor of Shear Reinf. , Φt =	0.75	
Clear Cover =	50 mm	
Density of Concrete =	2.5 Mton/m^3	
Density of Dry Soil =	1.9 Mton/m^3	
Density of Saturated Soil =	2 Mton/m^3	
Density of Water =	1 Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	1.8 m
Width of Foundation (Lz) =	1.8 m
Thickness of Base Raft =	0.35 m
Length of Column (lx) =	0.5 m
Width of Column (lz) =	0.4 m

Area , A = Lx*Lz
= 3.24 m^2
Zxx = Lx*Lz^2/6
= 0.972 m^3
Zzz = Lz*Lx^2/6
= 0.972 m^3

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WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	1.8X1.8X0.35X2.5 =	2.835
Column =	0.5X0.4X1.15X2.5 =	0.575
Soil =	((1.8X1.8)-(0.5X0.4))X1.15X1.9 =	6.642
	Total =	10.052

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK :

(Staad Group : F3)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	7	111 D+C+V	**0.159**	0.511	-0.139	-0.327	0	-0.374
Min Fx	2	110 D+C+V	**-0.159**	0.652	-0.169	-0.398	0	0.374
Max Fy	2	114 D+C+(	0	**1.984**	0.104	0.244	0	0
Min Fy	2	127 0.6*(D	0	**-0.131**	-0.164	-0.386	0	0
Max Fz	2	103 D+C+L	0	1.662	**0.293**	0.689	0	0
Min Fz	19	122 0.6*(D	0	-0.037	**-0.498**	-1.17	0	0
Max Mx	2	103 D+C+L	0	1.662	0.293	**0.689**	0	0
Min Mx	19	122 0.6*(D	0	-0.037	-0.498	**-1.17**	0	0
Max My	2	101 D+C+T	0	1.253	0.067	0.156	**0**	0
Min My	2	101 D+C+T	0	1.253	0.067	0.156	**0**	0
Max Mz	2	110 D+C+V	-0.159	0.652	-0.169	-0.398	0	**0.374**
Min Mz	7	111 D+C+V	0.159	0.511	-0.139	-0.327	0	**-0.374**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
111	10.563	0.211	0.5	5.282	25.03	1.5	O.K.
110	10.704	0.232	0.5	5.352	23.07	1.5	O.K.
114	12.036	0.104	0.5	6.018	57.87	1.5	O.K.
127	9.921	0.164	0.5	4.961	30.25	1.5	O.K.
103	11.714	0.293	0.5	5.857	19.99	1.5	O.K.
122	10.015	0.498	0.5	5.008	10.06	1.5	O.K.
103	11.714	0.293	0.5	5.857	19.99	1.5	O.K.
122	10.015	0.498	0.5	5.008	10.06	1.5	O.K.
101	11.305	0.067	0.5	5.653	84.37	1.5	O.K.
101	11.305	0.067	0.5	5.653	84.37	1.5	O.K.
110	10.704	0.232	0.5	5.352	23.07	1.5	O.K.
111	10.563	0.211	0.5	5.282	25.03	1.5	O.K.

OVERTURNING CHECK:

L/C : 127 0.6*(D+C)+W_LG2

Max Upword Force = 0.131 Mton

Self Wt. of Foundation = 10.052 Mton

Overturning Moment = .131 X 1.8/2 + 0.386 = 0.504 Mton-m

Resisting Moment = 10.052 X 0.9 X 1.8/2 = 8.142 Mton-m

F.O.S = 16.15 >. 2 O.K.

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BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A	Mx/Zxx	Mz/Zzz	BASE PRESSURE AT POINT				Remarks
	(Mton/m^2)	(Mton/m^2)	(Mton/m^2)	A	B	C	D	
111	3.26	0.336	0.385	3.981	3.211	3.309	2.539	O.K.
110	3.304	0.409	0.385	4.098	3.328	3.28	2.51	O.K.
114	3.715	0.251	0	3.966	3.966	3.464	3.464	O.K.
127	3.062	0.397	0	3.459	3.459	2.665	2.665	O.K.
103	3.615	0.709	0	4.324	4.324	2.906	2.906	O.K.
122	3.091	1.204	0	4.295	4.295	1.887	1.887	O.K.
103	3.615	0.709	0	4.324	4.324	2.906	2.906	O.K.
122	3.091	1.204	0	4.295	4.295	1.887	1.887	O.K.
101	3.489	0.16	0	3.649	3.649	3.329	3.329	O.K.
101	3.489	0.16	0	3.649	3.649	3.329	3.329	O.K.
110	3.304	0.409	0.385	4.098	3.328	3.28	2.51	O.K.
111	3.26	0.336	0.385	3.981	3.211	3.309	2.539	O.K.
			MAX =	4.324	4.324	3.464	3.464	
			MIN =	3.459	3.211	1.887	1.887	

SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

(Staad Group : F3)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	7	215 1.2*(D	**0.255**	0.522	-0.132	-0.311	0	-0.598
Min Fx	2	214 1.2*(D	**-0.255**	0.746	-0.181	-0.425	0	0.598
Max Fy	2	210 1.2*(D	0	**2.613**	-0.091	-0.213	0	0
Min Fy	2	223 0.9*(D	0	**-0.285**	-0.267	-0.626	0	0
Max Fz	2	206 1.2*(D	0	1.983	**0.383**	0.899	0	0
Min Fz	19	218 0.9*(D	0	-0.137	**-0.8**	-1.879	0	0
Max Mx	2	206 1.2*(D	0	1.983	0.383	**0.899**	0	0
Min Mx	19	218 0.9*(D	0	-0.137	-0.8	**-1.879**	0	0
Max My	2	201 1.4*(D	0	1.754	0.087	0.205	**0**	0
Min My	2	201 1.4*(D	0	1.754	0.087	0.205	**0**	0
Max Mz	2	214 1.2*(D	-0.255	0.746	-0.181	-0.425	0	**0.598**
Min Mz	7	215 1.2*(D	0.255	0.522	-0.132	-0.311	0	**-0.598**

STRESS CALCULATION FOR STRENGTH DESIGN :

L/C	(Wf+Fy)/A	Mx/Zx	Mz/Zz	BASE PRESSURE AT POINT			
	(Mton/m^2)	(Mton/m^2)	(Mton/m^2)	A	B	C	D
215	3.264	0.32	0.615	4.199	2.969	3.559	2.329
214	3.333	0.437	0.615	4.385	3.155	3.511	2.281
210	3.909	0.219	0	4.128	4.128	3.69	3.69
223	3.015	0.644	0	3.659	3.659	2.371	2.371
206	3.715	0.925	0	4.64	4.64	2.79	2.79
218	3.06	1.933	0	4.993	4.993	1.127	1.127
206	3.715	0.925	0	4.64	4.64	2.79	2.79
218	3.06	1.933	0	4.993	4.993	1.127	1.127
201	3.644	0.211	0	3.855	3.855	3.433	3.433
201	3.644	0.211	0	3.855	3.855	3.433	3.433
214	3.333	0.437	0.615	4.385	3.155	3.511	2.281
215	3.264	0.32	0.615	4.199	2.969	3.559	2.329

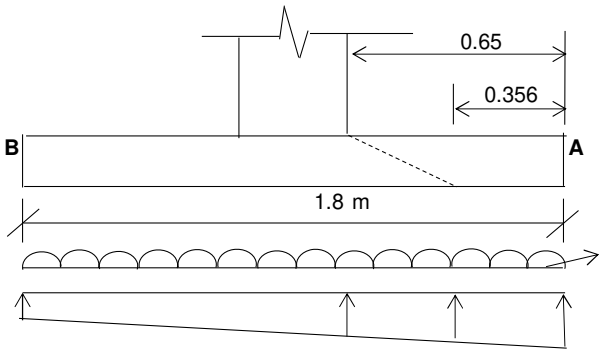
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PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point B (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
215	4.199	2.969	3.75	3.96	0.304	0.448
214	4.385	3.155	3.94	4.14	0.343	0.515
210	4.128	4.128	4.13	4.13	0.225	0.38
223	3.659	3.659	3.66	3.66	0.126	0.213
206	4.64	4.64	4.64	4.64	0.334	0.562
218	4.993	4.993	4.99	4.99	0.409	0.689
206	4.64	4.64	4.64	4.64	0.334	0.562
218	4.993	4.993	4.99	4.99	0.409	0.689
201	3.855	3.855	3.86	3.86	0.167	0.282
201	3.855	3.855	3.86	3.86	0.167	0.282
214	4.385	3.155	3.94	4.14	0.343	0.515
215	4.199	2.969	3.75	3.96	0.304	0.448
				MAX =	0.409	0.689

ALONG X - DIRECTION :

Mu = 0.409 Mton-m

Φ*Kn = Mu/bd^2 deff. = 294 mm

Mu/bd^2 = 0.409X10^7/(1000X294^2) = 0.047 N/mm^2 < 6.9 N/mm^2

Pt = 0.0001

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X294/100 = 5.29 cm^2

Provided **12** mm @ **150** C/C Provide Ast = 7.54 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0026

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ONE-WAY SHEAR CHECK

Vu = 0.689 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X0.689X10^4X294/0.409/10^7)X1000X294/10^4

= 28.473 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X294/10^4

= 29.129 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X294/10^4

= 50.44 Mton

HENCE, Vc = 28.473 Mton

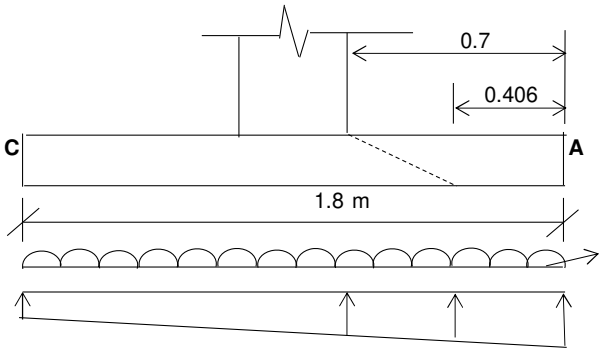
Φt*Vc = 0.75X28.473 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 21.355 Mton **> 0.689Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(28.473X10^4+0.66X35^0.5X1000X294)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 88.232 Mton **> Vu= 0.689Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point C (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
215	4.199	3.559	3.95	4.05	0.32	0.493
214	4.385	3.511	4.05	4.19	0.379	0.578
210	4.128	3.69	3.96	4.03	0.289	0.454
223	3.659	2.371	3.16	3.37	0.228	0.302
206	4.64	2.79	3.92	4.22	0.505	0.727
218	4.993	1.127	3.49	4.12	0.719	0.962
206	4.64	2.79	3.92	4.22	0.505	0.727
218	4.993	1.127	3.49	4.12	0.719	0.962
201	3.855	3.433	3.69	3.76	0.222	0.342
201	3.855	3.433	3.69	3.76	0.222	0.342
214	4.385	3.511	4.05	4.19	0.379	0.578
215	4.199	3.559	3.95	4.05	0.32	0.493
				MAX =	0.719	0.962

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ALONG Z - DIRECTION :

Mu = 0.719 T-m

$\Phi*Kn = Mu/bd^2$ deff. = 294 mm

Mu/bd^2 = 0.719X10^7/(1000X294^2) = 0.083 N/mm^2 < 6.9 N/mm^2

Pt = 0.0002

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X294/100 = 5.29 cm^2

Provided **12** mm @ **150** C/C Provide Ast = 7.54 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0026

ONE-WAY SHEAR CHECK

Vu = 0.962 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X0.962X10^4X294/0.719/10^7)X1000X294/10^4

= 28.34 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X294/10^4

= 29.129 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X294/10^4

= 50.44 Mton

HENCE, Vc = 28.34 Mton

Φt*Vc = 0.75X28.34 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 21.255 Mton **> 0.962Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(28.34X10^4+0.66X35^0.5X1000X294)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 88.222 Mton **> Vu= 0.962Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 3.24 m^2

Perimeter (bo) = 2 X ((0.5 + 0.294) + (0.4 + 0.294)) = 2.976 m

β = 1.25

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 4.324 $Mton/m^2$

Vu = ((3.24 - (0.5X0.4))X4.324

= 13.145 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X2.976X10^3X294/10^4

= 170.816 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1.25)X1X35^0.5X2.976X10^3X294/10^4

= 228.79 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X294/(2.976X10^3))X1X35^0.5X(2.976X10^3)X294/10^4

= 213.255 Mton

HENCE, Vc = 170.816 Mton

Φt*Vn = Φt*Vc = 0.75X170.816 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

128.112 Mton **> 13.145Mton , Hence O.K.**

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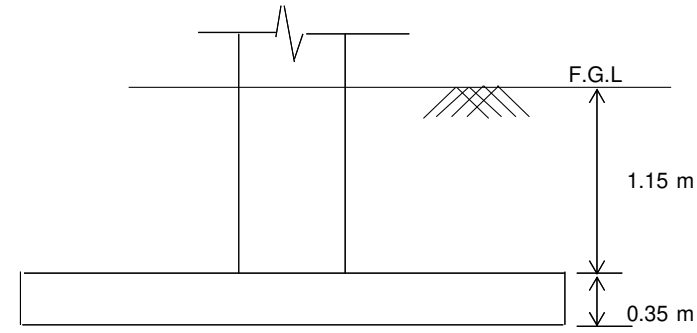
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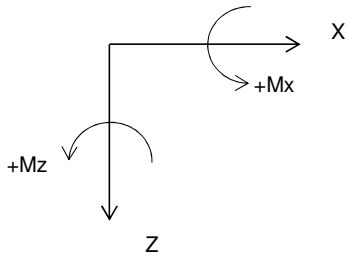
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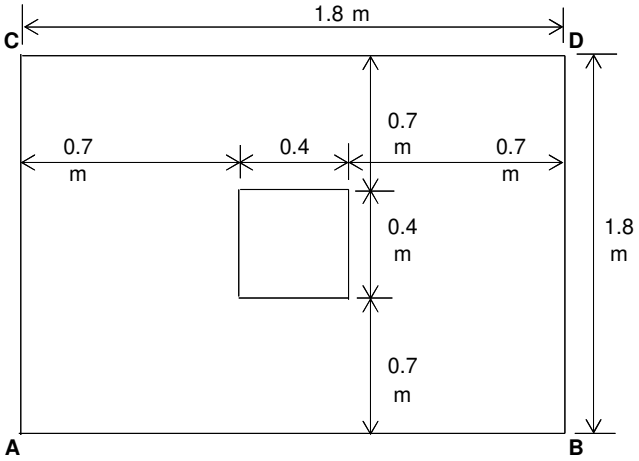
8.4 DESIGN OF FOUNDATION MKD. F4 :

(NODE NO. : 10 13 16)

Net Safe Bearing Capacity of Soil =	59	Mton/m^2	*(Ref. Nearest Bore Hole Mkd. BH-04)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	59 + 1.5 X 1.9 =	61.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	1.8	m
Width of Foundation (Lz) =	1.8	m
Thickness of Base Raft =	0.35	m
Length of Column (lx) =	0.4	m
Width of Column (lz) =	0.4	m

Area , A = Lx*Lz
= 3.24 m^2
Zxx = Lx*Lz^2/6
= 0.972 m^3
Zzz = Lz*Lx^2/6
= 0.972 m^3

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WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	1.8X1.8X0.35X2.5 =	2.835
Column =	0.4X0.4X1.15X2.5 =	0.46
Soil =	((1.8X1.8)-(0.4X0.4))X1.15X1.9 =	6.73
	Total =	10.025

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK :

(Staad Group : F4)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	10	101 D+C+T	**0**	0.903	0	0	0	0
Min Fx	10	101 D+C+T	**0**	0.903	0	0	0	0
Max Fy	10	101 D+C+T	0	**0.903**	0	0	0	0
Min Fy	10	122 0.6*(D	0	**0.542**	0.038	0.09	0	0
Max Fz	10	106 D+C+V	0	0.903	**0.038**	0.09	0	0
Min Fz	10	107 D+C+V	0	0.903	**-0.038**	-0.09	0	0
Max Mx	10	106 D+C+V	0	0.903	0.038	**0.09**	0	0
Min Mx	10	107 D+C+V	0	0.903	-0.038	**-0.09**	0	0
Max My	10	101 D+C+T	0	0.903	0	0	**0**	0
Min My	10	101 D+C+T	0	0.903	0	0	**0**	0
Max Mz	10	101 D+C+T	0	0.903	0	0	0	**0**
Min Mz	10	101 D+C+T	0	0.903	0	0	0	**0**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
101	10.928	0	0.5	5.464		1.5	O.K.
101	10.928	0	0.5	5.464		1.5	O.K.
101	10.928	0	0.5	5.464		1.5	O.K.
122	10.567	0.038	0.5	5.284	139.05	1.5	O.K.
106	10.928	0.038	0.5	5.464	143.79	1.5	O.K.
107	10.928	0.038	0.5	5.464	143.79	1.5	O.K.
106	10.928	0.038	0.5	5.464	143.79	1.5	O.K.
107	10.928	0.038	0.5	5.464	143.79	1.5	O.K.
101	10.928	0	0.5	5.464		1.5	O.K.
101	10.928	0	0.5	5.464		1.5	O.K.
101	10.928	0	0.5	5.464		1.5	O.K.
101	10.928	0	0.5	5.464		1.5	O.K.

OVERTURNING CHECK:

L/C : 122 0.6*(D+C)+W_L1

Max Upword Force = 0 Mton

Self Wt. of Foundation = 10.025 Mton

Overturning Moment = 0 X 1.8/2 + 0.09 = 0.09 Mton-m

Resisting Moment = 10.025 X 0.9 X 1.8/2 = 8.12 Mton-m

F.O.S = 90.22 >. 2 O.K.

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BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A	Mx/Zxx	Mz/Zzz	BASE PRESSURE AT POINT				Remarks
	(Mton/m^2)	(Mton/m^2)	(Mton/m^2)	A	B	C	D	
101	3.373	0	0	3.373	3.373	3.373	3.373	O.K.
101	3.373	0	0	3.373	3.373	3.373	3.373	O.K.
101	3.373	0	0	3.373	3.373	3.373	3.373	O.K.
122	3.261	0.093	0	3.354	3.354	3.168	3.168	O.K.
106	3.373	0.093	0	3.466	3.466	3.28	3.28	O.K.
107	3.373	0.093	0	3.466	3.466	3.28	3.28	O.K.
106	3.373	0.093	0	3.466	3.466	3.28	3.28	O.K.
107	3.373	0.093	0	3.466	3.466	3.28	3.28	O.K.
101	3.373	0	0	3.373	3.373	3.373	3.373	O.K.
101	3.373	0	0	3.373	3.373	3.373	3.373	O.K.
101	3.373	0	0	3.373	3.373	3.373	3.373	O.K.
101	3.373	0	0	3.373	3.373	3.373	3.373	O.K.
			MAX =	3.466	3.466	3.373	3.373	
			MIN =	3.354	3.354	3.168	3.168	

SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

(Staad Group : F4)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	10	201 1.4*(D	**0**	1.265	0	0	0	0
Min Fx	10	201 1.4*(D	**0**	1.265	0	0	0	0
Max Fy	10	201 1.4*(D	0	**1.265**	0	0	0	0
Min Fy	10	218 0.9*(D	0	**0.813**	0.062	0.145	0	0
Max Fz	10	210 1.2*(D	0	1.084	**0.062**	0.145	0	0
Min Fz	10	211 1.2*(D	0	1.084	**-0.062**	-0.145	0	0
Max Mx	10	210 1.2*(D	0	1.084	0.062	**0.145**	0	0
Min Mx	10	211 1.2*(D	0	1.084	-0.062	**-0.145**	0	0
Max My	10	201 1.4*(D	0	1.265	0	0	**0**	0
Min My	10	201 1.4*(D	0	1.265	0	0	**0**	0
Max Mz	10	201 1.4*(D	0	1.265	0	0	0	**0**
Min Mz	10	201 1.4*(D	0	1.265	0	0	0	**0**

STRESS CALCULATION FOR STRENGTH DESIGN :

L/C	(Wf+Fy)/A	Mx/Zx	Mz/Zz	BASE PRESSURE AT POINT			
	(Mton/m^2)	(Mton/m^2)	(Mton/m^2)	A	B	C	D
201	3.485	0	0	3.485	3.485	3.485	3.485
201	3.485	0	0	3.485	3.485	3.485	3.485
201	3.485	0	0	3.485	3.485	3.485	3.485
218	3.345	0.149	0	3.494	3.494	3.196	3.196
210	3.429	0.149	0	3.578	3.578	3.28	3.28
211	3.429	0.149	0	3.578	3.578	3.28	3.28
210	3.429	0.149	0	3.578	3.578	3.28	3.28
211	3.429	0.149	0	3.578	3.578	3.28	3.28
201	3.485	0	0	3.485	3.485	3.485	3.485
201	3.485	0	0	3.485	3.485	3.485	3.485
201	3.485	0	0	3.485	3.485	3.485	3.485
201	3.485	0	0	3.485	3.485	3.485	3.485

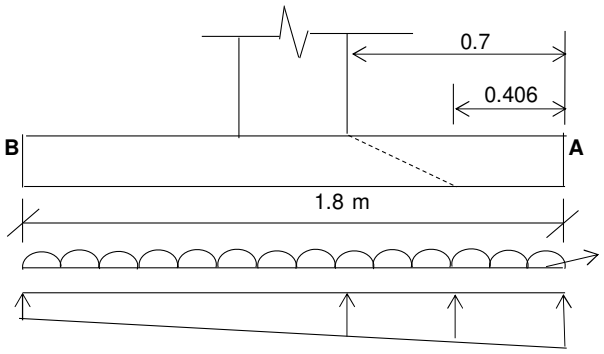
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PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point B (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
201	3.485	3.485	3.49	3.49	0.103	0.172
201	3.485	3.485	3.49	3.49	0.103	0.172
201	3.485	3.485	3.49	3.49	0.103	0.172
218	3.494	3.494	3.49	3.49	0.107	0.177
210	3.578	3.578	3.58	3.58	0.127	0.21
211	3.578	3.578	3.58	3.58	0.127	0.21
210	3.578	3.578	3.58	3.58	0.127	0.21
211	3.578	3.578	3.58	3.58	0.127	0.21
201	3.485	3.485	3.49	3.49	0.103	0.172
201	3.485	3.485	3.49	3.49	0.103	0.172
201	3.485	3.485	3.49	3.49	0.103	0.172
201	3.485	3.485	3.49	3.49	0.103	0.172
				MAX =	0.127	0.21

ALONG X - DIRECTION :

Mu = 0.127 Mton-m

Φ*Kn = Mu/bd^2 deff. = 294 mm

Mu/bd^2 = 0.127X10^7/(1000X294^2) = 0.015 N/mm^2 < 6.9 N/mm^2

Pt = 0.0000

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X294/100 = 5.29 cm^2

Provided **12** mm @ **150** C/C Provide Ast = 7.54 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0026

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ONE-WAY SHEAR CHECK

Vu = 0.21 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X0.21X10^4X294/0.127/10^7)X1000X294/10^4

= 28.461 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X294/10^4

= 29.129 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X294/10^4

= 50.44 Mton

HENCE, Vc = 28.461 Mton

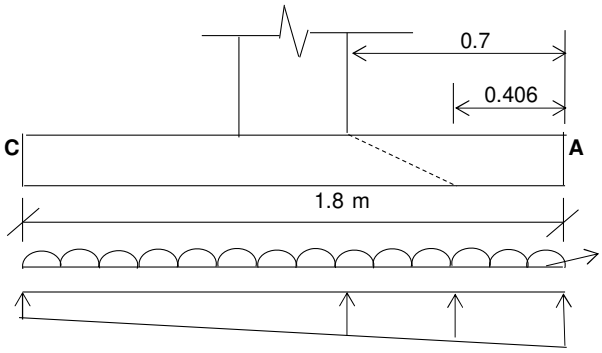
Φt*Vc = 0.75X28.461 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 21.346 Mton **> 0.21Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(28.461X10^4+0.66X35^0.5X1000X294)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 88.231 Mton **> Vu= 0.21Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point C (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
201	3.485	3.485	3.49	3.49	0.103	0.172
201	3.485	3.485	3.49	3.49	0.103	0.172
201	3.485	3.485	3.49	3.49	0.103	0.172
218	3.494	3.196	3.38	3.43	0.125	0.189
210	3.578	3.28	3.46	3.51	0.146	0.224
211	3.578	3.28	3.46	3.51	0.146	0.224
210	3.578	3.28	3.46	3.51	0.146	0.224
211	3.578	3.28	3.46	3.51	0.146	0.224
201	3.485	3.485	3.49	3.49	0.103	0.172
201	3.485	3.485	3.49	3.49	0.103	0.172
201	3.485	3.485	3.49	3.49	0.103	0.172
201	3.485	3.485	3.49	3.49	0.103	0.172
				MAX =	0.146	0.224

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ALONG Z - DIRECTION :

Mu = 0.146 T-m

$\Phi*Kn = Mu/bd^2$ deff. = 294 mm

Mu/bd^2 = 0.146X10^7/(1000X294^2) = 0.017 N/mm^2 < 6.9 N/mm^2

Pt = 0.0000

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X294/100 = 5.29 cm^2

Provided **12** mm @ **150** C/C Provide Ast = 7.54 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0026

ONE-WAY SHEAR CHECK

Vu = 0.224 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X0.224X10^4X294/0.146/10^7)X1000X294/10^4

= 28.415 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X294/10^4

= 29.129 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X294/10^4

= 50.44 Mton

HENCE, Vc = 28.415 Mton

Φt*Vc = 0.75X28.415 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 21.311 Mton **> 0.224Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(28.415X10^4+0.66X35^0.5X1000X294)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 88.228 Mton **> Vu= 0.224Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 3.24 m^2

Perimeter (bo) = 2 X ((0.4 + 0.294) + (0.4 + 0.294)) = 2.776 m

β = 1

αs = 40 (Interior Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 3.466 $Mton/m^2$

Vu = ((3.24 - (0.4X0.4))X3.466

= 10.675 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X2.776X10^3X294/10^4

= 159.336 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X2.776X10^3X294/10^4

= 246.247 Mton

[c] = $0.083*(2+\alpha s*d.eff/b_o)*\lambda*fc'^{0.5}*bo*d.eff$

= 0.083X(2+40X294/(2.776X10^3))X1X35^0.5X(2.776X10^3)X294/10^4

= 249.923 Mton

HENCE, Vc = 159.336 Mton

Φt*Vn = Φt*Vc = 0.75X159.336 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

119.502 Mton **> 10.675Mton , Hence O.K.**

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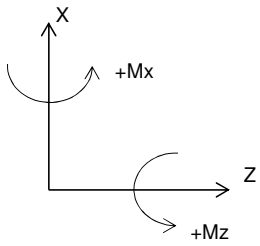
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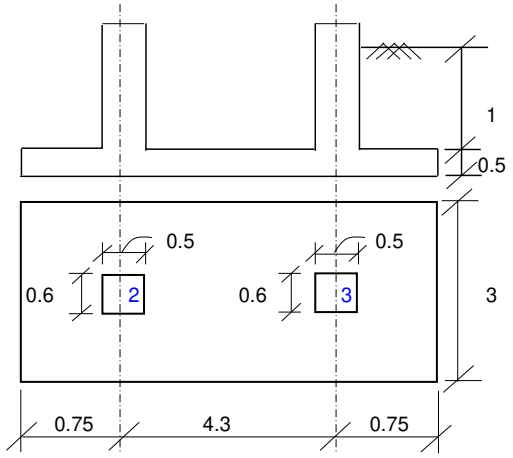
8.5 DESIGN OF FOUNDATION MKD. F5 :

(For Staad Node. No. - 2 & 3)

Net Safe Bearing Capacity of Soil =	32.5	Mton/m^2	*(Ref. Nearest Bore Hole Mkd. BH-04)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	32.5 + 1.5 X 1.9 =	35.35	Mton/m^2

Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinf. , Φ =	0.9	
Strength Reduction Factor of Shear Reinf. , Φt =	0.75	
Clear Cover =	75	mm
Density of Concrete =	2.5	Mton/m^3
Density of Dry Soil =	1.9	Mton/m^3
Density of Saturated Soil =	2	Mton/m^3
Density of Water =	1	Mton/m^3





FOUNDATION DIMENSION :

Cantilever from Node No. 2 =	0.75	m
Cantilever from Node No. 3 =	0.75	m
C/C Distance between Column =	4.3	m
Length of Foundation (Lx) =	5.8	m
Width of Foundation (Lz) =	3	m
Thickness of Raft =	0.5	m
Column size at Node No. 2 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.5	m
Height =	1.000	m
Column size at Node No. 3 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.5	m
Height =	1.000	m
Area , A = Lx*Lz =	17.4	m^2
Zxx = Lx*Lz^2/6 =	8.7	m^3
Zzz = Lz*Lx^2/6 =	16.82	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	5.8X3X0.5X2.5 =	21.75
Column =	(0.6X0.5+0.6X0.5)X1X2.5 =	1.5
Soil =	((5.8X3)-(0.6X0.5)-(0.6X0.5))X1X1.9 =	31.92
	Total =	55.17

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STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : SMBS BAG SPLITTER FOUNDATION_R0)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	2	0	4.259	0.081	0.166	0	3	0	4.259	0.081	0.166	0
102	2	-0.123	5.602	4.031	8.264	0.252	3	-0.123	5.602	4.031	8.264	0.252
103	2	0.027	3.827	0	0	-0.055	3	0.027	3.827	0	0	-0.055
104	2	0.016	4.414	0.061	0.125	-0.032	3	0.016	4.414	0.061	0.125	-0.032
105	2	-0.077	5.421	3.023	6.198	0.157	3	-0.077	5.421	3.023	6.198	0.157
106	2	-1.568	-0.13	0.58	1.189	3.214	3	-1.568	-0.13	0.58	1.189	3.214
107	2	-1.646	0.229	0.579	1.187	3.374	3	-1.646	0.229	0.579	1.187	3.374
108	2	0.514	6.09	0.158	0.324	-1.054	3	0.514	6.09	0.158	0.324	-1.054
109	2	0.45	6.389	0.068	0.139	-0.923	3	0.45	6.389	0.068	0.139	-0.923
110	2	0.157	-9.183	3.526	7.229	-0.322	3	0.157	2.609	6.004	12.309	-0.322
111	2	0.092	2.94	1.048	2.148	-0.189	3	0.092	-8.852	3.526	7.229	-0.189
112	2	-0.221	2.892	0.081	0.166	0.453	3	-0.221	2.892	0.081	0.166	0.453
113	2	0.088	3.705	0	0	-0.18	3	0.088	3.705	0	0	-0.18
114	2	0.009	3.652	0	0	-0.018	3	0.009	4.392	0.204	0.418	-0.018
115	2	-0.001	2.952	0.202	0.414	0.002	3	-0.001	2.212	0.406	0.832	0.002
116	2	-1.185	2.289	0.435	0.892	2.429	3	-1.185	2.289	0.435	0.892	2.429
117	2	-1.243	2.559	0.434	0.89	2.549	3	-1.243	2.559	0.434	0.89	2.549
118	2	0.377	6.954	0.119	0.243	-0.772	3	0.377	6.954	0.119	0.243	-0.772
119	2	0.329	7.179	0.051	0.105	-0.673	3	0.329	7.179	0.051	0.105	-0.673
120	2	0.109	-4.5	2.645	5.421	-0.223	3	0.109	4.344	4.503	9.231	-0.223
121	2	0.06	4.592	0.786	1.611	-0.123	3	0.06	-4.252	2.645	5.421	-0.123
122	2	-0.123	4.648	0.043	0.087	0.252	3	-0.123	4.648	0.043	0.087	0.252
123	2	0.039	5.075	0	0	-0.08	3	0.039	5.075	0	0	-0.08
124	2	-0.002	5.047	0	0	0.005	3	-0.002	5.435	0.107	0.22	0.005
125	2	-0.008	4.679	0.106	0.217	0.015	3	-0.008	4.291	0.213	0.437	0.015
126	2	-1.572	-1.45	0.58	1.189	3.222	3	-1.572	-1.45	0.58	1.189	3.222
127	2	-1.65	-1.091	0.579	1.187	3.382	3	-1.65	-1.091	0.579	1.187	3.382
128	2	0.51	4.77	0.158	0.324	-1.046	3	0.51	4.77	0.158	0.324	-1.046
129	2	0.446	5.069	0.068	0.139	-0.915	3	0.446	5.069	0.068	0.139	-0.915
130	2	0.153	-10.503	3.526	7.229	-0.314	3	0.153	1.289	6.004	12.309	-0.314
131	2	0.088	1.62	1.048	2.148	-0.181	3	0.088	-10.172	3.526	7.229	-0.181
132	2	-0.156	1.694	0.057	0.116	0.319	3	-0.156	1.694	0.057	0.116	0.319
133	2	0.061	2.263	0	0	-0.124	3	0.061	2.263	0	0	-0.124
134	2	0.005	2.226	0	0	-0.011	3	0.005	2.744	0.143	0.293	-0.011
135	2	-0.002	1.736	0.141	0.29	0.003	3	-0.002	1.218	0.284	0.583	0.003

CHECK FOR OVER TURNING

L/C : 130 *(Max. Tension at Node No. 2)*

O.T.M. = (10.503 X 5.05) + (7.229 + 12.309)
= 72.578 Mton-m
R.M. = (1.289X0.75) + (55.17X0.9X2.9)
= 144.96 Mton-m
F.O.S. = 2.000 > 1.5 **SAFE**

L/C : 131 *(Max. Tension at Node No. 3)*

O.T.M. = (10.172 X 5.05) + (2.148 + 7.229)
= 60.746 Mton-m
R.M. = (1.62X0.75) + (55.17X0.9X2.9)
= 145.209 Mton-m
F.O.S. = 2.390 > 1.5 **SAFE**

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CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 110

Max. Horizontal shear , ΣFx^2+ΣFz^2 = 9.54 Mton
Corresponding vertical load , ΣFy = -6.574 Mton
90% Wt. if Foundation , 0.9*Wf = 49.653 Mton
Total Downward load , V = 43.079 Mton
Horizontal frictional coeff. μ = 0.5 *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*
Resisting shear force = μ * V = 21.540 Mton
F.O.S. = 2.260 > 1.5 **SAFE** *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

BASE PRESSURE :

						BASE PRESSURE		
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.	REMARKS
101	0.000	8.518	0.162	0.332	0.000	3.68	3.64	O.K.
102	-0.246	11.204	8.062	16.528	0.504	4.855	2.774	O.K.
103	0.054	7.654	0.000	0.000	-0.110	3.623	3.598	O.K.
104	0.032	8.828	0.122	0.250	-0.064	3.7	3.656	O.K.
105	-0.154	10.842	6.046	12.396	0.314	4.567	3.021	O.K.
106	-3.136	-0.260	1.160	2.378	6.428	4.036	2.276	O.K.
107	-3.292	0.458	1.158	2.374	6.748	4.114	2.28	O.K.
108	1.028	12.180	0.316	0.648	-2.108	4.152	3.59	O.K.
109	0.900	12.778	0.136	0.278	-1.846	4.134	3.676	O.K.
110	0.314	-6.574	9.530	44.891	-0.644	5.536	0.05	O.K.
111	0.184	-5.912	4.574	-15.976	-0.378	3.824	1.838	O.K.
112	-0.442	5.784	0.162	0.332	0.906	3.627	3.379	O.K.
113	0.176	7.410	0.000	0.000	-0.360	3.638	3.555	O.K.
114	0.018	8.044	0.204	2.009	-0.036	3.757	3.509	O.K.
115	-0.002	5.164	0.608	-0.345	0.004	3.488	3.447	O.K.
116	-2.370	4.578	0.870	1.784	4.858	4.098	2.769	O.K.
117	-2.486	5.118	0.868	1.780	5.098	4.157	2.773	O.K.
118	0.754	13.908	0.238	0.486	-1.544	4.176	3.764	O.K.
119	0.658	14.358	0.102	0.210	-1.346	4.163	3.829	O.K.
120	0.218	-0.156	7.148	33.667	-0.446	5.215	1.109	O.K.
121	0.120	0.340	3.431	-11.983	-0.246	3.931	2.45	O.K.
122	-0.246	9.296	0.086	0.174	0.504	3.773	3.637	O.K.
123	0.078	10.150	0.000	0.000	-0.160	3.772	3.736	O.K.
124	-0.004	10.482	0.107	1.054	0.010	3.837	3.709	O.K.
125	-0.016	8.970	0.319	-0.180	0.030	3.7	3.672	O.K.
126	-3.144	-2.900	1.160	2.378	6.444	3.886	2.122	O.K.
127	-3.300	-2.182	1.158	2.374	6.764	3.964	2.127	O.K.
128	1.020	9.540	0.316	0.648	-2.092	3.998	3.44	O.K.
129	0.892	10.138	0.136	0.278	-1.830	3.98	3.526	O.K.
130	0.306	-9.214	9.530	44.891	-0.628	5.382	**-0.1**	O.K.
131	0.176	-8.552	4.574	-15.976	-0.362	3.671	1.688	O.K.
132	-0.312	3.388	0.114	0.232	0.638	3.453	3.278	O.K.
133	0.122	4.526	0.000	0.000	-0.248	3.459	3.402	O.K.
134	0.010	4.970	0.143	1.407	-0.022	3.543	3.37	O.K.
135	-0.004	2.954	0.425	-0.241	0.006	3.355	3.325	O.K.
MAX. BASE PRESSURE FOR , L/C : 110						5.536		
MIN. BASE PRESSURE FOR , L/C : 130							-0.100	

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STAAD OUTPUT FOR STRENGTH DESIGN : : (Staad File : SMBS BAG SPLITTER FOUNDATION_R0)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	2	0.013	4.62	0	0	-0.026	3	0.013	4.62	0	0	-0.026
202	2	-0.004	5.996	0.097	0.199	0.009	3	-0.004	5.996	0.097	0.199	0.009
203	2	-0.152	7.608	4.837	9.917	0.311	3	-0.152	7.608	4.837	9.917	0.311
204	2	0.023	5.581	0	0	-0.047	3	0.023	5.581	0	0	-0.047
205	2	-1.222	2.059	0.464	0.951	2.505	3	-1.222	2.059	0.464	0.951	2.505
206	2	-1.284	2.346	0.463	0.95	2.633	3	-1.284	2.346	0.463	0.95	2.633
207	2	0.444	7.035	0.126	0.259	-0.909	3	0.444	7.035	0.126	0.259	-0.909
208	2	0.392	7.274	0.054	0.112	-0.804	3	0.392	7.274	0.054	0.112	-0.804
209	2	0.158	-5.184	2.821	5.783	-0.324	3	0.158	4.25	4.803	9.847	-0.324
210	2	0.106	4.515	0.838	1.719	-0.217	3	0.106	-4.919	2.821	5.783	-0.217
211	2	-2.52	-0.487	0.928	1.902	5.166	3	-2.52	-0.487	0.928	1.902	5.166
212	2	-2.645	0.087	0.926	1.899	5.422	3	-2.645	0.087	0.926	1.899	5.422
213	2	0.811	9.465	0.253	0.518	-1.663	3	0.811	9.465	0.253	0.518	-1.663
214	2	0.709	9.943	0.109	0.223	-1.453	3	0.709	9.943	0.109	0.223	-1.453
215	2	0.24	-14.972	5.642	11.566	-0.492	3	0.24	3.895	9.606	19.694	-0.492
216	2	0.136	4.425	1.677	3.438	-0.279	3	0.136	-14.442	5.642	11.566	-0.279
217	2	-0.236	4.33	0.081	0.166	0.483	3	-0.236	4.33	0.081	0.166	0.483
218	2	0.073	5.143	0	0	-0.15	3	0.073	5.143	0	0	-0.15
219	2	-0.006	5.09	0	0	0.012	3	-0.006	5.83	0.204	0.418	0.012
220	2	-0.016	4.39	0.202	0.414	0.032	3	-0.016	3.65	0.406	0.832	0.032
221	2	-2.515	-2.518	0.928	1.902	5.156	3	-2.515	-2.518	0.928	1.902	5.156
222	2	-2.64	-1.944	0.926	1.899	5.412	3	-2.64	-1.944	0.926	1.899	5.412
223	2	0.816	7.434	0.253	0.518	-1.673	3	0.816	7.434	0.253	0.518	-1.673
224	2	0.714	7.912	0.109	0.223	-1.463	3	0.714	7.912	0.109	0.223	-1.463
225	2	0.245	-17.003	5.642	11.566	-0.502	3	0.245	1.864	9.606	19.694	-0.502
226	2	0.141	2.394	1.677	3.438	-0.289	3	0.141	-16.474	5.642	11.566	-0.289
227	2	-0.222	2.562	0.081	0.166	0.455	3	-0.222	2.562	0.081	0.166	0.455
228	2	0.087	3.375	0	0	-0.179	3	0.087	3.375	0	0	-0.179
229	2	0.008	3.322	0	0	-0.017	3	0.008	4.062	0.204	0.418	-0.017
230	2	-0.002	2.622	0.202	0.414	0.004	3	-0.002	1.882	0.406	0.832	0.004

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STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	0.026	9.240	0.000	0.000	-0.052	3.708	3.696
202	-0.008	11.992	0.194	0.398	0.018	3.886	3.834
203	-0.304	15.216	9.674	19.834	0.622	5.296	2.794
204	0.046	11.162	0.000	0.000	-0.094	3.823	3.801
205	-2.444	4.118	0.928	1.902	5.010	4.096	2.718
206	-2.568	4.692	0.926	1.900	5.266	4.159	2.722
207	0.888	14.070	0.252	0.518	-1.818	4.219	3.74
208	0.784	14.548	0.108	0.224	-1.608	4.205	3.809
209	0.316	-0.934	7.624	35.913	-0.648	5.327	0.907
210	0.212	-0.404	3.659	-12.781	-0.434	3.957	2.338
211	-5.040	-0.974	1.856	3.804	10.332	4.528	1.701
212	-5.290	0.174	1.852	3.798	10.844	4.653	1.708
213	1.622	18.930	0.506	1.036	-3.326	4.703	3.815
214	1.418	19.886	0.218	0.446	-2.906	4.674	3.953
215	0.480	-11.077	15.248	71.824	-0.984	6.917	-1.849
216	0.272	-10.017	7.319	-25.560	-0.558	4.179	1.011
217	-0.472	8.660	0.162	0.332	0.966	3.799	3.538
218	0.146	10.286	0.000	0.000	-0.300	3.796	3.727
219	-0.012	10.920	0.204	2.009	0.024	3.92	3.676
220	-0.032	8.040	0.608	-0.345	0.064	3.661	3.605
221	-5.030	-5.036	1.856	3.804	10.312	4.293	1.47
222	-5.280	-3.888	1.852	3.798	10.824	4.417	1.477
223	1.632	14.868	0.506	1.036	-3.346	4.471	3.579
224	1.428	15.824	0.218	0.446	-2.926	4.443	3.717
225	0.490	-15.139	15.248	71.824	-1.004	6.686	-2.085
226	0.282	-14.080	7.319	-25.562	-0.578	3.948	0.775
227	-0.444	5.124	0.162	0.332	0.910	3.59	3.341
228	0.174	6.750	0.000	0.000	-0.358	3.6	3.517
229	0.016	7.384	0.204	2.009	-0.034	3.718	3.472
230	-0.004	4.504	0.608	-0.345	0.008	3.451	3.408

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DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	3.708	0.558	1.007	0.828	0.07	0.046	0.402	0.437
202	3.886	0.736	1.328	1.091	0.092	0.061	0.53	0.576
203	5.296	2.146	3.874	3.183	0.268	0.178	1.545	1.68
204	3.823	0.673	1.215	0.998	0.084	0.056	0.485	0.527
205	4.096	0.946	1.708	1.403	0.118	0.079	0.681	0.741
206	4.159	1.009	1.821	1.496	0.126	0.084	0.726	0.79
207	4.219	1.069	1.93	1.585	0.134	0.089	0.77	0.837
208	4.205	1.055	1.904	1.565	0.132	0.088	0.76	0.826
209	5.327	2.177	3.929	3.228	0.272	0.181	1.567	1.705
210	3.957	0.807	1.457	1.197	0.101	0.067	0.581	0.632
211	4.528	1.378	2.487	2.044	0.172	0.114	0.992	1.079
212	4.653	1.503	2.713	2.229	0.188	0.125	1.082	1.177
213	4.703	1.553	2.803	2.303	0.194	0.129	1.118	1.216
214	4.674	1.524	2.751	2.26	0.191	0.126	1.097	1.193
215	6.917	3.767	6.799	5.586	0.471	0.313	2.712	2.95
216	4.179	1.029	1.857	1.526	0.129	0.085	0.741	0.806
217	3.799	0.649	1.171	0.962	0.081	0.054	0.467	0.508
218	3.796	0.646	1.166	0.958	0.081	0.054	0.465	0.506
219	3.92	0.77	1.39	1.142	0.096	0.064	0.554	0.603
220	3.661	0.511	0.922	0.758	0.064	0.042	0.368	0.4
221	4.293	1.143	2.063	1.695	0.143	0.095	0.823	0.895
222	4.417	1.267	2.287	1.879	0.158	0.105	0.912	0.992
223	4.471	1.321	2.384	1.959	0.165	0.11	0.951	1.034
224	4.443	1.293	2.334	1.918	0.162	0.107	0.931	1.012
225	6.686	3.536	6.382	5.244	0.442	0.293	2.546	2.769
226	3.948	0.798	1.44	1.183	0.1	0.066	0.575	0.625
227	3.59	0.44	0.794	0.653	0.055	0.037	0.317	0.345
228	3.6	0.45	0.812	0.667	0.056	0.037	0.324	0.352
229	3.718	0.568	1.025	0.842	0.071	0.047	0.409	0.445
230	3.451	0.301	0.543	0.446	0.038	0.025	0.217	0.236
		MAX. =	**6.799**	**5.586**	**0.471**	**0.313**	**2.712**	**2.95**

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TOP LONGITUDINAL REINFORCEMENT :

Mu = 6.799 Mton-m
Φ*Kn = Mu/bd^2 deff. = 417 mm
Mu/bd^2 = 6.799X10^7/(1000X417^2) = 0.391 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X417/100 = 7.51 cm^2
Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm^2 at Top - Longitudinal **O.K.**
Total Provide Pt = 0.0032

ONE-WAY SHEAR CHECK

Vu = 5.586 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X5.586X10^4X417/6.799/10^7)X1000X417/10^4
= 40.249 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X417/10^4
= 41.741 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X417/10^4
= 71.543 Mton
HENCE, Vc = 40.249 Mton
Φt*Vc = 0.75X40.249 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 30.187 Mton **> 5.586Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(40.249X10^4+0.66X35^0.5X1000X417)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 125.135 Mton **> Vu= 5.586Mton , CROSS SECTION O.K.**

BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 0.471 Mton-m
Φ*Kn = Mu/bd^2 deff. = 417 mm
Mu/bd^2 = 0.471X10^7/(1000X417^2) = 0.027 N/mm^2 < 6.9 N/mm^2
Pt = 0.0001
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X417/100 = 7.51 cm^2
Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm^2 at Bottom - Longitudinal **O.K.**
Total Provide Pt = 0.0032

ONE-WAY SHEAR CHECK

Vu = 0.313 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X0.313X10^4X417/0.471/10^7)X1000X417/10^4
= 40.101 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X417/10^4
= 41.741 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X417/10^4
= 71.543 Mton
HENCE, Vc = 40.101 Mton
Φt*Vc = 0.75X40.101 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 30.076 Mton **> 0.313Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(40.101X10^4+0.66X35^0.5X1000X417)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 125.124 Mton **> Vu= 0.313Mton , CROSS SECTION O.K.**

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BOTTOM CROSS REINFORCEMENT :

Mu = 2.712 Mton-m
Φ*Kn = Mu/bd^2 deff. = 417 mm
Mu/bd^2 = 2.712X10^7/(1000X417^2) = 0.156 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X417/100 = 7.51 cm^2
Provided **16** mm @ **200** C/C Provide Ast = 10.05 cm^2 at Bottom - Cross **O.K.**
Total Provide Pt = 0.0024

ONE-WAY SHEAR CHECK

Vu = 2.95 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0024X2.95X10^4X417/2.712/10^7)X1000X417/10^4
= 40.244 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0024)X1000X417/10^4
= 41.173 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X417/10^4
= 71.543 Mton
HENCE, Vc = 40.244 Mton
Φt*Vc = 0.75X40.244 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 30.183 Mton **> 2.95Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(40.244X10^4+0.66X35^0.5X1000X417)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 125.135 Mton **> Vu= 2.95Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 17.4 m^2
Perimeter (bo) = 2 X ((0.5+0.417) + (0.6+0.417) + (0.5+0.417) + (0.6 + 0.417)) = 7.736 m
β = 1.2
αs = 20 (Corner Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*
Pmax = 5.536 $Mton/m^2$
Vu = ((17.4 - (0.5X0.6) - (0.5X0.6))X5.536
= 93.005 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*
[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1
= 0.33X1X35^0.5X7.736X10^3X417/10^4
= 629.797 Mton
[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff
= 0.17X(1+2/1.2)X1X35^0.5X7.736X10^3X417/10^4
= 865.175 Mton
[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff
= 0.083X(2+20X417/(7.736X10^3))X1X35^0.5X(7.736X10^3)X417/10^4
= 487.578 Mton
HENCE, Vc = 487.578 Mton
Φt*Vn = Φt*Vc = 0.75X487.578 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
365.684 Mton **> 93.005Mton , Hence O.K.**

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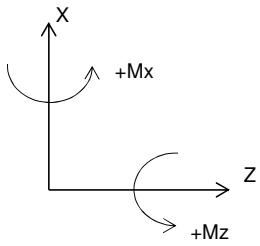
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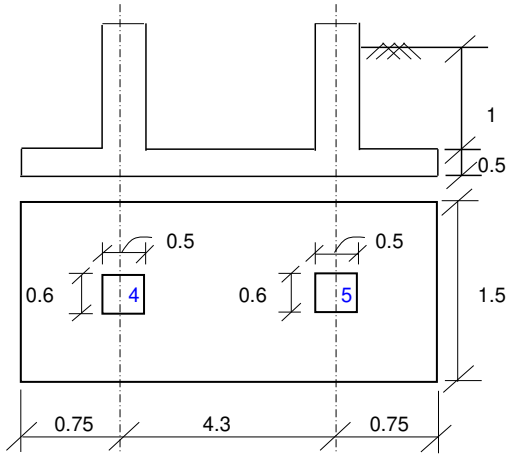
8.6 DESIGN OF FOUNDATION MKD. F6 :

(For Staad Node. No. - 4 & 5)

Net Safe Bearing Capacity of Soil =	32.5 Mton/m^2	*(Ref. Nearest Bore Hole Mkd. BH-04)*
Depth of Foundation from FGL =	1.5 m	
Gross Allowable Bearing Capacity of Soil =	32.5 + 1.5 X 1.9 = 35.35 Mton/m^2	

Compressive Strength of Concrete , fc' =	35 N/mm^2
Yield Strength of Main Reinf. , fy =	420 N/mm^2
Yield Strength of Shear Reinf. , fyt =	420 N/mm^2
Strength Reduction Factor of Main Reinf. , Φ =	0.9
Strength Reduction Factor of Shear Reinf. , Φt =	0.75
Clear Cover =	75 mm
Density of Concrete =	2.5 Mton/m^3
Density of Dry Soil =	1.9 Mton/m^3
Density of Saturated Soil =	2 Mton/m^3
Density of Water =	1 Mton/m^3





FOUNDATION DIMENSION :

Cantilever from Node No. 4 =	0.75 m
Cantilever from Node No. 5 =	0.75 m
C/C Distance between Column =	4.3 m
Length of Foundation (Lx) =	5.8 m
Width of Foundation (Lz) =	1.5 m
Thickness of Raft =	0.5 m
Column size at Node No. 4 :	
Length along X-Direction =	0.6 m
Length along Z-Direction =	0.5 m
Height =	1.000 m
Column size at Node No. 5 :	
Length along X-Direction =	0.6 m
Length along Z-Direction =	0.5 m
Height =	1.000 m
Area , A = Lx*Lz =	8.7 m^2
Zxx = Lx*Lz^2/6 =	2.175 m^3
Zzz = Lz*Lx^2/6 =	8.41 m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	5.8X1.5X0.5X2.5 =	10.875
Column =	(0.6X0.5+0.6X0.5)X1X2.5 =	1.5
Soil =	((5.8X1.5)-(0.6X0.5)-(0.6X0.5))X1X1.9 =	15.39
	Total =	27.765

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STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : SMBS BAG SPLITTER FOUNDATION_R0)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	4	-0.041	3.642	-0.035	-0.072	0.084	5	-0.041	3.642	-0.035	-0.072	0.084
102	4	0.808	1.372	-4.311	-8.838	-1.656	5	0.808	1.372	-4.311	-8.838	-1.656
103	4	-0.135	3.534	0	0	0.277	5	-0.135	3.534	0	0	0.277
104	4	-0.137	3.805	-0.026	-0.053	0.28	5	-0.137	3.805	-0.026	-0.053	0.28
105	4	0.5	2.103	-3.233	-6.627	-1.026	5	0.5	2.103	-3.233	-6.627	-1.026
106	4	-1.075	5.472	0.022	0.045	2.204	5	-1.075	5.472	0.022	0.045	2.204
107	4	-1.009	5.757	0.012	0.025	2.069	5	-1.009	5.757	0.012	0.025	2.069
108	4	1.534	0.348	-0.742	-1.521	-3.145	5	1.534	0.348	-0.742	-1.521	-3.145
109	4	1.584	0.714	-0.636	-1.304	-3.247	5	1.584	0.714	-0.636	-1.304	-3.247
110	4	-0.136	3.244	0.042	0.086	0.279	5	-0.136	15.036	2.52	5.166	0.279
111	4	-0.077	15.36	-2.435	-4.992	0.158	5	-0.077	3.568	0.043	0.088	0.158
112	4	-0.135	3.534	0	0	0.277	5	-0.135	3.534	0	0	0.277
113	4	0.215	2.788	-0.081	-0.166	-0.441	5	0.215	2.788	-0.081	-0.166	-0.441
114	4	0.01	2.07	-0.409	-0.838	-0.021	5	0.01	2.81	-0.205	-0.42	-0.021
115	4	0.009	4.241	-0.203	-0.416	-0.018	5	0.009	3.501	0.001	0.002	-0.018
116	4	-0.758	4.904	-0.043	-0.088	1.553	5	-0.758	4.904	-0.043	-0.088	1.553
117	4	-0.708	5.118	-0.051	-0.104	1.451	5	-0.708	5.118	-0.051	-0.104	1.451
118	4	1.199	1.061	-0.616	-1.263	-2.459	5	1.199	1.061	-0.616	-1.263	-2.459
119	4	1.237	1.335	-0.537	-1.1	-2.535	5	1.237	1.335	-0.537	-1.1	-2.535
120	4	-0.053	3.233	-0.028	-0.057	0.109	5	-0.053	12.077	1.83	3.753	0.109
121	4	-0.009	12.32	-1.886	-3.866	0.018	5	-0.009	3.476	-0.027	-0.056	0.018
122	4	-0.02	3.365	-0.06	-0.122	0.041	5	-0.02	3.365	-0.06	-0.122	0.041
123	4	0.164	2.973	-0.102	-0.21	-0.335	5	0.164	2.973	-0.102	-0.21	-0.335
124	4	0.056	2.596	-0.274	-0.563	-0.115	5	0.056	2.985	-0.167	-0.343	-0.115
125	4	0.056	3.736	-0.166	-0.341	-0.114	5	0.056	3.347	-0.059	-0.121	-0.114
126	4	-1.079	4.211	0.022	0.046	2.211	5	-1.079	4.211	0.022	0.046	2.211
127	4	-1.013	4.496	0.012	0.025	2.076	5	-1.013	4.496	0.012	0.025	2.076
128	4	1.53	-0.913	-0.742	-1.52	-3.137	5	1.53	-0.913	-0.742	-1.52	-3.137
129	4	1.58	-0.547	-0.636	-1.303	-3.24	5	1.58	-0.547	-0.636	-1.303	-3.24
130	4	-0.14	1.983	0.042	0.087	0.286	5	-0.14	13.775	2.52	5.167	0.286
131	4	-0.081	14.099	-2.435	-4.991	0.165	5	-0.081	2.307	0.043	0.089	0.165
132	4	-0.095	2.158	0	0	0.196	5	-0.095	2.158	0	0	0.196
133	4	0.15	1.636	-0.057	-0.116	-0.307	5	0.15	1.636	-0.057	-0.116	-0.307
134	4	0.006	1.133	-0.286	-0.587	-0.013	5	0.006	1.651	-0.143	-0.294	-0.013
135	4	0.005	2.653	-0.142	-0.291	-0.011	5	0.005	2.135	0.001	0.002	-0.011

CHECK FOR OVER TURNING

L/C : 128 *(Max. Tension at Node No. 4)*

O.T.M. = (0.913 X 5.05) + (1.52 + 1.52)
= 7.651 Mton-m
R.M. = (-0.913X0.75) + (27.765X0.9X2.9)
= 71.782 Mton-m
F.O.S. = 9.380 > 1.5 **SAFE**

L/C : 128 *(Max. Tension at Node No. 5)*

O.T.M. = (0.913 X 5.05) + (1.52 + 1.52)
= 7.651 Mton-m
R.M. = (-0.913X0.75) + (27.765X0.9X2.9)
= 71.782 Mton-m
F.O.S. = 9.380 > 1.5 **SAFE**

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CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 102

Max. Horizontal shear , ΣFx^2+ΣFz^2 =	8.77 Mton		
Corresponding vertical load , ΣFy =	2.744 Mton		
90% Wt. if Foundation , 0.9*Wf =	24.989 Mton		
Total Downward load , V =	27.733 Mton		
Horizontal frictional coeff. μ =	0.5	*(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*	
Resisting shear force = μ * V =	13.867 Mton		
F.O.S. =	1.580 >	1.5 **SAFE**	*(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

BASE PRESSURE :

						BASE PRESSURE		
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.	REMARKS
101	-0.082	7.284	-0.070	-0.144	0.168	4.123	3.934	O.K.
102	1.616	2.744	-8.622	-17.676	-3.312	7.131	**-0.118**	O.K.
103	-0.270	7.068	0.000	0.000	0.554	4.259	3.749	O.K.
104	-0.274	7.610	-0.052	-0.106	0.560	4.336	3.796	O.K.
105	1.000	4.206	-6.466	-13.254	-2.052	6.194	1.155	O.K.
106	-2.150	10.944	0.044	0.090	4.408	6.487	2.412	O.K.
107	-2.018	11.514	0.024	0.050	4.138	6.423	2.606	O.K.
108	3.068	0.696	-1.484	-3.042	-6.290	6.525	0.018	O.K.
109	3.168	1.428	-1.272	-2.608	-6.494	6.651	0.06	O.K.
110	-0.272	18.280	2.562	30.605	0.558	9.188	1.397	O.K.
111	-0.154	18.928	-2.392	-30.257	0.316	9.11	1.624	O.K.
112	-0.270	7.068	0.000	0.000	0.554	4.259	3.749	O.K.
113	0.430	5.576	-0.162	-0.332	-0.882	4.277	3.387	O.K.
114	0.020	4.880	-0.614	0.333	-0.042	3.811	3.693	O.K.
115	0.018	7.742	-0.202	-2.005	-0.036	4.336	3.826	O.K.
116	-1.516	9.808	-0.086	-0.176	3.106	5.768	2.87	O.K.
117	-1.416	10.236	-0.102	-0.208	2.902	5.727	3.009	O.K.
118	2.398	2.122	-1.232	-2.526	-4.918	5.997	0.874	O.K.
119	2.474	2.670	-1.074	-2.200	-5.070	6.091	0.906	O.K.
120	-0.106	15.310	1.802	22.711	0.218	7.752	2.15	O.K.
121	-0.018	15.796	-1.913	-22.937	0.036	7.751	2.263	O.K.
122	-0.040	6.730	-0.120	-0.244	0.082	4.032	3.898	O.K.
123	0.328	5.946	-0.204	-0.420	-0.670	4.233	3.517	O.K.
124	0.112	5.581	-0.441	-0.070	-0.230	3.947	3.719	O.K.
125	0.112	7.083	-0.225	-1.298	-0.228	4.265	3.746	O.K.
126	-2.158	8.422	0.044	0.092	4.422	6.203	2.115	O.K.
127	-2.026	8.992	0.024	0.050	4.152	6.14	2.31	O.K.
128	3.060	-1.826	-1.484	-3.040	-6.274	6.228	**-0.265**	O.K.
129	3.160	-1.094	-1.272	-2.606	-6.480	6.355	**-0.224**	O.K.
130	-0.280	15.758	2.562	30.607	0.572	8.905	1.1	O.K.
131	-0.162	16.406	-2.392	-30.255	0.330	8.826	1.328	O.K.
132	-0.190	4.316	0.000	0.000	0.392	3.868	3.507	O.K.
133	0.300	3.272	-0.114	-0.232	-0.614	3.877	3.258	O.K.
134	0.012	2.784	-0.429	0.233	-0.026	3.551	3.472	O.K.
135	0.010	4.788	-0.141	-1.403	-0.022	3.919	3.565	O.K.
MAX. BASE PRESSURE FOR , L/C : 110						9.188		
MIN. BASE PRESSURE FOR , L/C : 128							-0.265	

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO	
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STAAD OUTPUT FOR STRENGTH DESIGN : : (Staad File : SMBS BAG SPLITTER FOUNDATION_R0)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	4	0.013	4.414	-0.001	-0.003	-0.026	5	0.013	4.414	-0.001	-0.003	-0.026
202	4	-0.039	4.414	-0.074	-0.151	0.08	5	-0.039	4.414	-0.074	-0.151	0.08
203	4	0.98	1.69	-5.205	-10.67	-2.009	5	0.98	1.69	-5.205	-10.67	-2.009
204	4	-0.117	4.21	-0.04	-0.081	0.239	5	-0.117	4.21	-0.04	-0.081	0.239
205	4	-1.087	6.248	0.019	0.039	2.228	5	-1.087	6.248	0.019	0.039	2.228
206	4	-1.034	6.476	0.011	0.022	2.12	5	-1.034	6.476	0.011	0.022	2.12
207	4	1	2.149	-0.592	-1.214	-2.051	5	1	2.149	-0.592	-1.214	-2.051
208	4	1.04	2.442	-0.508	-1.041	-2.133	5	1.04	2.442	-0.508	-1.041	-2.133
209	4	-0.336	4.466	0.035	0.071	0.688	5	-0.336	13.899	2.017	4.135	0.688
210	4	-0.288	14.158	-1.947	-3.991	0.591	5	-0.288	4.725	0.036	0.073	0.591
211	4	-1.693	7.502	-0.004	-0.008	3.47	5	-1.693	7.502	-0.004	-0.008	3.47
212	4	-1.587	7.958	-0.02	-0.041	3.253	5	-1.587	7.958	-0.02	-0.041	3.253
213	4	2.482	-0.697	-1.226	-2.514	-5.088	5	2.482	-0.697	-1.226	-2.514	-5.088
214	4	2.562	-0.111	-1.057	-2.166	-5.252	5	2.562	-0.111	-1.057	-2.166	-5.252
215	4	-0.19	3.937	0.028	0.058	0.39	5	-0.19	22.804	3.993	8.186	0.39
216	4	-0.096	23.322	-3.935	-8.067	0.196	5	-0.096	4.455	0.03	0.061	0.196
217	4	-0.03	3.982	-0.04	-0.082	0.062	5	-0.03	3.982	-0.04	-0.082	0.062
218	4	0.32	3.236	-0.121	-0.248	-0.656	5	0.32	3.236	-0.121	-0.248	-0.656
219	4	0.115	2.518	-0.449	-0.921	-0.235	5	0.115	3.258	-0.245	-0.503	-0.235
220	4	0.114	4.689	-0.243	-0.499	-0.233	5	0.114	3.949	-0.039	-0.08	-0.233
221	4	-1.726	6.548	0.036	0.074	3.539	5	-1.726	6.548	0.036	0.074	3.539
222	4	-1.621	7.004	0.02	0.041	3.323	5	-1.621	7.004	0.02	0.041	3.323
223	4	2.448	-1.651	-1.187	-2.432	-5.019	5	2.448	-1.651	-1.187	-2.432	-5.019
224	4	2.528	-1.065	-1.017	-2.085	-5.183	5	2.528	-1.065	-1.017	-2.085	-5.183
225	4	-0.224	2.983	0.068	0.139	0.459	5	-0.224	21.85	4.033	8.267	0.459
226	4	-0.13	22.369	-3.895	-7.986	0.265	5	-0.13	3.501	0.07	0.142	0.265
227	4	-0.136	3.218	0	0	0.279	5	-0.136	3.218	0	0	0.279
228	4	0.214	2.472	-0.081	-0.166	-0.439	5	0.214	2.472	-0.081	-0.166	-0.439
229	4	0.009	1.754	-0.409	-0.838	-0.019	5	0.009	2.494	-0.205	-0.42	-0.019
230	4	0.008	3.925	-0.203	-0.416	-0.017	5	0.008	3.185	0.001	0.002	-0.017

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STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	0.026	8.828	-0.002	-0.006	-0.052	4.231	4.181
202	-0.078	8.828	-0.148	-0.302	0.160	4.316	4.097
203	1.960	3.380	-10.410	-21.340	-4.018	7.965	-0.805
204	-0.234	8.420	-0.080	-0.162	0.478	4.398	3.92
205	-2.174	12.496	0.038	0.078	4.456	6.686	2.57
206	-2.068	12.952	0.022	0.044	4.240	6.635	2.725
207	2.000	4.298	-1.184	-2.428	-4.102	5.86	1.511
208	2.080	4.884	-1.016	-2.082	-4.266	5.962	1.544
209	-0.672	18.365	2.052	24.487	1.376	8.847	1.758
210	-0.576	18.883	-1.911	-24.199	1.182	8.783	1.941
211	-3.386	15.004	-0.008	-0.016	6.940	8.109	1.723
212	-3.174	15.916	-0.040	-0.082	6.506	8.022	2.02
213	4.964	-1.394	-2.452	-5.028	-10.176	8.308	-2.245
214	5.124	-0.222	-2.114	-4.332	-10.504	8.51	-2.179
215	-0.380	26.741	4.021	48.808	0.780	12.427	0.103
216	-0.192	27.777	-3.905	-48.570	0.392	12.34	0.429
217	-0.060	7.964	-0.080	-0.164	0.124	4.183	4.03
218	0.640	6.472	-0.242	-0.496	-1.312	4.597	3.273
219	0.230	5.776	-0.694	0.167	-0.470	4.091	3.619
220	0.228	8.638	-0.282	-2.170	-0.466	4.657	3.712
221	-3.452	13.096	0.072	0.148	7.078	7.969	1.425
222	-3.242	14.008	0.040	0.082	6.646	7.867	1.736
223	4.896	-3.302	-2.374	-4.864	-10.038	8.005	-2.382
224	5.056	-2.130	-2.034	-4.170	-10.366	8.208	-2.315
225	-0.448	24.833	4.101	48.970	0.918	12.291	-0.199
226	-0.260	25.870	-3.825	-48.410	0.530	12.165	0.165
227	-0.272	6.436	0.000	0.000	0.558	4.188	3.675
228	0.428	4.944	-0.162	-0.332	-0.878	4.203	3.317
229	0.018	4.248	-0.614	0.333	-0.038	3.737	3.623
230	0.016	7.110	-0.202	-2.005	-0.034	4.263	3.755

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DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	4.231	1.081	1.951	1.603	0.135	0.09	0.109	0.036
202	4.316	1.166	2.105	1.729	0.146	0.097	0.118	0.038
203	7.965	4.815	8.691	7.141	0.602	0.4	0.488	0.159
204	4.398	1.248	2.253	1.851	0.156	0.104	0.126	0.041
205	6.686	3.536	6.382	5.244	0.442	0.293	0.358	0.117
206	6.635	3.485	6.29	5.168	0.436	0.289	0.353	0.115
207	5.86	2.71	4.892	4.019	0.339	0.225	0.274	0.089
208	5.962	2.812	5.076	4.17	0.352	0.233	0.285	0.093
209	8.847	5.697	10.283	8.449	0.712	0.473	0.577	0.188
210	8.783	5.633	10.168	8.354	0.704	0.468	0.57	0.186
211	8.109	4.959	8.951	7.354	0.62	0.412	0.502	0.164
212	8.022	4.872	8.794	7.225	0.609	0.404	0.493	0.161
213	8.308	5.158	9.31	7.649	0.645	0.428	0.522	0.17
214	8.51	5.36	9.675	7.949	0.67	0.445	0.543	0.177
215	12.427	9.277	16.745	13.758	1.16	0.77	0.939	0.306
216	12.34	9.19	16.588	13.629	1.149	0.763	0.93	0.303
217	4.183	1.033	1.865	1.532	0.129	0.086	0.105	0.034
218	4.597	1.447	2.612	2.146	0.181	0.12	0.147	0.048
219	4.091	0.941	1.699	1.396	0.118	0.078	0.095	0.031
220	4.657	1.507	2.72	2.235	0.188	0.125	0.153	0.05
221	7.969	4.819	8.698	7.147	0.602	0.4	0.488	0.159
222	7.867	4.717	8.514	6.995	0.59	0.392	0.478	0.156
223	8.005	4.855	8.763	7.2	0.607	0.403	0.492	0.16
224	8.208	5.058	9.13	7.501	0.632	0.42	0.512	0.167
225	12.291	9.141	16.5	13.556	1.143	0.759	0.926	0.302
226	12.165	9.015	16.272	13.369	1.127	0.748	0.913	0.297
227	4.188	1.038	1.874	1.539	0.13	0.086	0.105	0.034
228	4.203	1.053	1.901	1.562	0.132	0.087	0.107	0.035
229	3.737	0.587	1.06	0.871	0.073	0.049	0.059	0.019
230	4.263	1.113	2.009	1.651	0.139	0.092	0.113	0.037
		MAX. =	**16.745**	**13.758**	**1.16**	**0.77**	**0.939**	**0.306**

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO
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TOP LONGITUDINAL REINFORCEMENT :

Mu = 16.745 Mton-m
Φ*Kn = Mu/bd^2 deff. = 417 mm
Mu/bd^2 = 16.745X10^7/(1000X417^2) = 0.963 N/mm^2 < 6.9 N/mm^2
Pt = 0.0026
Min. Pt = 0.0018 Reqd. Ast = 0.0026X1000X417/100 = 10.84 cm^2
Provided 16 mm @ 150 C/C Provide Ast = 13.4 cm^2 at Top - Longitudinal **O.K.**
Total Provide Pt = 0.0032

ONE-WAY SHEAR CHECK

Vu = 13.758 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X13.758X10^4X417/16.745/10^7)X1000X417/10^4
= 40.249 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X417/10^4
= 41.741 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X417/10^4
= 71.543 Mton
HENCE, Vc = 40.249 Mton
Φt*Vc = 0.75X40.249 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 30.187 Mton **> 13.758Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(40.249X10^4+0.66X35^0.5X1000X417)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 125.135 Mton **> Vu= 13.758Mton , CROSS SECTION O.K.**

BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 1.16 Mton-m
Φ*Kn = Mu/bd^2 deff. = 417 mm
Mu/bd^2 = 1.16X10^7/(1000X417^2) = 0.067 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X417/100 = 7.51 cm^2
Provided 16 mm @ 150 C/C Provide Ast = 13.4 cm^2 at Bottom - Longitudinal **O.K.**
Total Provide Pt = 0.0032

ONE-WAY SHEAR CHECK

Vu = 0.77 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X0.77X10^4X417/1.16/10^7)X1000X417/10^4
= 40.1 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X417/10^4
= 41.741 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X417/10^4
= 71.543 Mton
HENCE, Vc = 40.1 Mton
Φt*Vc = 0.75X40.1 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 30.075 Mton **> 0.77Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(40.1X10^4+0.66X35^0.5X1000X417)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 125.124 Mton **> Vu= 0.77Mton , CROSS SECTION O.K.**

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BOTTOM CROSS REINFORCEMENT :

Mu = 0.939 Mton-m
Φ*Kn = Mu/bd^2 deff. = 417 mm
Mu/bd^2 = 0.939X10^7/(1000X417^2) = 0.054 N/mm^2 < 6.9 N/mm^2
Pt = 0.0001
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X417/100 = 7.51 cm^2
Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Bottom - Cross **O.K.**
Total Provide Pt = 0.0024

ONE-WAY SHEAR CHECK

Vu = 0.306 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0024X0.306X10^4X417/0.939/10^7)X1000X417/10^4
= 39.703 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0024)X1000X417/10^4
= 41.173 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X417/10^4
= 71.543 Mton
HENCE, Vc = 39.703 Mton
Φt*Vc = 0.75X39.703 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 29.777 Mton **> 0.306Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(39.703X10^4+0.66X35^0.5X1000X417)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 125.094 Mton **> Vu= 0.306Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 8.7 m^2
Perimeter (bo) = 2 X ((0.5+0.417) + (0.6+0.417) + (0.5+0.417) + (0.6 + 0.417)) = 7.736 m
β = 1.2
αs = 20 (Corner Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*
Pmax = 9.188 $Mton/m^2$
Vu = ((8.7 - (0.5X0.6) - (0.5X0.6))X9.188
= 74.423 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*
[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1
= 0.33X1X35^0.5X7.736X10^3X417/10^4
= 629.797 Mton
[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff
= 0.17X(1+2/1.2)X1X35^0.5X7.736X10^3X417/10^4
= 865.175 Mton
[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff
= 0.083X(2+20X417/(7.736X10^3))X1X35^0.5X(7.736X10^3)X417/10^4
= 487.578 Mton
HENCE, Vc = 487.578 Mton
Φt*Vn = Φt*Vc = 0.75X487.578 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
365.684 Mton **> 74.423Mton , Hence O.K.**

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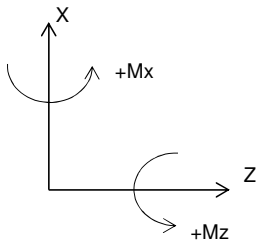
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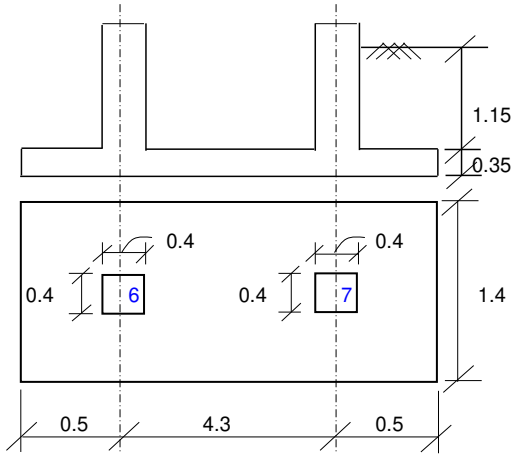
8.7 DESIGN OF FOUNDATION MKD. F7 :

(For Staad Node. No. - 6 & 7)

Net Safe Bearing Capacity of Soil =	32.5	Mton/m^2	*(Ref. Nearest Bore Hole Mkd. BH-04)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	32.5 + 1.5 X 1.9 =	35.35	Mton/m^2

Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinf. , Φ =	0.9	
Strength Reduction Factor of Shear Reinf. , Φt =	0.75	
Clear Cover =	75	mm
Density of Concrete =	2.5	Mton/m^3
Density of Dry Soil =	1.9	Mton/m^3
Density of Saturated Soil =	2	Mton/m^3
Density of Water =	1	Mton/m^3





FOUNDATION DIMENSION :

Cantilever from Node No. 6 =	0.5	m
Cantilever from Node No. 7 =	0.5	m
C/C Distance between Column =	4.3	m
Length of Foundation (Lx) =	5.3	m
Width of Foundation (Lz) =	1.4	m
Thickness of Raft =	0.35	m
Column size at Node No. 6 :		
Length along X-Direction =	0.4	m
Length along Z-Direction =	0.4	m
Height =	1.150	m
Column size at Node No. 7 :		
Length along X-Direction =	0.4	m
Length along Z-Direction =	0.4	m
Height =	1.150	m
Area , A = Lx*Lz =	7.42	m^2
Zxx = Lx*Lz^2/6 =	1.731	m^3
Zzz = Lz*Lx^2/6 =	6.554	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	5.3X1.4X0.35X2.5 =	6.493
Column =	(0.4X0.4+0.4X0.4)X1.15X2.5 =	0.92
Soil =	((5.3X1.4)-(0.4X0.4)-(0.4X0.4))X1.15X1.9 =	15.514
	Total =	22.927

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STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : SMBS BAG SPLITTER FOUNDATION_R0)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	6	-0.012	1.966	0	0	0.025	7	-0.012	1.966	0	0	0.025
102	6	0	2.22	0	0	0	7	0	2.22	0	0	0
103	6	0	1.398	0	0	0	7	0	1.398	0	0	0
104	6	-0.009	1.824	0	0	0.018	7	-0.009	1.824	0	0	0.018
105	6	0	2.015	0	0	0	7	0	2.015	0	0	0
106	6	0.004	4.984	0	0	-0.008	7	0.004	4.984	0	0	-0.008
107	6	0.004	5.009	0	0	-0.008	7	0.004	5.009	0	0	-0.008
108	6	-1.097	-2.254	0	0	2.249	7	-1.097	-2.254	0	0	2.249
109	6	-1.106	-2.287	0	0	2.267	7	-1.106	-2.287	0	0	2.267
110	6	-0.019	1.358	0	0	0.039	7	-0.019	1.358	0	0	0.039
111	6	-0.018	1.356	0	0	0.037	7	-0.018	1.356	0	0	0.037
112	6	0.001	1.908	0	0	-0.002	7	0.001	1.908	0	0	-0.002
113	6	-0.155	0.881	0	0	0.318	7	-0.155	0.881	0	0	0.318
114	6	0	1.398	0	0	0	7	0	1.398	0	0	0
115	6	-0.002	1.393	0	0	0.004	7	-0.002	1.393	0	0	0.004
116	6	0.003	4.548	0	0	-0.006	7	0.003	4.548	0	0	-0.006
117	6	0.003	4.567	0	0	-0.006	7	0.003	4.567	0	0	-0.006
118	6	-0.823	-0.88	0	0	1.687	7	-0.823	-0.88	0	0	1.687
119	6	-0.829	-0.905	0	0	1.701	7	-0.829	-0.905	0	0	1.701
120	6	-0.014	1.829	0	0	0.029	7	-0.014	1.829	0	0	0.029
121	6	-0.013	1.827	0	0	0.028	7	-0.013	1.827	0	0	0.028
122	6	0.001	2.126	0	0	-0.001	7	0.001	2.126	0	0	-0.001
123	6	-0.081	1.587	0	0	0.167	7	-0.081	1.587	0	0	0.167
124	6	0	1.859	0	0	0	7	0	1.859	0	0	0
125	6	-0.001	1.856	0	0	0.002	7	-0.001	1.856	0	0	0.002
126	6	0.004	4.425	0	0	-0.008	7	0.004	4.425	0	0	-0.008
127	6	0.004	4.45	0	0	-0.008	7	0.004	4.45	0	0	-0.008
128	6	-1.097	-2.813	0	0	2.249	7	-1.097	-2.813	0	0	2.249
129	6	-1.106	-2.846	0	0	2.267	7	-1.106	-2.846	0	0	2.267
130	6	-0.019	0.799	0	0	0.039	7	-0.019	0.799	0	0	0.039
131	6	-0.018	0.797	0	0	0.037	7	-0.018	0.797	0	0	0.037
132	6	0.001	1.196	0	0	-0.001	7	0.001	1.196	0	0	-0.001
133	6	-0.109	0.477	0	0	0.222	7	-0.109	0.477	0	0	0.222
134	6	0	0.839	0	0	0	7	0	0.839	0	0	0
135	6	-0.001	0.835	0	0	0.003	7	-0.001	0.835	0	0	0.003

CHECK FOR OVER TURNING

L/C : 129 *(Max. Tension at Node No. 6)*

O.T.M. = (2.846 X 4.8) + (0 + 0)
= 13.661 Mton-m
R.M. = (-2.846X0.5) + (22.927X0.9X2.65)
= 53.258 Mton-m
F.O.S. = 3.900 > 1.5 **SAFE**

L/C : 129 *(Max. Tension at Node No. 7)*

O.T.M. = (2.846 X 4.8) + (0 + 0)
= 13.661 Mton-m
R.M. = (-2.846X0.5) + (22.927X0.9X2.65)
= 53.258 Mton-m
F.O.S. = 3.900 > 1.5 **SAFE**

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CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 109

Max. Horizontal shear , ΣFx^2+ΣFz^2 =	2.21 Mton		
Corresponding vertical load , ΣFy =	-4.574 Mton		
90% Wt. if Foundation , 0.9*Wf =	20.634 Mton		
Total Downward load , V =	16.060 Mton		
Horizontal frictional coeff. μ =	0.5	*(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*	
Resisting shear force = μ * V =	8.030 Mton		
F.O.S. =	3.630 >	1.5 **SAFE**	*(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

BASE PRESSURE :

						BASE PRESSURE		
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.	REMARKS
101	-0.024	3.932	0.000	0.000	0.050	3.649	3.591	O.K.
102	0.000	4.440	0.000	0.000	0.000	3.688	3.688	O.K.
103	0.000	2.796	0.000	0.000	0.000	3.467	3.467	O.K.
104	-0.018	3.648	0.000	0.000	0.036	3.602	3.561	O.K.
105	0.000	4.030	0.000	0.000	0.000	3.633	3.633	O.K.
106	0.008	9.968	0.000	0.000	-0.016	4.443	4.424	O.K.
107	0.008	10.018	0.000	0.000	-0.016	4.449	4.431	O.K.
108	-2.194	-4.508	0.000	0.000	4.498	5.081	**-0.116**	O.K.
109	-2.212	-4.574	0.000	0.000	4.534	5.093	**-0.146**	O.K.
110	-0.038	2.716	0.000	0.000	0.078	3.501	3.411	O.K.
111	-0.036	2.712	0.000	0.000	0.074	3.498	3.413	O.K.
112	0.002	3.816	0.000	0.000	-0.004	3.606	3.602	O.K.
113	-0.310	1.762	0.000	0.000	0.636	3.695	2.96	O.K.
114	0.000	2.796	0.000	0.000	0.000	3.467	3.467	O.K.
115	-0.004	2.786	0.000	0.000	0.008	3.47	3.461	O.K.
116	0.006	9.096	0.000	0.000	-0.012	4.323	4.309	O.K.
117	0.006	9.134	0.000	0.000	-0.012	4.328	4.314	O.K.
118	-1.646	-1.760	0.000	0.000	3.374	4.802	0.904	O.K.
119	-1.658	-1.810	0.000	0.000	3.402	4.811	0.881	O.K.
120	-0.028	3.658	0.000	0.000	0.058	3.616	3.549	O.K.
121	-0.026	3.654	0.000	0.000	0.056	3.615	3.55	O.K.
122	0.002	4.252	0.000	0.000	-0.002	3.664	3.662	O.K.
123	-0.162	3.174	0.000	0.000	0.334	3.711	3.325	O.K.
124	0.000	3.718	0.000	0.000	0.000	3.591	3.591	O.K.
125	-0.002	3.712	0.000	0.000	0.004	3.592	3.588	O.K.
126	0.008	8.850	0.000	0.000	-0.016	4.292	4.273	O.K.
127	0.008	8.900	0.000	0.000	-0.016	4.299	4.28	O.K.
128	-2.194	-5.626	0.000	0.000	4.498	4.93	**-0.267**	O.K.
129	-2.212	-5.692	0.000	0.000	4.534	4.942	**-0.297**	O.K.
130	-0.038	1.598	0.000	0.000	0.078	3.35	3.26	O.K.
131	-0.036	1.594	0.000	0.000	0.074	3.347	3.262	O.K.
132	0.002	2.392	0.000	0.000	-0.002	3.413	3.411	O.K.
133	-0.218	0.954	0.000	0.000	0.444	3.475	2.962	O.K.
134	0.000	1.678	0.000	0.000	0.000	3.316	3.316	O.K.
135	-0.002	1.670	0.000	0.000	0.006	3.318	3.311	O.K.
MAX. BASE PRESSURE FOR , L/C : 109						5.093		
MIN. BASE PRESSURE FOR , L/C : 129							-0.297	

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STAAD OUTPUT FOR STRENGTH DESIGN : : (Staad File : SMBS BAG SPLITTER FOUNDATION_R0)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	6	0	1.957	0	0	0	7	0	1.957	0	0	0
202	6	-0.014	2.605	0	0	0.03	7	-0.014	2.605	0	0	0.03
203	6	0	2.91	0	0	0	7	0	2.91	0	0	0
204	6	0	1.985	0	0	0	7	0	1.985	0	0	0
205	6	0.003	4.546	0	0	-0.007	7	0.003	4.546	0	0	-0.007
206	6	0.003	4.566	0	0	-0.007	7	0.003	4.566	0	0	-0.007
207	6	-0.878	-1.244	0	0	1.799	7	-0.878	-1.244	0	0	1.799
208	6	-0.885	-1.27	0	0	1.814	7	-0.885	-1.27	0	0	1.814
209	6	-0.015	1.646	0	0	0.031	7	-0.015	1.646	0	0	0.031
210	6	-0.014	1.644	0	0	0.03	7	-0.014	1.644	0	0	0.03
211	6	0.006	7.722	0	0	-0.013	7	0.006	7.722	0	0	-0.013
212	6	0.006	7.762	0	0	-0.013	7	0.006	7.762	0	0	-0.013
213	6	-1.755	-3.859	0	0	3.598	7	-1.755	-3.859	0	0	3.598
214	6	-1.77	-3.911	0	0	3.628	7	-1.77	-3.911	0	0	3.628
215	6	-0.03	1.921	0	0	0.062	7	-0.03	1.921	0	0	0.062
216	6	-0.029	1.917	0	0	0.059	7	-0.029	1.917	0	0	0.059
217	6	0.001	2.495	0	0	-0.002	7	0.001	2.495	0	0	-0.002
218	6	-0.155	1.468	0	0	0.318	7	-0.155	1.468	0	0	0.318
219	6	0	1.985	0	0	0	7	0	1.985	0	0	0
220	6	-0.002	1.98	0	0	0.004	7	-0.002	1.98	0	0	0.004
221	6	0.006	6.996	0	0	-0.013	7	0.006	6.996	0	0	-0.013
222	6	0.006	7.036	0	0	-0.013	7	0.006	7.036	0	0	-0.013
223	6	-1.755	-4.585	0	0	3.598	7	-1.755	-4.585	0	0	3.598
224	6	-1.77	-4.638	0	0	3.628	7	-1.77	-4.638	0	0	3.628
225	6	-0.03	1.194	0	0	0.062	7	-0.03	1.194	0	0	0.062
226	6	-0.029	1.191	0	0	0.059	7	-0.029	1.191	0	0	0.059
227	6	0.001	1.768	0	0	-0.002	7	0.001	1.768	0	0	-0.002
228	6	-0.155	0.741	0	0	0.318	7	-0.155	0.741	0	0	0.318
229	6	0	1.258	0	0	0	7	0	1.258	0	0	0
230	6	-0.002	1.253	0	0	0.004	7	-0.002	1.253	0	0	0.004

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STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	0.000	3.914	0.000	0.000	0.000	3.617	3.617
202	-0.028	5.210	0.000	0.000	0.060	3.827	3.757
203	0.000	5.820	0.000	0.000	0.000	3.874	3.874
204	0.000	3.970	0.000	0.000	0.000	3.625	3.625
205	0.006	9.092	0.000	0.000	-0.014	4.323	4.307
206	0.006	9.132	0.000	0.000	-0.014	4.329	4.313
207	-1.756	-2.488	0.000	0.000	3.598	4.833	0.676
208	-1.770	-2.540	0.000	0.000	3.628	4.843	0.652
209	-0.030	3.292	0.000	0.000	0.062	3.569	3.498
210	-0.028	3.288	0.000	0.000	0.060	3.568	3.498
211	0.012	15.444	0.000	0.000	-0.026	5.186	5.156
212	0.012	15.524	0.000	0.000	-0.026	5.197	5.167
213	-3.510	-7.718	0.000	0.000	7.196	6.207	-2.107
214	-3.540	-7.822	0.000	0.000	7.256	6.228	-2.156
215	-0.060	3.842	0.000	0.000	0.124	3.679	3.536
216	-0.058	3.834	0.000	0.000	0.118	3.675	3.538
217	0.002	4.990	0.000	0.000	-0.004	3.765	3.76
218	-0.310	2.936	0.000	0.000	0.636	3.853	3.118
219	0.000	3.970	0.000	0.000	0.000	3.625	3.625
220	-0.004	3.960	0.000	0.000	0.008	3.628	3.619
221	0.012	13.992	0.000	0.000	-0.026	4.991	4.961
222	0.012	14.072	0.000	0.000	-0.026	5.001	4.971
223	-3.510	-9.170	0.000	0.000	7.196	6.011	-2.303
224	-3.540	-9.276	0.000	0.000	7.256	6.032	-2.352
225	-0.060	2.388	0.000	0.000	0.124	3.483	3.34
226	-0.058	2.382	0.000	0.000	0.118	3.479	3.343
227	0.002	3.536	0.000	0.000	-0.004	3.569	3.564
228	-0.310	1.482	0.000	0.000	0.636	3.657	2.922
229	0.000	2.516	0.000	0.000	0.000	3.429	3.429
230	-0.004	2.506	0.000	0.000	0.008	3.432	3.423

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DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	3.617	0.557	1.059	0.936	0.025	0.017	0.07	0.129
202	3.827	0.767	1.458	1.289	0.035	0.024	0.096	0.177
203	3.874	0.814	1.548	1.368	0.037	0.025	0.102	0.188
204	3.625	0.565	1.074	0.95	0.025	0.018	0.071	0.131
205	4.323	1.263	2.401	2.123	0.057	0.039	0.158	0.292
206	4.329	1.269	2.413	2.133	0.057	0.039	0.159	0.293
207	4.833	1.773	3.371	2.98	0.08	0.055	0.222	0.41
208	4.843	1.783	3.39	2.997	0.08	0.055	0.223	0.412
209	3.569	0.509	0.968	0.856	0.023	0.016	0.064	0.118
210	3.568	0.508	0.966	0.854	0.023	0.016	0.064	0.117
211	5.186	2.126	4.042	3.574	0.096	0.066	0.266	0.491
212	5.197	2.137	4.063	3.592	0.096	0.066	0.267	0.494
213	6.207	3.147	5.983	5.29	0.142	0.098	0.393	0.727
214	6.228	3.168	6.023	5.325	0.143	0.098	0.396	0.732
215	3.679	0.619	1.177	1.041	0.028	0.019	0.077	0.143
216	3.675	0.615	1.169	1.034	0.028	0.019	0.077	0.142
217	3.765	0.705	1.34	1.185	0.032	0.022	0.088	0.163
218	3.853	0.793	1.508	1.333	0.036	0.025	0.099	0.183
219	3.625	0.565	1.074	0.95	0.025	0.018	0.071	0.131
220	3.628	0.568	1.08	0.955	0.026	0.018	0.071	0.131
221	4.991	1.931	3.671	3.246	0.087	0.06	0.241	0.446
222	5.001	1.941	3.69	3.263	0.087	0.06	0.243	0.448
223	6.011	2.951	5.611	4.961	0.133	0.091	0.369	0.682
224	6.032	2.972	5.651	4.996	0.134	0.092	0.372	0.687
225	3.483	0.423	0.804	0.711	0.019	0.013	0.053	0.098
226	3.479	0.419	0.797	0.704	0.019	0.013	0.052	0.097
227	3.569	0.509	0.968	0.856	0.023	0.016	0.064	0.118
228	3.657	0.597	1.135	1.004	0.027	0.019	0.075	0.138
229	3.429	0.369	0.702	0.62	0.017	0.011	0.046	0.085
230	3.432	0.372	0.707	0.625	0.017	0.012	0.047	0.086
		MAX. =	**6.023**	**5.325**	**0.143**	**0.098**	**0.396**	**0.732**

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TOP LONGITUDINAL REINFORCEMENT :

Mu = 6.023 Mton-m
Φ*Kn = Mu/bd^2 deff. = 269 mm
Mu/bd^2 = 6.023X10^7/(1000X269^2) = 0.832 N/mm^2 < 6.9 N/mm^2
Pt = 0.0022
Min. Pt = 0.0018 Reqd. Ast = 0.0022X1000X269/100 = 5.92 cm^2
Provided 12 mm @ 150 C/C Provide Ast = 7.54 cm^2 at Top - Longitudinal **O.K.**
Total Provide Pt = 0.0028

ONE-WAY SHEAR CHECK

Vu = 5.325 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0028X5.325X10^4X269/6.023/10^7)X1000X269/10^4
= 25.767 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0028)X1000X269/10^4
= 26.743 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X269/10^4
= 46.151 Mton
HENCE, Vc = 25.767 Mton
Φt*Vc = 0.75X25.767 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 19.325 Mton **> 5.325Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(25.767X10^4+0.66X35^0.5X1000X269)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 80.708 Mton **> Vu= 5.325Mton , CROSS SECTION O.K.**

BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 0.143 Mton-m
Φ*Kn = Mu/bd^2 deff. = 269 mm
Mu/bd^2 = 0.143X10^7/(1000X269^2) = 0.02 N/mm^2 < 6.9 N/mm^2
Pt = 0.0001
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X269/100 = 4.84 cm^2
Provided 12 mm @ 150 C/C Provide Ast = 7.54 cm^2 at Bottom - Longitudinal **O.K.**
Total Provide Pt = 0.0028

ONE-WAY SHEAR CHECK

Vu = 0.098 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0028X0.098X10^4X269/0.143/10^7)X1000X269/10^4
= 25.699 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0028)X1000X269/10^4
= 26.743 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X269/10^4
= 46.151 Mton
HENCE, Vc = 25.699 Mton
Φt*Vc = 0.75X25.699 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 19.274 Mton **> 0.098Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(25.699X10^4+0.66X35^0.5X1000X269)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 80.703 Mton **> Vu= 0.098Mton , CROSS SECTION O.K.**

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BOTTOM CROSS REINFORCEMENT :

Mu = 0.396 Mton-m
Φ*Kn = Mu/bd^2 deff. = 269 mm
Mu/bd^2 = 0.396X10^7/(1000X269^2) = 0.055 N/mm^2 < 6.9 N/mm^2
Pt = 0.0001
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X269/100 = 4.84 cm^2
Provided 12 mm @ 150 C/C Provide Ast = 7.54 cm^2 at Bottom - Cross **O.K.**
Total Provide Pt = 0.0028

ONE-WAY SHEAR CHECK

Vu = 0.732 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0028X0.732X10^4X269/0.396/10^7)X1000X269/10^4
= 26.099 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0028)X1000X269/10^4
= 26.743 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X269/10^4
= 46.151 Mton
HENCE, Vc = 26.099 Mton
Φt*Vc = 0.75X26.099 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 19.574 Mton **> 0.732Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(26.099X10^4+0.66X35^0.5X1000X269)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 80.733 Mton **> Vu= 0.732Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 7.42 m^2
Perimeter (bo) = 2 X ((0.4+0.269) + (0.4+0.269) + (0.4+0.269) + (0.4 + 0.269)) = 5.352 m
β = 1
αs = 20 (Corner Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*
Pmax = 5.093 $Mton/m^2$
Vu = ((7.42 - (0.4X0.4) - (0.4X0.4))X5.093
= 36.16 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*
[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1
= 0.33X1X35^0.5X5.352X10^3X269/10^4
= 281.071 Mton
[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff
= 0.17X(1+2/1)X1X35^0.5X5.352X10^3X269/10^4
= 434.383 Mton
[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff
= 0.083X(2+20X269/(5.352X10^3))X1X35^0.5X(5.352X10^3)X269/10^4
= 212.451 Mton
HENCE, Vc = 212.451 Mton
Φt*Vn = Φt*Vc = 0.75X212.451 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
159.338 Mton **> 36.16Mton , Hence O.K.**

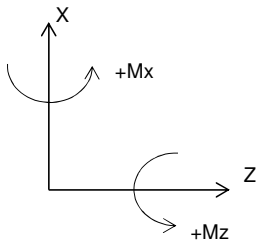
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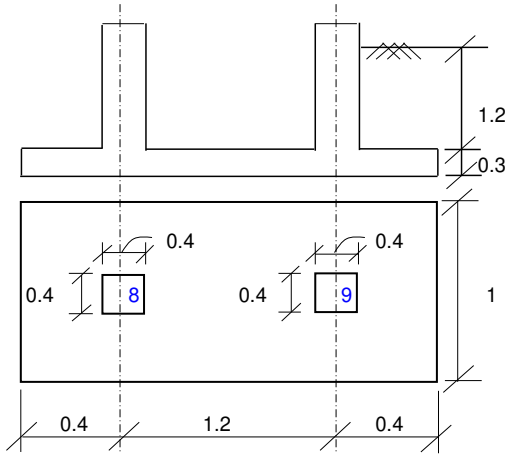
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8.8 DESIGN OF FOUNDATION MKD. F8 :

(For Staad Node. No. - 8 & 9)

Net Safe Bearing Capacity of Soil =	45 Mton/m^2	*(Ref. Nearest Bore Hole Mkd. BH-04)*
Depth of Foundation from FGL =	1.5 m	
Gross Allowable Bearing Capacity of Soil =	45 + 1.5 X 1.9 = 47.85 Mton/m^2	
Compressive Strength of Concrete , fc' =	35 N/mm^2	
Yield Strength of Main Reinf. , fy =	420 N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420 N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9	
Strength Reduction Factor of Shear Reinf. , Φt =	0.75	
Clear Cover =	75 mm	
Density of Concrete =	2.5 Mton/m^3	
Density of Dry Soil =	1.9 Mton/m^3	
Density of Saturated Soil =	2 Mton/m^3	
Density of Water =	1 Mton/m^3	





FOUNDATION DIMENSION :

Cantilever from Node No. 8 =	0.4 m
Cantilever from Node No. 9 =	0.4 m
C/C Distance between Column =	1.2 m
Length of Foundation (Lx) =	2 m
Width of Foundation (Lz) =	1 m
Thickness of Raft =	0.3 m
Column size at Node No. 8 :	
Length along X-Direction =	0.4 m
Length along Z-Direction =	0.4 m
Height =	1.200 m
Column size at Node No. 9 :	
Length along X-Direction =	0.4 m
Length along Z-Direction =	0.4 m
Height =	1.200 m
Area , A = Lx*Lz =	2 m^2
Zxx = Lx*Lz^2/6 =	0.333 m^3
Zzz = Lz*Lx^2/6 =	0.667 m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	2X1X0.3X2.5 =	1.5
Column =	(0.4X0.4+0.4X0.4)X1.2X2.5 =	0.96
Soil =	((2X1)-(0.4X0.4)-(0.4X0.4))X1.2X1.9 =	3.83
	Total =	6.29

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STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : SMBS BAG SPLITTER FOUNDATION_R0)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	8	0	1.797	0	0	0	9	0	1.797	0	0	0
102	8	0	1.815	0	0	0	9	0	1.815	0	0	0
103	8	0	1.135	0	0	0	9	0	1.135	0	0	0
104	8	0	1.631	0	0	0	9	0	1.631	0	0	0
105	8	0	1.645	0	0	0	9	0	1.645	0	0	0
106	8	0	1.072	0	0	0	9	0	1.072	0	0	0
107	8	0	1.072	0	0	0	9	0	1.072	0	0	0
108	8	0	1.196	0	0	0	9	0	1.196	0	0	0
109	8	0	1.197	0	0	0	9	0	1.197	0	0	0
110	8	0	1.134	0	0	0	9	0	1.134	0	0	0
111	8	0	1.134	0	0	0	9	0	1.134	0	0	0
112	8	0	1.126	0	0	0	9	0	1.126	0	0	0
113	8	0	1.144	0	0	0	9	0	1.144	0	0	0
114	8	0	1.135	0	0	0	9	0	1.135	0	0	0
115	8	0	1.135	0	0	0	9	0	1.135	0	0	0
116	8	0	1.592	0	0	0	9	0	1.592	0	0	0
117	8	0	1.592	0	0	0	9	0	1.592	0	0	0
118	8	0	1.685	0	0	0	9	0	1.685	0	0	0
119	8	0	1.685	0	0	0	9	0	1.685	0	0	0
120	8	0	1.638	0	0	0	9	0	1.638	0	0	0
121	8	0	1.638	0	0	0	9	0	1.638	0	0	0
122	8	0	1.634	0	0	0	9	0	1.634	0	0	0
123	8	0	1.644	0	0	0	9	0	1.644	0	0	0
124	8	0	1.639	0	0	0	9	0	1.639	0	0	0
125	8	0	1.639	0	0	0	9	0	1.639	0	0	0
126	8	0	0.618	0	0	0	9	0	0.618	0	0	0
127	8	0	0.618	0	0	0	9	0	0.618	0	0	0
128	8	0	0.742	0	0	0	9	0	0.742	0	0	0
129	8	0	0.743	0	0	0	9	0	0.743	0	0	0
130	8	0	0.68	0	0	0	9	0	0.68	0	0	0
131	8	0	0.68	0	0	0	9	0	0.68	0	0	0
132	8	0	0.675	0	0	0	9	0	0.675	0	0	0
133	8	0	0.687	0	0	0	9	0	0.687	0	0	0
134	8	0	0.681	0	0	0	9	0	0.681	0	0	0
135	8	0	0.681	0	0	0	9	0	0.681	0	0	0

CHECK FOR OVER TURNING

L/C : 126 *(Max. Tension at Node No. 8)*

O.T.M. = (0 X 1.6) + (0 + 0)
= 0 Mton-m
R.M. = (0.618X0.4) + (6.29X0.9X1)
= 5.908 Mton-m
F.O.S. = > 1.5 **SAFE**

L/C : 126 *(Max. Tension at Node No. 9)*

O.T.M. = (0 X 1.6) + (0 + 0)
= 0 Mton-m
R.M. = (0.618X0.4) + (6.29X0.9X1)
= 5.908 Mton-m
F.O.S. = > 1.5 **SAFE**

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CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 101

Max. Horizontal shear , ΣFx^2+ΣFz^2 =	0.00 Mton		
Corresponding vertical load , ΣFy =	3.594 Mton		
90% Wt. if Foundation , 0.9*Wf =	5.661 Mton		
Total Downward load , V =	9.255 Mton		
Horizontal frictional coeff. μ =	0.5	*(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*	
Resisting shear force = μ * V =	4.628 Mton		
F.O.S. =	> 1.5	**SAFE**	*(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

BASE PRESSURE :

L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	BASE PRESSURE P max.	BASE PRESSURE P min.	REMARKS
101	0.000	3.594	0.000	0.000	0.000	4.942	4.942	O.K.
102	0.000	3.630	0.000	0.000	0.000	4.96	4.96	O.K.
103	0.000	2.270	0.000	0.000	0.000	4.28	4.28	O.K.
104	0.000	3.262	0.000	0.000	0.000	4.776	4.776	O.K.
105	0.000	3.290	0.000	0.000	0.000	4.79	4.79	O.K.
106	0.000	2.144	0.000	0.000	0.000	4.217	4.217	O.K.
107	0.000	2.144	0.000	0.000	0.000	4.217	4.217	O.K.
108	0.000	2.392	0.000	0.000	0.000	4.341	4.341	O.K.
109	0.000	2.394	0.000	0.000	0.000	4.342	4.342	O.K.
110	0.000	2.268	0.000	0.000	0.000	4.279	4.279	O.K.
111	0.000	2.268	0.000	0.000	0.000	4.279	4.279	O.K.
112	0.000	2.252	0.000	0.000	0.000	4.271	4.271	O.K.
113	0.000	2.288	0.000	0.000	0.000	4.289	4.289	O.K.
114	0.000	2.270	0.000	0.000	0.000	4.28	4.28	O.K.
115	0.000	2.270	0.000	0.000	0.000	4.28	4.28	O.K.
116	0.000	3.184	0.000	0.000	0.000	4.737	4.737	O.K.
117	0.000	3.184	0.000	0.000	0.000	4.737	4.737	O.K.
118	0.000	3.370	0.000	0.000	0.000	4.83	4.83	O.K.
119	0.000	3.370	0.000	0.000	0.000	4.83	4.83	O.K.
120	0.000	3.276	0.000	0.000	0.000	4.783	4.783	O.K.
121	0.000	3.276	0.000	0.000	0.000	4.783	4.783	O.K.
122	0.000	3.268	0.000	0.000	0.000	4.779	4.779	O.K.
123	0.000	3.288	0.000	0.000	0.000	4.789	4.789	O.K.
124	0.000	3.278	0.000	0.000	0.000	4.784	4.784	O.K.
125	0.000	3.278	0.000	0.000	0.000	4.784	4.784	O.K.
126	0.000	1.236	0.000	0.000	0.000	3.763	3.763	O.K.
127	0.000	1.236	0.000	0.000	0.000	3.763	3.763	O.K.
128	0.000	1.484	0.000	0.000	0.000	3.887	3.887	O.K.
129	0.000	1.486	0.000	0.000	0.000	3.888	3.888	O.K.
130	0.000	1.360	0.000	0.000	0.000	3.825	3.825	O.K.
131	0.000	1.360	0.000	0.000	0.000	3.825	3.825	O.K.
132	0.000	1.350	0.000	0.000	0.000	3.82	3.82	O.K.
133	0.000	1.374	0.000	0.000	0.000	3.832	3.832	O.K.
134	0.000	1.362	0.000	0.000	0.000	3.826	3.826	O.K.
135	0.000	1.362	0.000	0.000	0.000	3.826	3.826	O.K.
MAX. BASE PRESSURE FOR , L/C : 102						4.960		
MIN. BASE PRESSURE FOR , L/C : 126							3.763	

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STAAD OUTPUT FOR STRENGTH DESIGN : : (Staad File : SMBS BAG SPLITTER FOUNDATION_R0)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	8	0	1.589	0	0	0	9	0	1.589	0	0	0
202	8	0	2.425	0	0	0	9	0	2.425	0	0	0
203	8	0	2.447	0	0	0	9	0	2.447	0	0	0
204	8	0	1.698	0	0	0	9	0	1.698	0	0	0
205	8	0	1.311	0	0	0	9	0	1.311	0	0	0
206	8	0	1.311	0	0	0	9	0	1.311	0	0	0
207	8	0	1.411	0	0	0	9	0	1.411	0	0	0
208	8	0	1.411	0	0	0	9	0	1.411	0	0	0
209	8	0	1.361	0	0	0	9	0	1.361	0	0	0
210	8	0	1.361	0	0	0	9	0	1.361	0	0	0
211	8	0	1.597	0	0	0	9	0	1.597	0	0	0
212	8	0	1.597	0	0	0	9	0	1.597	0	0	0
213	8	0	1.795	0	0	0	9	0	1.795	0	0	0
214	8	0	1.797	0	0	0	9	0	1.797	0	0	0
215	8	0	1.696	0	0	0	9	0	1.696	0	0	0
216	8	0	1.696	0	0	0	9	0	1.696	0	0	0
217	8	0	1.689	0	0	0	9	0	1.689	0	0	0
218	8	0	1.707	0	0	0	9	0	1.707	0	0	0
219	8	0	1.698	0	0	0	9	0	1.698	0	0	0
220	8	0	1.698	0	0	0	9	0	1.698	0	0	0
221	8	0	0.921	0	0	0	9	0	0.921	0	0	0
222	8	0	0.921	0	0	0	9	0	0.921	0	0	0
223	8	0	1.119	0	0	0	9	0	1.119	0	0	0
224	8	0	1.121	0	0	0	9	0	1.121	0	0	0
225	8	0	1.02	0	0	0	9	0	1.02	0	0	0
226	8	0	1.02	0	0	0	9	0	1.02	0	0	0
227	8	0	1.012	0	0	0	9	0	1.012	0	0	0
228	8	0	1.03	0	0	0	9	0	1.03	0	0	0
229	8	0	1.021	0	0	0	9	0	1.021	0	0	0
230	8	0	1.021	0	0	0	9	0	1.021	0	0	0

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STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	0.000	3.178	0.000	0.000	0.000	4.734	4.734
202	0.000	4.850	0.000	0.000	0.000	5.57	5.57
203	0.000	4.894	0.000	0.000	0.000	5.592	5.592
204	0.000	3.396	0.000	0.000	0.000	4.843	4.843
205	0.000	2.622	0.000	0.000	0.000	4.456	4.456
206	0.000	2.622	0.000	0.000	0.000	4.456	4.456
207	0.000	2.822	0.000	0.000	0.000	4.556	4.556
208	0.000	2.822	0.000	0.000	0.000	4.556	4.556
209	0.000	2.722	0.000	0.000	0.000	4.506	4.506
210	0.000	2.722	0.000	0.000	0.000	4.506	4.506
211	0.000	3.194	0.000	0.000	0.000	4.742	4.742
212	0.000	3.194	0.000	0.000	0.000	4.742	4.742
213	0.000	3.590	0.000	0.000	0.000	4.94	4.94
214	0.000	3.594	0.000	0.000	0.000	4.942	4.942
215	0.000	3.392	0.000	0.000	0.000	4.841	4.841
216	0.000	3.392	0.000	0.000	0.000	4.841	4.841
217	0.000	3.378	0.000	0.000	0.000	4.834	4.834
218	0.000	3.414	0.000	0.000	0.000	4.852	4.852
219	0.000	3.396	0.000	0.000	0.000	4.843	4.843
220	0.000	3.396	0.000	0.000	0.000	4.843	4.843
221	0.000	1.842	0.000	0.000	0.000	4.066	4.066
222	0.000	1.842	0.000	0.000	0.000	4.066	4.066
223	0.000	2.238	0.000	0.000	0.000	4.264	4.264
224	0.000	2.242	0.000	0.000	0.000	4.266	4.266
225	0.000	2.040	0.000	0.000	0.000	4.165	4.165
226	0.000	2.040	0.000	0.000	0.000	4.165	4.165
227	0.000	2.024	0.000	0.000	0.000	4.157	4.157
228	0.000	2.060	0.000	0.000	0.000	4.175	4.175
229	0.000	2.042	0.000	0.000	0.000	4.166	4.166
230	0.000	2.042	0.000	0.000	0.000	4.166	4.166

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DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	4.734	1.704	0.136	0.308	0.034	0	0.077	0.138
202	5.57	2.54	0.203	0.46	0.051	0	0.114	0.206
203	5.592	2.562	0.205	0.464	0.051	0	0.115	0.208
204	4.843	1.813	0.145	0.328	0.036	0	0.082	0.147
205	4.456	1.426	0.114	0.258	0.029	0	0.064	0.116
206	4.456	1.426	0.114	0.258	0.029	0	0.064	0.116
207	4.556	1.526	0.122	0.276	0.031	0	0.069	0.124
208	4.556	1.526	0.122	0.276	0.031	0	0.069	0.124
209	4.506	1.476	0.118	0.267	0.03	0	0.066	0.12
210	4.506	1.476	0.118	0.267	0.03	0	0.066	0.12
211	4.742	1.712	0.137	0.31	0.034	0	0.077	0.139
212	4.742	1.712	0.137	0.31	0.034	0	0.077	0.139
213	4.94	1.91	0.153	0.346	0.038	0	0.086	0.155
214	4.942	1.912	0.153	0.346	0.038	0	0.086	0.155
215	4.841	1.811	0.145	0.328	0.036	0	0.081	0.147
216	4.841	1.811	0.145	0.328	0.036	0	0.081	0.147
217	4.834	1.804	0.144	0.327	0.036	0	0.081	0.146
218	4.852	1.822	0.146	0.33	0.036	0	0.082	0.148
219	4.843	1.813	0.145	0.328	0.036	0	0.082	0.147
220	4.843	1.813	0.145	0.328	0.036	0	0.082	0.147
221	4.066	1.036	0.083	0.188	0.021	0	0.047	0.084
222	4.066	1.036	0.083	0.188	0.021	0	0.047	0.084
223	4.264	1.234	0.099	0.223	0.025	0	0.056	0.1
224	4.266	1.236	0.099	0.224	0.025	0	0.056	0.1
225	4.165	1.135	0.091	0.205	0.023	0	0.051	0.092
226	4.165	1.135	0.091	0.205	0.023	0	0.051	0.092
227	4.157	1.127	0.09	0.204	0.023	0	0.051	0.091
228	4.175	1.145	0.092	0.207	0.023	0	0.052	0.093
229	4.166	1.136	0.091	0.206	0.023	0	0.051	0.092
230	4.166	1.136	0.091	0.206	0.023	0	0.051	0.092
		MAX. =	**0.205**	**0.464**	**0.051**	**0**	**0.115**	**0.208**

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TOP LONGITUDINAL REINFORCEMENT :

Mu = 0.205 Mton-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 0.205X10^7/(1000X219^2) = 0.043 N/mm^2 < 6.9 N/mm^2
Pt = 0.0001
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X219/100 = 3.94 cm^2
Provided 12 mm @ 200 C/C Provide Ast = 5.65 cm^2 at Top - Longitudinal **O.K.**
Total Provide Pt = 0.0026

ONE-WAY SHEAR CHECK

Vu = 0.464 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0026X0.464X10^4X219/0.205/10^7)X1000X219/10^4
= 21.21 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0026)X1000X219/10^4
= 21.698 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X219/10^4
= 37.573 Mton
HENCE, Vc = 21.21 Mton
Φt*Vc = 0.75X21.21 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 15.908 Mton **> 0.464Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(21.21X10^4+0.66X35^0.5X1000X219)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 65.724 Mton **> Vu= 0.464Mton , CROSS SECTION O.K.**

BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 0.051 Mton-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 0.051X10^7/(1000X219^2) = 0.011 N/mm^2 < 6.9 N/mm^2
Pt = 0.0000
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X219/100 = 3.94 cm^2
Provided 12 mm @ 200 C/C Provide Ast = 5.65 cm^2 at Bottom - Longitudinal **O.K.**
Total Provide Pt = 0.0026

ONE-WAY SHEAR CHECK

Vu = 0 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0026X0X10^4X219/0.051/10^7)X1000X219/10^4
= 20.73 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0026)X1000X219/10^4
= 21.698 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X219/10^4
= 37.573 Mton
HENCE, Vc = 20.73 Mton
Φt*Vc = 0.75X20.73 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 15.548 Mton **> 0Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(20.73X10^4+0.66X35^0.5X1000X219)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 65.688 Mton **> Vu= 0Mton , CROSS SECTION O.K.**

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BOTTOM CROSS REINFORCEMENT :

Mu = 0.115 Mton-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 0.115X10^7/(1000X219^2) = 0.024 N/mm^2 < 6.9 N/mm^2
Pt = 0.0001
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X219/100 = 3.94 cm^2
Provided 12 mm @ 200 C/C Provide Ast = 5.65 cm^2 at Bottom - Cross **O.K.**
Total Provide Pt = 0.0026

ONE-WAY SHEAR CHECK

Vu = 0.208 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0026X0.208X10^4X219/0.115/10^7)X1000X219/10^4
= 21.113 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0026)X1000X219/10^4
= 21.698 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X219/10^4
= 37.573 Mton
HENCE, Vc = 21.113 Mton
Φt*Vc = 0.75X21.113 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 15.835 Mton **> 0.208Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(21.113X10^4+0.66X35^0.5X1000X219)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 65.717 Mton **> Vu= 0.208Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 2 m^2
Perimeter (bo) = 2 X ((0.4+0.219) + (0.4+0.219) + (0.4+0.219) + (0.4 + 0.219)) = 4.952 m
β = 1
αs = 20 (Corner Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*
Pmax = 4.960 $Mton/m^2$
Vu = ((2 - (0.4X0.4) - (0.4X0.4))X4.96
= 8.333 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*
[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1
= 0.33X1X35^0.5X4.952X10^3X219/10^4
= 211.725 Mton
[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff
= 0.17X(1+2/1)X1X35^0.5X4.952X10^3X219/10^4
= 327.212 Mton
[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff
= 0.083X(2+20X219/(4.952X10^3))X1X35^0.5X(4.952X10^3)X219/10^4
= 153.605 Mton
HENCE, Vc = 153.605 Mton
Φt*Vn = Φt*Vc = 0.75X153.605 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
115.204 Mton **> 8.333Mton , Hence O.K.**

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9.0 DESIGN OF COLUMN

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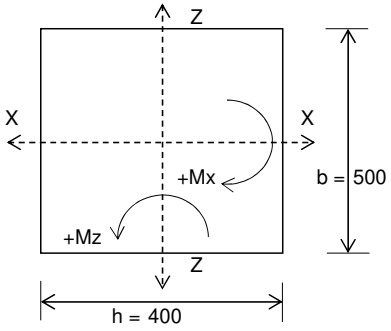
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9.1 DESIGN OF COLUMN MKD. C1

Design Data :

Depth Parallel to X-X axis, h =	400	mm
Depth Parallel to Z-Z axis, b =	500	mm
Clear Cover =	50	mm
d' =	70	mm
Y'_h = (h-d')/h =	0.7	mm
Y'_b = (b-d')/b =	0.7	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.7	M, Hence	k*lux/b =	4.25 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.7	M, Hence	k*luz/h =	3.4 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	8	Nos	20	Dia Bar Equally Distributed In Four Side
Provided	0	NOS	16	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	25.13	cm^2		
Total Provide Pt =	0.013	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 555 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^\alpha + (Mnz/Mnz1)^\alpha$
1	1.753	-0.628	0	2.697	0.966	0	0.004	**0.06**	21	0.06	16.8	0.01
26	-0.575	0	0	-0.885	0	0	-0.001	0.06	21	0.06	16.8	0
1	1.674	-0.004	0	2.575	0.006	0	0.004	0.06	21	0.06	16.8	0
1	1.674	-0.004	0	2.575	0.006	0	0.004	0.06	21	0.06	16.8	0
1	1.366	-0.636	0	2.102	0.978	0	0.003	0.06	21	0.06	16.8	0.01
18	1.441	0.473	0	2.217	0.728	0	0.003	0.06	21	0.06	16.8	0
1	1.674	-0.004	0	2.575	0.006	0	0.004	0.06	21	0.06	16.8	0
1	1.674	-0.004	0	2.575	0.006	0	0.004	0.06	21	0.06	16.8	0
18	1.441	0.473	0	2.217	0.728	0	0.003	0.06	21	0.06	16.8	0
1	1.366	-0.636	0	2.102	0.978	0	0.003	0.06	21	0.06	16.8	0.01
1	1.674	-0.004	0	2.575	0.006	0	0.004	0.06	21	0.06	16.8	0
1	1.674	-0.004	0	2.575	0.006	0	0.004	0.06	21	0.06	16.8	0
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

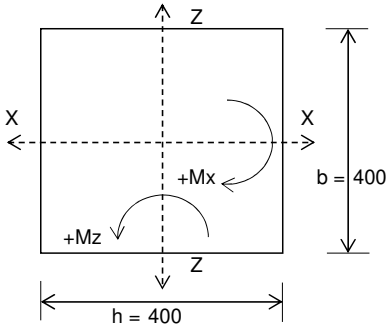
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9.2 DESIGN OF COLUMN MKD. C2

Design Data :

Depth Parallel to X-X axis, h =	400	mm
Depth Parallel to Z-Z axis, b =	400	mm
Clear Cover =	50	mm
d' =	70	mm
Y'_h = (h-d')/h =	0.7	mm
Y'_b = (b-d')/b =	0.7	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.7	M, Hence	k*lux/b =	4.25 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.7	M, Hence	k*luz/h =	4.25 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	8	Nos	20	Dia Bar Equally Distributed In Four Side
Provided	0	NOS	16	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	25.13	cm^2		
Total Provide Pt =	0.016	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 460 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^\alpha + (Mnz/Mnz1)^\alpha$
9	1.43	-0.007	0	2.2	0.011	0	0.004	**0.07**	15.68	0.07	15.68	0
40	-0.371	0	0	-0.571	0	0	-0.001	0.07	15.68	0.07	15.68	0
12	0.846	0.009	0.653	1.302	0.014	1.005	0.002	0.07	15.68	0.07	15.68	0.01
12	0.959	0.009	-0.63	1.475	0.014	0.969	0.003	0.07	15.68	0.07	15.68	0.01
9	1.357	-0.007	0	2.088	0.011	0	0.004	0.07	15.68	0.07	15.68	0
12	1.336	0.01	0	2.055	0.015	0	0.004	0.07	15.68	0.07	15.68	0
12	1.336	0.01	0	2.055	0.015	0	0.004	0.07	15.68	0.07	15.68	0
12	1.336	0.01	0	2.055	0.015	0	0.004	0.07	15.68	0.07	15.68	0
12	1.336	0.01	0	2.055	0.015	0	0.004	0.07	15.68	0.07	15.68	0
9	1.357	-0.007	0	2.088	0.011	0	0.004	0.07	15.68	0.07	15.68	0
12	0.846	0.009	0.653	1.302	0.014	1.005	0.002	0.07	15.68	0.07	15.68	0.01
12	0.959	0.009	-0.63	1.475	0.014	0.969	0.003	0.07	15.68	0.07	15.68	0.01
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

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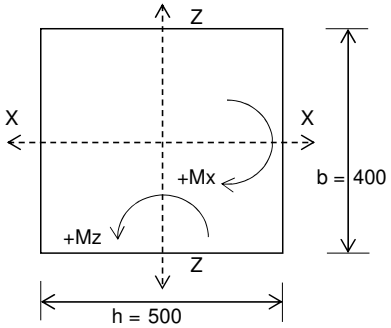
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9.3 DESIGN OF COLUMN MKD. C3

Design Data :

Depth Parallel to X-X axis, h =	500	mm
Depth Parallel to Z-Z axis, b =	400	mm
Clear Cover =	50	mm
d' =	70	mm
Y'_h = (h-d')/h =	0.7	mm
Y'_b = (b-d')/b =	0.7	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.7	M, Hence	k*lux/b =	3.4 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.7	M, Hence	k*luz/h =	4.25 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	8	Nos	20	Dia Bar Equally Distributed In Four Side
Provided	0	NOS	16	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	25.13	cm^2		
Total Provide Pt =	0.013	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 555 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^α$ + $(Mnz/Mnz1)^α$
2	2.613	0.213	0	4.02	0.328	0	0.006	**0.06**	16.8	0.06	21	0
27	-1.301	0	0	-2.002	0	0	-0.003	0.06	16.8	0.06	21	0
2	0.746	0.425	0.598	1.148	0.654	0.92	0.002	0.06	16.8	0.06	21	0.01
7	0.522	0.311	-0.598	0.803	0.478	0.92	0.001	0.06	16.8	0.06	21	0.01
2	1.983	-0.899	0	3.051	1.383	0	0.004	0.06	16.8	0.06	21	0.02
19	-0.137	1.879	0	-0.211	2.891	0	0	0.06	16.8	0.06	21	0.06
2	1.754	-0.205	0	2.698	0.315	0	0.004	0.06	16.8	0.06	21	0
2	1.754	-0.205	0	2.698	0.315	0	0.004	0.06	16.8	0.06	21	0
19	-0.137	1.879	0	-0.211	2.891	0	0	0.06	16.8	0.06	21	0.06
2	1.983	-0.899	0	3.051	1.383	0	0.004	0.06	16.8	0.06	21	0.02
2	0.746	0.425	0.598	1.148	0.654	0.92	0.002	0.06	16.8	0.06	21	0.01
7	0.522	0.311	-0.598	0.803	0.478	0.92	0.001	0.06	16.8	0.06	21	0.01
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

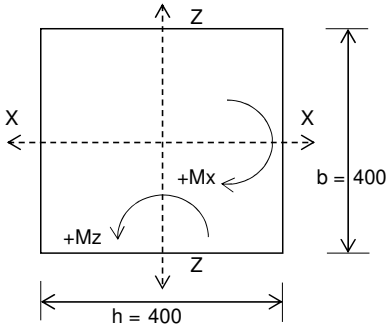
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9.4 DESIGN OF COLUMN MKD. C4

Design Data :

Depth Parallel to X-X axis, h =	400	mm
Depth Parallel to Z-Z axis, b =	400	mm
Clear Cover =	50	mm
d' =	70	mm
$Y'_h = (h-d')/h$ =	0.7	mm
$Y'_b = (b-d')/b$ =	0.7	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.7	M, Hence	k*lux/b =	4.25 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.7	M, Hence	k*luz/h =	4.25 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	8	Nos	20	Dia Bar Equally Distributed In Four Side
Provided	0	NOS	16	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	25.13	cm^2		
Total Provide Pt =	0.016	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 460 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^\alpha + (Mnz/Mnz1)^\alpha$
201	1.265	0.000	0	1.946	0	0	0.003	0.07	15.68	0.07	15.68	0
201	0	0	0	0	0	0	0	0.07	15.68	0.07	15.68	0
201	1.265	0	0	1.946	0	0	0.003	0.07	15.68	0.07	15.68	0
201	1.265	0	0	1.946	0	0	0.003	0.07	15.68	0.07	15.68	0
210	1.084	-0.145	0	1.668	0.223	0	0.003	0.07	15.68	0.07	15.68	0
211	1.084	0.145	0	1.668	0.223	0	0.003	0.07	15.68	0.07	15.68	0
201	1.265	0	0	1.946	0	0	0.003	0.07	15.68	0.07	15.68	0
201	1.265	0	0	1.946	0	0	0.003	0.07	15.68	0.07	15.68	0
211	1.084	0.145	0	1.668	0.223	0	0.003	0.07	15.68	0.07	15.68	0
210	1.084	-0.145	0	1.668	0.223	0	0.003	0.07	15.68	0.07	15.68	0
201	1.265	0	0	1.946	0	0	0.003	0.07	15.68	0.07	15.68	0
201	1.265	0	0	1.946	0	0	0.003	0.07	15.68	0.07	15.68	0
												SAFE

Stirrups

Provided 10 Dia. Stirrups @ 250 mm C/C O.K. *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

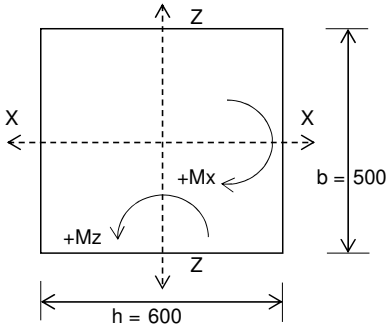
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9.5 DESIGN OF COLUMN MKD. C5

Design Data :

Depth Parallel to X-X axis, h =	600	mm
Depth Parallel to Z-Z axis, b =	500	mm
Clear Cover =	50	mm
d' =	70	mm
Y'_h = (h-d')/h =	0.8	mm
Y'_b = (b-d')/b =	0.7	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.7	M, Hence	k*lux/b =	2.833 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.7	M, Hence	k*luz/h =	3.4 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	12	Nos	20	Dia Bar Equally Distributed In Four Side
Provided	0	NOS	16	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	37.70	cm^2		
Total Provide Pt =	0.013	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 832 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	(Mnx/Mnx1)^α + (Mnz/Mnz1)^α
216	22.968	-8.067	-0.196	35.335	12.411	0.302	0.034	0.065	34.125	0.065	40.95	0.2
225	-18.333	0	0	-28.205	0	0	-0.027	0.065	34.125	0.065	40.95	0
214	-0.466	-2.166	5.252	-0.717	3.332	8.08	-0.001	0.065	34.125	0.065	40.95	0.1
212	-0.267	1.899	-5.422	-0.411	2.922	8.342	0	0.065	34.125	0.065	40.95	0.1
203	1.336	-10.67	2.009	2.055	16.415	3.091	0.002	0.065	34.125	0.065	40.95	0.32
215	3.541	19.694	0.492	5.448	30.298	0.757	0.005	0.065	34.125	0.065	40.95	0.83
201	4.206	0	0.026	6.471	0	0.04	0.006	0.065	34.125	0.065	40.95	0
201	4.206	0	0.026	6.471	0	0.04	0.006	0.065	34.125	0.065	40.95	0
215	3.541	19.694	0.492	5.448	30.298	0.757	0.005	0.065	34.125	0.065	40.95	0.83
203	1.336	-10.67	2.009	2.055	16.415	3.091	0.002	0.065	34.125	0.065	40.95	0.32
214	-0.466	-2.166	5.252	-0.717	3.332	8.08	-0.001	0.065	34.125	0.065	40.95	0.1
212	-0.267	1.899	-5.422	-0.411	2.922	8.342	0	0.065	34.125	0.065	40.95	0.1
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

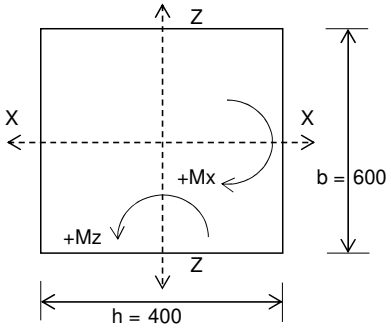
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MD-216-3890-EG-CV-CAL-1442	Mansourah Massarah Gold Project	Page OF
900707_GAU01_AA04_CED01_38903_OU100688672_R2	CALCULATION REAGENTS – SMBS & PEROXIDE STORAGE AND MIXING STATION DESIGN CALCULATION OF FOUNDATION	Date 30-11-2023
O19045_450_C_K_24178_001_R2	AS BUILT	Rev. D01

9.6 DESIGN OF COLUMN MKD. C5

Design Data :

Depth Parallel to X-X axis, h =	400	mm
Depth Parallel to Z-Z axis, b =	600	mm
Clear Cover =	50	mm
d' =	70	mm
$Y'_h = (h-d')/h$ =	0.7	mm
$Y'_b = (b-d')/b$ =	0.8	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.7	M, Hence	k*lux/b =	4.25 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.7	M, Hence	k*luz/h =	2.833 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	8	Nos	20	Dia Bar Equally Distributed In Four Side
Provided	0	NOS	16	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	25.13	cm^2		
Total Provide Pt =	0.01	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 650 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^\alpha + (Mnz/Mnz1)^\alpha$
202	8.641	0.064	0.011	13.294	0.098	0.017	0.016	0.06	30.24	0.06	20.16	0
221	0.806	0	0	1.24	0	0	0.001	0.06	30.24	0.06	20.16	0
213	4.579	0.04	0.054	7.045	0.062	0.083	0.008	0.06	30.24	0.06	20.16	0
221	2.136	0.009	-0.047	3.286	0.014	0.072	0.004	0.06	30.24	0.06	20.16	0
222	2.145	0.006	-0.047	3.3	0.009	0.072	0.004	0.06	30.24	0.06	20.16	0
203	7.002	0.084	0.004	10.772	0.129	0.006	0.013	0.06	30.24	0.06	20.16	0
201	3.472	0.014	0.003	5.342	0.022	0.005	0.006	0.06	30.24	0.06	20.16	0
201	3.472	0.014	0.003	5.342	0.022	0.005	0.006	0.06	30.24	0.06	20.16	0
203	7.002	0.084	0.004	10.772	0.129	0.006	0.013	0.06	30.24	0.06	20.16	0
221	0.806	0	0	1.24	0	0	0.001	0.06	30.24	0.06	20.16	0
213	4.579	0.04	0.054	7.045	0.062	0.083	0.008	0.06	30.24	0.06	20.16	0
221	2.136	0.009	-0.047	3.286	0.014	0.072	0.004	0.06	30.24	0.06	20.16	0
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

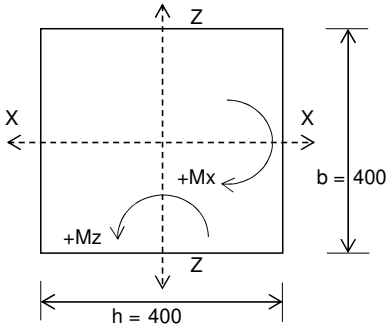
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MD-216-3890-EG-CV-CAL-1442	Mansourah Massarah Gold Project	Page OF
900707_GAU01_AA04_CED01_38903_OU100688672_R2	CALCULATION REAGENTS – SMBS & PEROXIDE STORAGE AND MIXING STATION DESIGN CALCULATION OF FOUNDATION	Date 30-11-2023
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9.7 DESIGN OF COLUMN MKD. C7 & C7a

Design Data :

Depth Parallel to X-X axis, h =	400	mm
Depth Parallel to Z-Z axis, b =	400	mm
Clear Cover =	50	mm
d' =	68	mm
Y'_h = (h-d')/h =	0.7	mm
Y'_b = (b-d')/b =	0.7	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.7	M, Hence	k*lux/b =	4.25 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.7	M, Hence	k*luz/h =	4.25 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	8	Nos	16	Dia Bar Equally Distributed In Four Side
Provided	0	NOS	16	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	16.08	cm^2		
Total Provide Pt =	0.01	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 432 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^α$ + $(Mnz/Mnz1)^α$
212	7.762	0.000	0.013	11.942	0	0.02	0.021	**0.06**	13.44	0.06	13.44	0
224	-5.347	0	0	-8.226	0	0	-0.015	0.06	13.44	0.06	13.44	0
211	7.722	0	0.013	11.88	0	0.02	0.021	0.06	13.44	0.06	13.44	0
214	-3.911	0	-3.628	-6.017	0	5.582	-0.011	0.06	13.44	0.06	13.44	0.24
201	1.589	0	0	2.445	0	0	0.004	0.06	13.44	0.06	13.44	0
201	1.589	0	0	2.445	0	0	0.004	0.06	13.44	0.06	13.44	0
201	1.589	0	0	2.445	0	0	0.004	0.06	13.44	0.06	13.44	0
201	1.589	0	0	2.445	0	0	0.004	0.06	13.44	0.06	13.44	0
201	1.589	0	0	2.445	0	0	0.004	0.06	13.44	0.06	13.44	0
201	1.589	0	0	2.445	0	0	0.004	0.06	13.44	0.06	13.44	0
211	7.722	0	0.013	11.88	0	0.02	0.021	0.06	13.44	0.06	13.44	0
214	-3.911	0	-3.628	-6.017	0	5.582	-0.011	0.06	13.44	0.06	13.44	0.24
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

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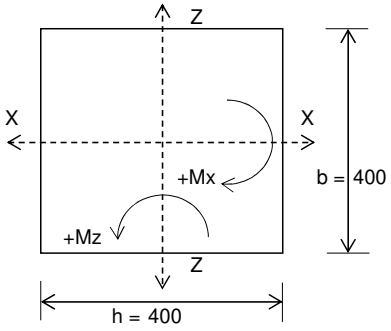
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O19045_450_C_K_24178_001_R2	AS BUILT	Rev.	D01

9.8 DESIGN OF COLUMN MKD. C8

Design Data :

Depth Parallel to X-X axis, h =	400	mm
Depth Parallel to Z-Z axis, b =	400	mm
Clear Cover =	50	mm
d' =	68	mm
$Y'_h = (h-d')/h$ =	0.7	mm
$Y'_b = (b-d')/b$ =	0.7	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.7	M, Hence	k*lux/b =	4.25 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.7	M, Hence	k*luz/h =	4.25 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	8	Nos	16	Dia Bar Equally Distributed In Four Side
Provided	0	NOS	16	Dia Bar Equally Distributed In Four Side

Total Provided Ast = 16.08 cm^2 **SAFE**

Total Provide Pt = 0.01 **SAFE**

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 432 Mton **SAFE**

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^α + (Mnz/Mnz1)^α$
202	1.568	0.000	0.147	2.412	0	0.226	0.004	**0.06**	13.44	0.06	13.44	0
224	-0.039	0	0	-0.06	0	0	0	0.06	13.44	0.06	13.44	0
211	1.418	0	0.597	2.182	0	0.918	0.004	0.06	13.44	0.06	13.44	0.01
224	0.497	0	-0.525	0.765	0	0.808	0.001	0.06	13.44	0.06	13.44	0.01
201	1.162	0	0.033	1.788	0	0.051	0.003	0.06	13.44	0.06	13.44	0
201	1.162	0	0.033	1.788	0	0.051	0.003	0.06	13.44	0.06	13.44	0
201	1.162	0	0.033	1.788	0	0.051	0.003	0.06	13.44	0.06	13.44	0
201	1.162	0	0.033	1.788	0	0.051	0.003	0.06	13.44	0.06	13.44	0
201	1.162	0	0.033	1.788	0	0.051	0.003	0.06	13.44	0.06	13.44	0
201	1.162	0	0.033	1.788	0	0.051	0.003	0.06	13.44	0.06	13.44	0
211	1.418	0	0.597	2.182	0	0.918	0.004	0.06	13.44	0.06	13.44	0.01
224	0.497	0	-0.525	0.765	0	0.808	0.001	0.06	13.44	0.06	13.44	0.01
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

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ATTACHMENT - 1 : ISOMETRIC VIEW OF STAAD MODEL

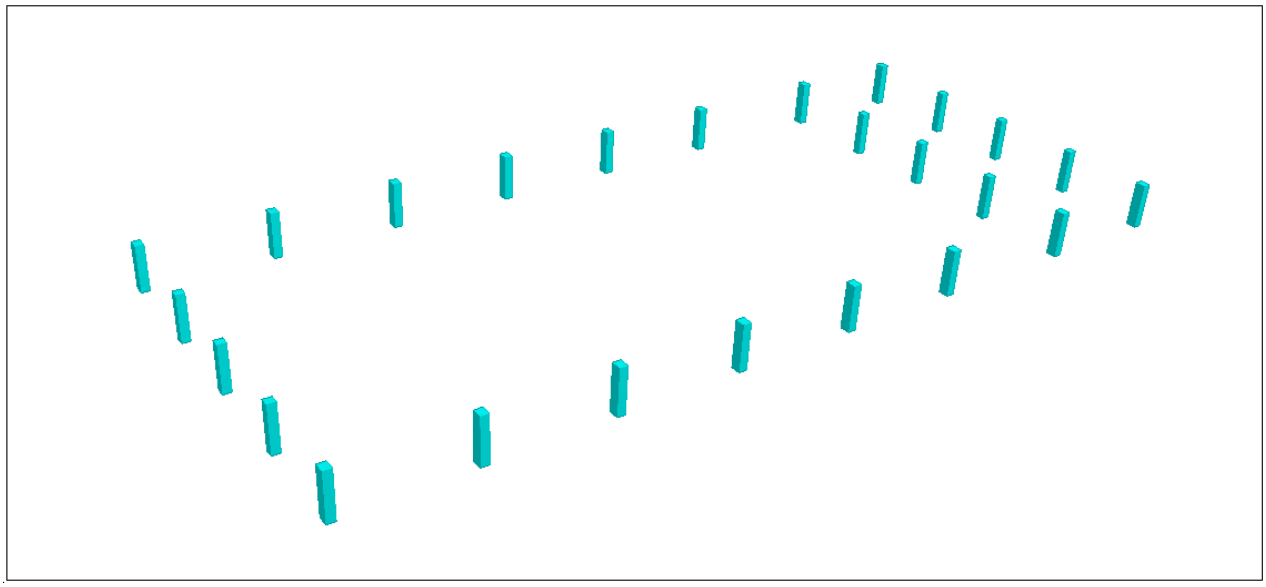
SMBS & PEROXIDE SHED FOUNDATION

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	Part		
Job Title	Ref		
	By	Date 12-Jun-20	Chd
Client	File SMBS_PEROXIDE FOUI	Date/Time	12-Jun-2020 16:37



3D Rendered View

ATTACHMENT - 2 : ISOMETRIC VIEW OF STAAD MODEL

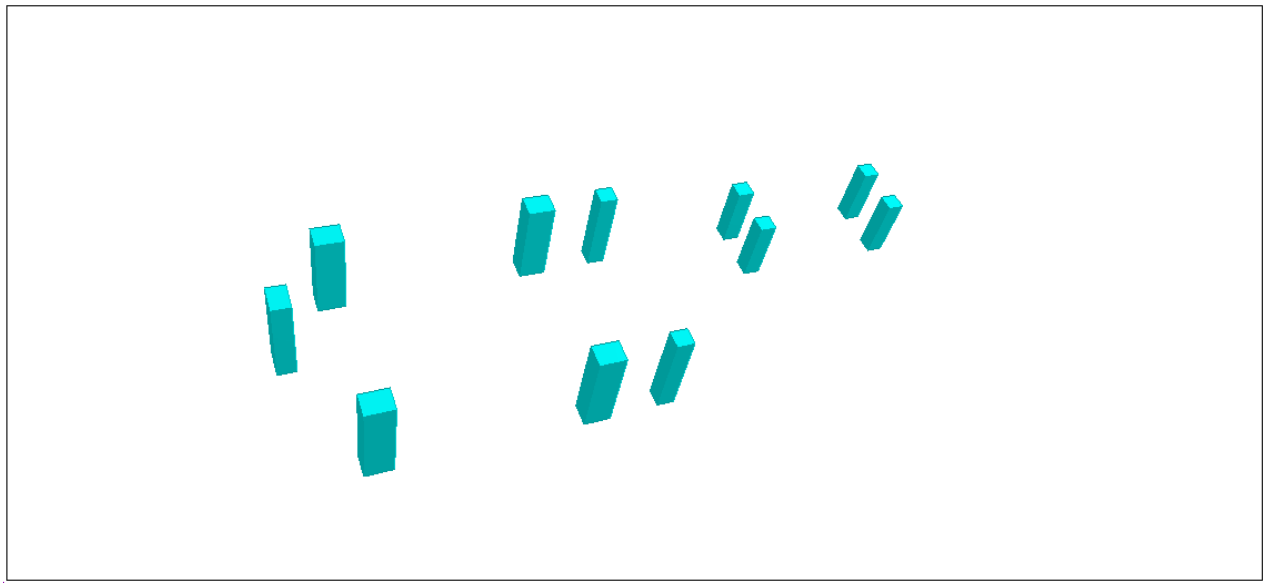
SMBS BAG SPLITTER FOUNDATION

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	Part		
Job Title MMGP	Ref		
	By SUE	Date 6/24/2020	Chd BDA
Client MAADEN	File SMBS BAG SPLITTER F	Date/Time 15-Jun-2020 23:57	



3D Rendered View

ATTACHMENT - 3 : STAAD INPUT

SMBS & PEROXIDE SHED FOUNDATION

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	Part		
Job Title	Ref		
	By	Date 12-Jun-20	Chd
Client	File SMBS_PEROXIDE FOUN	Date/Time 12-Jun-2020 16:37	

Job Information

	Engineer	Checked	Approved
Name:			
Date:	12-Jun-20		

Project ID	
Project Name	

Structure Type	SPACE FRAME

Number of Nodes	50	Highest Node	50
Number of Elements	25	Highest Beam	25

Number of Basic Load Cases	13
Number of Combination Load Cases	54

Included in this printout are data for:

All	The Whole Structure

Included in this printout are results for load cases:

Type	L/C	Name
Primary	1	DEAD LOAD (D)
Primary	2	COLLATERAL (C)
Primary	3	ROOF LIVE (LR)
Primary	4	WIND LEFT-1 (W_L1)
Primary	5	WIND RIGHT-1 (W_R1)
Primary	6	WIND LEFT-2 (W_L2)
Primary	7	WIND RIGHT-2 (W_R2)
Primary	8	WIND LONG-1 (W_LG1)
Primary	9	WIND LONG-2 (W_LG2)
Primary	10	SEISMIC_L (E_L)
Primary	11	SEISMIC_R (E_R)
Primary	12	TEMPERATURE (+)25 (T_(+)25)
Primary	13	TEMPERATURE (-)25 (T_(-)25)
Combination	101	D+C+T_(+)25
Combination	102	D+C+T_(-)25
Combination	103	D+C+LR
Combination	104	D+C+0.75*T_(+)25+0.75*LR
Combination	105	D+C+0.75*T_(-)25+0.75*LR
Combination	106	D+C+W_L1
Combination	107	D+C+W_R1
Combination	108	D+C+W_L2
Combination	109	D+C+W_R2
Combination	110	D+C+W_LG1
Combination	111	D+C+W_LG2
Combination	112	D+C+0.7*E_L
Combination	113	D+C+0.7*E_R
Combination	114	D+C+0.75*W_L1+0.75*LR
Combination	115	D+C+0.75*W_R1+0.75*LR
Combination	116	D+C+0.75*W_L2+0.75*LR

Job Information Cont...

Type	L/C	Name
Combination	120	D+C+0.75*(0.7*E_L)+0.75*LR
Combination	121	D+C+0.75*(0.7*E_R)+0.75*LR
Combination	122	0.6*(D+C)+W_L1
Combination	123	0.6*(D+C)+W_R1
Combination	124	0.6*(D+C)+W_L2
Combination	125	0.6*(D+C)+W_R2
Combination	126	0.6*(D+C)+W_LG1
Combination	127	0.6*(D+C)+W_LG2
Combination	128	0.6*(D+C)+0.7*E_L
Combination	129	0.6*(D+C)+0.7*E_R
Combination	201	1.4*(D+C)
Combination	202	1.4*(D+C+T_(+)25)+0.5*LR
Combination	203	1.4*(D+C+T_(-)25)+0.5*LR
Combination	204	1.2*(D+C)+1.6*LR+0.8*W_L1
Combination	205	1.2*(D+C)+1.6*LR+0.8*W_R1
Combination	206	1.2*(D+C)+1.6*LR+0.8*W_L2
Combination	207	1.2*(D+C)+1.6*LR+0.8*W_R2
Combination	208	1.2*(D+C)+1.6*LR+0.8*W_LG1
Combination	209	1.2*(D+C)+1.6*LR+0.8*W_LG2
Combination	210	1.2*(D+C)+1.6*W_L1+0.5*LR
Combination	211	1.2*(D+C)+1.6*W_R1+0.5*LR
Combination	212	1.2*(D+C)+1.6*W_L2+0.5*LR
Combination	213	1.2*(D+C)+1.6*W_R2+0.5*LR
Combination	214	1.2*(D+C)+1.6*W_LG1+0.5*LR
Combination	215	1.2*(D+C)+1.6*W_LG2+0.5*LR
Combination	216	1.2*(D+C)+E_L
Combination	217	1.2*(D+C)+E_R
Combination	218	0.9*(D+C)+1.6*W_L1
Combination	219	0.9*(D+C)+1.6*W_R1
Combination	220	0.9*(D+C)+1.6*W_L2
Combination	221	0.9*(D+C)+1.6*W_R2
Combination	222	0.9*(D+C)+1.6*W_LG1
Combination	223	0.9*(D+C)+1.6*W_LG2
Combination	224	0.9*(D+C)+E_L
Combination	225	0.9*(D+C)+E_R

Nodes

Node	X (m)	Y (m)	Z (m)
1	0	910.400	0
2	6.250	910.400	0
3	12.500	910.400	0
4	18.750	910.400	0
5	25.000	910.400	0
6	31.250	910.400	0
7	38.850	910.400	0
8	45.100	910.400	0
9	0	910.400	5.000
10	38.850	910.400	5.000
11	45.100	910.400	5.000
12	0	910.400	9.500
13	38.850	910.400	9.500
14	45.100	910.400	9.500
15	0	910.400	14.290
16	38.850	910.400	14.290
17	45.100	910.400	14.290
18	0	910.400	19.000
19	6.250	910.400	19.000
20	12.500	910.400	19.000
21	18.750	910.400	19.000
22	25.000	910.400	19.000
23	31.250	910.400	19.000
24	38.850	910.400	19.000
25	45.100	910.400	19.000
26	0	912.750	0
27	6.250	912.750	0
28	12.500	912.750	0
29	18.750	912.750	0
30	25.000	912.750	0
31	31.250	912.750	0
32	38.850	912.750	0
33	45.100	912.750	0
34	0	912.750	5.000
35	38.850	912.750	5.000
36	45.100	912.750	5.000
37	0	912.750	9.500
38	38.850	912.750	9.500
39	45.100	912.750	9.500
40	0	912.750	14.290
41	38.850	912.750	14.290
42	45.100	912.750	14.290
43	0	912.750	19.000
44	6.250	912.750	19.000
45	12.500	912.750	19.000
46	18.750	912.750	19.000
47	25.000	912.750	19.000
48	31.250	912.750	19.000
49	38.850	912.750	19.000

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
50	45.100	912.750	19.000

Beams

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1	1	26	2.350	1	0
2	2	27	2.350	1	0
3	3	28	2.350	1	0
4	4	29	2.350	1	0
5	5	30	2.350	1	0
6	6	31	2.350	1	0
7	7	32	2.350	1	0
8	8	33	2.350	1	0
9	12	37	2.350	2	0
10	15	40	2.350	2	0
11	18	43	2.350	1	0
12	9	34	2.350	2	0
13	19	44	2.350	1	0
14	20	45	2.350	1	0
15	21	46	2.350	1	0
16	22	47	2.350	1	0
17	23	48	2.350	1	0
18	24	49	2.350	1	0
19	25	50	2.350	1	0
20	14	39	2.350	2	0
21	17	42	2.350	2	0
22	11	36	2.350	2	0
23	13	38	2.350	2	0
24	16	41	2.350	2	0
25	10	35	2.350	2	0

Section Properties

Prop	Section	Area (cm^2)	I_{yy} (cm^4)	I_{zz} (cm^4)	J (cm^4)	Material
1	Rect 0.40x0.50	2E+3	417E+3	267E+3	547E+3	CONCRETE
2	Rect 0.40x0.40	1.6E+3	213E+3	213E+3	360E+3	CONCRETE

Materials

Mat	Name	E (kN/mm^2)	ν	Density (kg/m^3)	α (/°C)
1	CONCRETE	21.718	0.170	2.4E+3	10E -6
2	ALUMINUM	68.948	0.330	2.71E+3	23E -6
3	STEEL_50_KSI	199.948	0.300	7.83E+3	11.7E -6
4	STAINLESSSTEEL	197.930	0.300	7.83E+3	18E -6
5	STEEL_36_KSI	199.948	0.300	7.83E+3	11.7E -6
6	STEEL_275_NMM2	205.000	0.300	7.85E+3	12E -6
7	STEEL	205.000	0.300	7.83E+3	12E -6
8	STEEL_355_NMM2	205.000	0.300	7.85E+3	12E -6

Supports

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
1	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
2	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
3	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
4	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
5	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
6	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
7	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
8	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
9	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
10	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
11	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
12	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
13	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
14	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
15	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
16	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
17	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
18	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
19	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
20	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
21	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
22	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
23	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
24	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
25	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Releases

There is no data of this type.

Primary Load Cases

Number	Name	Type
1	DEAD LOAD (D)	None
2	COLLATERAL (C)	None
3	ROOF LIVE (LR)	None
4	WIND LEFT-1 (W_L1)	None
5	WIND RIGHT-1 (W_R1)	None
6	WIND LEFT-2 (W_L2)	None
7	WIND RIGHT-2 (W_R2)	None
8	WIND LONG-1 (W_LG1)	None
9	WIND LONG-2 (W_LG2)	None
10	SEISMIC_L (E_L)	None
11	SEISMIC_R (E_R)	None
12	TEMPERATURE (+)25 (T_(+)25)	None
13	TEMPERATURE (-)25 (T_(-)25)	None

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
101	D+C+T_(+)25	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		12	TEMPERATURE (+)25 (T_(+)25)	1.00
102	D+C+T_(-)25	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		13	TEMPERATURE (-)25 (T_(-)25)	1.00
103	D+C+LR	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		3	ROOF LIVE (LR)	1.00
104	D+C+0.75*T_(+)25+0.75*LR	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		12	TEMPERATURE (+)25 (T_(+)25)	0.75
		3	ROOF LIVE (LR)	0.75
105	D+C+0.75*T_(-)25+0.75*LR	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		13	TEMPERATURE (-)25 (T_(-)25)	0.75
		3	ROOF LIVE (LR)	0.75
106	D+C+W_L1	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		4	WIND LEFT-1 (W_L1)	1.00
107	D+C+W_R1	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		5	WIND RIGHT-1 (W_R1)	1.00
108	D+C+W_L2	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		6	WIND LEFT-2 (W_L2)	1.00
109	D+C+W_R2	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		7	WIND RIGHT-2 (W_R2)	1.00
110	D+C+W_LG1	1	DEAD LOAD (D)	1.00

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		2	COLLATERAL (C)	1.00
		8	WIND LONG-1 (W_LG1)	1.00
111	D+C+W_LG2	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		9	WIND LONG-2 (W_LG2)	1.00
112	D+C+0.7*E_L	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		10	SEISMIC_L (E_L)	1.00
113	D+C+0.7*E_R	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		11	SEISMIC_R (E_R)	1.00
114	D+C+0.75*W_L1+0.75*LR	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		4	WIND LEFT-1 (W_L1)	0.75
		3	ROOF LIVE (LR)	0.75
115	D+C+0.75*W_R1+0.75*LR	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		5	WIND RIGHT-1 (W_R1)	0.75
		3	ROOF LIVE (LR)	0.75
116	D+C+0.75*W_L2+0.75*LR	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		6	WIND LEFT-2 (W_L2)	0.75
		3	ROOF LIVE (LR)	0.75
117	D+C+0.75*W_R2+0.75*LR	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		7	WIND RIGHT-2 (W_R2)	0.75
		3	ROOF LIVE (LR)	0.75
118	D+C+0.75*W_LG1+0.75*LR	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		8	WIND LONG-1 (W_LG1)	0.75
		3	ROOF LIVE (LR)	0.75
119	D+C+0.75*W_LG2+0.75*LR	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		9	WIND LONG-2 (W_LG2)	0.75
		3	ROOF LIVE (LR)	0.75
120	D+C+0.75*(0.7*E_L)+0.75*LR	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		10	SEISMIC_L (E_L)	0.52
		3	ROOF LIVE (LR)	0.75
121	D+C+0.75*(0.7*E_R)+0.75*LR	1	DEAD LOAD (D)	1.00
		2	COLLATERAL (C)	1.00
		11	SEISMIC_R (E_R)	0.52
		3	ROOF LIVE (LR)	0.75
122	0.6*(D+C)+W_L1	1	DEAD LOAD (D)	0.60
		2	COLLATERAL (C)	0.60
		4	WIND LEFT-1 (W_L1)	1.00
123	0.6*(D+C)+W_R1	1	DEAD LOAD (D)	0.60
		2	COLLATERAL (C)	0.60
		5	WIND RIGHT-1 (W_R1)	1.00

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
124	0.6*(D+C)+W_L2	1	DEAD LOAD (D)	0.60
		2	COLLATERAL (C)	0.60
		6	WIND LEFT-2 (W_L2)	1.00
125	0.6*(D+C)+W_R2	1	DEAD LOAD (D)	0.60
		2	COLLATERAL (C)	0.60
		7	WIND RIGHT-2 (W_R2)	1.00
126	0.6*(D+C)+W_LG1	1	DEAD LOAD (D)	0.60
		2	COLLATERAL (C)	0.60
		8	WIND LONG-1 (W_LG1)	1.00
127	0.6*(D+C)+W_LG2	1	DEAD LOAD (D)	0.60
		2	COLLATERAL (C)	0.60
		9	WIND LONG-2 (W_LG2)	1.00
128	0.6*(D+C)+0.7*E_L	1	DEAD LOAD (D)	0.60
		2	COLLATERAL (C)	0.60
		10	SEISMIC_L (E_L)	0.70
129	0.6*(D+C)+0.7*E_R	1	DEAD LOAD (D)	0.60
		2	COLLATERAL (C)	0.60
		11	SEISMIC_R (E_R)	0.70
201	1.4*(D+C)	1	DEAD LOAD (D)	1.40
		2	COLLATERAL (C)	1.40
202	1.4*(D+C+T_(+)25)+0.5*LR	1	DEAD LOAD (D)	1.40
		2	COLLATERAL (C)	1.40
		12	TEMPERATURE (+)25 (T_(+)25)	1.40
		3	ROOF LIVE (LR)	0.50
203	1.4*(D+C+T_(-)25)+0.5*LR	1	DEAD LOAD (D)	1.40
		2	COLLATERAL (C)	1.40
		13	TEMPERATURE (-)25 (T_(-)25)	1.40
		3	ROOF LIVE (LR)	0.50
204	1.2*(D+C)+1.6*LR+0.8*W_L1	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		3	ROOF LIVE (LR)	1.60
		4	WIND LEFT-1 (W_L1)	0.80
205	1.2*(D+C)+1.6*LR+0.8*W_R1	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		3	ROOF LIVE (LR)	1.60
		5	WIND RIGHT-1 (W_R1)	0.80
206	1.2*(D+C)+1.6*LR+0.8*W_L2	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		3	ROOF LIVE (LR)	1.60
		6	WIND LEFT-2 (W_L2)	0.80
207	1.2*(D+C)+1.6*LR+0.8*W_R2	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		3	ROOF LIVE (LR)	1.60
		7	WIND RIGHT-2 (W_R2)	0.80
208	1.2*(D+C)+1.6*LR+0.8*W_LG1	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		3	ROOF LIVE (LR)	1.60
		8	WIND LONG-1 (W_LG1)	0.80
209	1.2*(D+C)+1.6*LR+0.8*W_LG2	1	DEAD LOAD (D)	1.20

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		2	COLLATERAL (C)	1.20
		3	ROOF LIVE (LR)	1.60
		9	WIND LONG-2 (W_LG2)	0.80
210	1.2*(D+C)+1.6*W_L1+0.5*LR	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		4	WIND LEFT-1 (W_L1)	1.60
		3	ROOF LIVE (LR)	0.50
211	1.2*(D+C)+1.6*W_R1+0.5*LR	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		5	WIND RIGHT-1 (W_R1)	1.60
		3	ROOF LIVE (LR)	0.50
212	1.2*(D+C)+1.6*W_L2+0.5*LR	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		6	WIND LEFT-2 (W_L2)	1.60
		3	ROOF LIVE (LR)	0.50
213	1.2*(D+C)+1.6*W_R2+0.5*LR	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		7	WIND RIGHT-2 (W_R2)	1.60
		3	ROOF LIVE (LR)	0.50
214	1.2*(D+C)+1.6*W_LG1+0.5*LR	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		8	WIND LONG-1 (W_LG1)	1.60
		3	ROOF LIVE (LR)	0.50
215	1.2*(D+C)+1.6*W_LG2+0.5*LR	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		9	WIND LONG-2 (W_LG2)	1.60
		3	ROOF LIVE (LR)	0.50
216	1.2*(D+C)+E_L	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		10	SEISMIC_L (E_L)	1.00
217	1.2*(D+C)+E_R	1	DEAD LOAD (D)	1.20
		2	COLLATERAL (C)	1.20
		11	SEISMIC_R (E_R)	1.00
218	0.9*(D+C)+1.6*W_L1	1	DEAD LOAD (D)	0.90
		2	COLLATERAL (C)	0.90
		4	WIND LEFT-1 (W_L1)	1.60
219	0.9*(D+C)+1.6*W_R1	1	DEAD LOAD (D)	0.90
		2	COLLATERAL (C)	0.90
		5	WIND RIGHT-1 (W_R1)	1.60
220	0.9*(D+C)+1.6*W_L2	1	DEAD LOAD (D)	0.90
		2	COLLATERAL (C)	0.90
		6	WIND LEFT-2 (W_L2)	1.60
221	0.9*(D+C)+1.6*W_R2	1	DEAD LOAD (D)	0.90
		2	COLLATERAL (C)	0.90
		7	WIND RIGHT-2 (W_R2)	1.60
222	0.9*(D+C)+1.6*W_LG1	1	DEAD LOAD (D)	0.90
		2	COLLATERAL (C)	0.90
		8	WIND LONG-1 (W_LG1)	1.60
223	0.9*(D+C)+1.6*W_LG2	1	DEAD LOAD (D)	0.90

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		2	COLLATERAL (C)	0.90
		9	WIND LONG-2 (W_LG2)	1.60
224	0.9*(D+C)+E_L	1	DEAD LOAD (D)	0.90
		2	COLLATERAL (C)	0.90
		10	SEISMIC_L (E_L)	1.00
225	0.9*(D+C)+E_R	1	DEAD LOAD (D)	0.90
		2	COLLATERAL (C)	0.90
		11	SEISMIC_R (E_R)	1.00

Load Generators

There is no data of this type.

1 DEAD LOAD (D) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	-0.029	0.002	-	-	-
27	-	-0.122	-0.059	-	-	-
28	-	-0.122	-0.059	-	-	-
29	-	-0.122	-0.059	-	-	-
30	-	-0.122	-0.059	-	-	-
31	-	-0.122	-0.059	-	-	-
32	-	-0.122	-0.059	-	-	-
33	-	-0.029	0.002	-	-	-
34	-	-0.036	-	-	-	-
36	-	-0.036	-	-	-	-
37	-	-0.036	-	-	-	-
39	-	-0.036	-	-	-	-
40	-	-0.036	-	-	-	-
42	-	-0.036	-	-	-	-
43	-	-0.028	0.002	-	-	-
44	-	-0.122	-0.059	-	-	-
45	-	-0.122	-0.059	-	-	-
46	-	-0.122	-0.059	-	-	-
47	-	-0.122	-0.059	-	-	-
48	-	-0.122	-0.059	-	-	-
49	-	-0.122	-0.059	-	-	-
50	-	-0.028	0.002	-	-	-

1 DEAD LOAD (D) : Selfweight

Direction	Factor	Assigned Geometry
Y	-1.000	ALL

2 COLLATERAL (C) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	-0.037	-0.003	-	-	-
27	-	-0.002	-0.003	-	-	-
28	-	-0.002	-0.003	-	-	-
29	-	-0.002	-0.003	-	-	-
30	-	-0.002	-0.003	-	-	-
31	-	-0.002	-0.003	-	-	-
32	-	-0.002	-0.003	-	-	-
33	-	-0.037	-0.003	-	-	-
34	-	-0.029	-0.002	-	-	-
36	-	-0.029	-0.002	-	-	-
37	-	-0.015	0.003	-	-	-
39	-	-0.015	0.003	-	-	-
40	-	0.008	-	-	-	-
42	-	0.008	-	-	-	-
43	-	-0.055	-0.001	-	-	-
44	-	-0.046	0.003	-	-	-
45	-	-0.046	0.003	-	-	-
46	-	-0.046	0.003	-	-	-
47	-	-0.046	0.003	-	-	-
48	-	-0.046	0.003	-	-	-
49	-	-0.046	0.003	-	-	-
50	-	-0.055	-0.001	-	-	-

3 ROOF LIVE (LR) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	-0.056	0.007	-	-	-
27	-	-0.409	-0.231	-	-	-
28	-	-0.409	-0.231	-	-	-
29	-	-0.409	-0.231	-	-	-
30	-	-0.409	-0.231	-	-	-
31	-	-0.409	-0.231	-	-	-
32	-	-0.409	-0.231	-	-	-
33	-	-0.056	0.007	-	-	-
34	-	-0.098	-	-	-	-
36	-	-0.098	-	-	-	-
37	-	-0.098	-	-	-	-
39	-	-0.098	-	-	-	-
40	-	-0.098	-	-	-	-
42	-	-0.098	-	-	-	-
43	-	-0.055	0.007	-	-	-
44	-	-0.409	-0.231	-	-	-
45	-	-0.409	-0.231	-	-	-
46	-	-0.409	-0.231	-	-	-
47	-	-0.409	-0.231	-	-	-
48	-	-0.409	-0.231	-	-	-
49	-	-0.409	-0.231	-	-	-

3 ROOF LIVE (LR) : Node Loads Cont...

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
50	-	-0.055	0.007	-	-	-

4 WIND LEFT-1 (W L1) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	0.245	-0.137	-	-	-
27	-	-0.566	0.176	-	-	-
28	-	-0.566	0.176	-	-	-
29	-	-0.566	0.176	-	-	-
30	-	-0.566	0.176	-	-	-
31	-	-0.566	0.176	-	-	-
32	-	-0.566	0.176	-	-	-
33	-	0.245	-0.137	-	-	-
34	0.169	0.246	-	-	-	-
35	-	-	-0.038	-	-	-
36	0.169	0.246	-	-	-	-
37	0.174	0.217	-	-	-	-
38	-	-	-0.038	-	-	-
39	0.174	0.217	-	-	-	-
40	0.169	0.206	-	-	-	-
41	-	-	-0.038	-	-	-
42	0.169	0.206	-	-	-	-
43	-	0.026	0.123	-	-	-
44	-	0.815	0.531	-	-	-
45	-	0.815	0.531	-	-	-
46	-	0.815	0.531	-	-	-
47	-	0.815	0.531	-	-	-
48	-	0.815	0.531	-	-	-
49	-	0.815	0.531	-	-	-
50	-	0.026	0.123	-	-	-

5 WIND RIGHT-1 (W_R1) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	0.024983	0.124	-	-	-
27	-	0.815	0.531	-	-	-
28	-	0.815	0.531	-	-	-
29	-	0.815	0.531	-	-	-
30	-	0.815	0.531	-	-	-
31	-	0.815	0.531	-	-	-
32	-	0.815	0.531	-	-	-
33	-	0.024983	0.124	-	-	-
34	0.169	0.207	-	-	-	-
35	-	-	0.038	-	-	-
36	0.169	0.207	-	-	-	-
37	0.174	0.217	-	-	-	-
38	-	-	0.038	-	-	-
39	0.174	0.217	-	-	-	-
40	0.169	0.247	-	-	-	-
41	-	-	0.038	-	-	-
42	0.169	0.247	-	-	-	-
43	-	0.243	-0.136	-	-	-
44	-	0.566	0.175	-	-	-
45	-	0.566	0.175	-	-	-
46	-	0.566	0.175	-	-	-
47	-	0.566	0.175	-	-	-
48	-	0.566	0.175	-	-	-
49	-	0.566	0.175	-	-	-
50	-	0.243	-0.136	-	-	-

6 WIND LEFT-2 (W_L2) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	-0.181	-0.168	-	-	-
27	-	0.217	0.077	-	-	-
28	-	0.217	0.077	-	-	-
29	-	0.217	0.077	-	-	-
30	-	0.217	0.077	-	-	-
31	-	0.217	0.077	-	-	-
32	-	0.217	0.077	-	-	-
33	-	-0.181	-0.168	-	-	-
34	-0.154	0.176	-	-	-	-
36	-0.154	0.176	-	-	-	-
37	-0.157	0.143	-	-	-	-
39	-0.157	0.143	-	-	-	-
40	-0.154	0.134	-	-	-	-
42	-0.154	0.134	-	-	-	-
43	-	-0.038	0.091	-	-	-
44	-	0.467	0.433	-	-	-
45	-	0.467	0.433	-	-	-
46	-	0.467	0.433	-	-	-

6 WIND LEFT-2 (W_L2) : Node Loads Cont...

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
47	-	0.467	0.433	-	-	-
48	-	0.467	0.433	-	-	-
49	-	0.467	0.433	-	-	-
50	-	-0.038	0.091	-	-	-

7 WIND RIGHT-2 (W_R2) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	-0.039	0.093	-	-	-
27	-	0.467	0.432	-	-	-
28	-	0.467	0.432	-	-	-
29	-	0.467	0.432	-	-	-
30	-	0.467	0.432	-	-	-
31	-	0.467	0.432	-	-	-
32	-	0.467	0.432	-	-	-
33	-	-0.039	0.093	-	-	-
34	-0.154	0.135	-	-	-	-
36	-0.154	0.135	-	-	-	-
37	-0.157	0.143	-	-	-	-
39	-0.157	0.143	-	-	-	-
40	-0.154	0.176	-	-	-	-
42	-0.154	0.176	-	-	-	-
43	-	0.180	-0.167	-	-	-
44	-	0.217	0.076	-	-	-
45	-	0.217	0.076	-	-	-
46	-	0.217	0.076	-	-	-
47	-	0.217	0.076	-	-	-
48	-	0.217	0.076	-	-	-
49	-	0.217	0.076	-	-	-
50	-	0.180	-0.167	-	-	-

8 WIND LONG-1 (W LG1) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	0.141	-	-	-	-
	-	0.256	0.061	-	-	-
27	-	0.742	0.232	-	-	-
	0.159	-0.141	-	-	-	-
28	-	0.742	0.232	-	-	-
29	-	0.742	0.232	-	-	-
30	-	0.742	0.232	-	-	-
31	-	0.742	0.232	-	-	-
32	-	0.742	0.232	-	-	-
33	-	0.256	0.061	-	-	-
34	0.169	0.086268	-	-	-	-
36	0.169	0.086268	-	-	-	-
37	0.174	0.147	-	-	-	-
39	0.174	0.147	-	-	-	-
40	0.169	0.201	-	-	-	-
42	0.169	0.201	-	-	-	-
43	-	0.141	-	-	-	-
	-	0.243	0.046	-	-	-
44	-	0.883	0.201	-	-	-
	0.159	-0.141	-	-	-	-
45	-	0.883	0.201	-	-	-
46	-	0.883	0.201	-	-	-
47	-	0.883	0.201	-	-	-
48	-	0.883	0.201	-	-	-
49	-	0.883	0.201	-	-	-
50	-	0.243	0.046	-	-	-

9 WIND LONG-2 (W LG2) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	0.243	0.046	-	-	-
27	-	0.883	0.202	-	-	-
28	-	0.883	0.202	-	-	-
29	-	0.883	0.202	-	-	-
30	-	0.883	0.202	-	-	-
31	-	0.883	0.202	-	-	-
32	-	0.883	0.202	-	-	-
	-0.159	-0.141	-	-	-	-
33	-	0.141	-	-	-	-
	-	0.243	0.046	-	-	-
34	-0.154	0.201	-	-	-	-
36	-0.154	0.201	-	-	-	-
37	-0.167	0.147	-	-	-	-
39	-0.167	0.147	-	-	-	-
40	-0.154	0.086268	-	-	-	-
42	-0.154	0.086268	-	-	-	-
43	-	0.256	0.061	-	-	-

9 WIND LONG-2 (W LG2) : Node Loads Cont...

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
44	-	0.742	0.233	-	-	-
45	-	0.742	0.233	-	-	-
46	-	0.742	0.233	-	-	-
47	-	0.742	0.233	-	-	-
48	-	0.742	0.233	-	-	-
49	-	0.742	0.233	-	-	-
	-0.159	-0.141	-	-	-	-
50	-	0.141	-	-	-	-
	-	0.256	0.061	-	-	-

10 SEISMIC L (E L) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	-0.004	-0.003	-	-	-
	-	0.017	-	-	-	-
27	-	-0.002	-0.004	-	-	-
	0.019	-0.017	-	-	-	-
28	-	-0.002	-0.004	-	-	-
29	-	-0.002	-0.004	-	-	-
30	-	-0.002	-0.004	-	-	-
31	-	-0.002	-0.004	-	-	-
32	-	-0.002	-0.004	-	-	-
33	-	-0.004	-0.003	-	-	-
34	-	0.004	-	-	-	-
36	-	0.004	-	-	-	-
37	-	0.004	-	-	-	-
39	-	0.004	-	-	-	-
40	-	-0.004	-	-	-	-
42	-	-0.004	-	-	-	-
43	-	0.004	0.003	-	-	-
	-	0.017	-	-	-	-
44	-	0.002	0.004	-	-	-
	0.019	-0.017	-	-	-	-
45	-	0.002	0.004	-	-	-
46	-	0.002	0.004	-	-	-
47	-	0.002	0.004	-	-	-
48	-	0.002	0.004	-	-	-
49	-	0.002	0.004	-	-	-
50	-	0.004	0.003	-	-	-

11 SEISMIC R (E R) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	0.004	0.003	-	-	-
27	-	0.002	0.004	-	-	-
28	-	0.002	0.004	-	-	-
29	-	0.002	0.004	-	-	-
30	-	0.002	0.004	-	-	-
31	-	0.002	0.004	-	-	-
32	-	0.002	0.004	-	-	-
	0.019	-0.017	-	-	-	-
33	-	0.004	0.003	-	-	-
	-	0.017	-	-	-	-
34	-	-0.004	-	-	-	-
36	-	-0.004	-	-	-	-
37	-	-	-	-	-	-
39	-	-	-	-	-	-
40	-	0.004	-	-	-	-
42	-	0.004	-	-	-	-
43	-	-0.004	-0.003	-	-	-
44	-	-0.002	-0.004	-	-	-
45	-	-0.002	-0.004	-	-	-
46	-	-0.002	-0.004	-	-	-
47	-	-0.002	-0.004	-	-	-
48	-	-0.002	-0.004	-	-	-
49	-	-0.002	-0.004	-	-	-
	0.019	-0.017	-	-	-	-
50	-	-0.004	-0.003	-	-	-
	-	0.017	-	-	-	-

12 TEMPERATURE (+)25 (T (+)25) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	0.016	0.010	-	-	-
27	-	-	-0.004	-	-	-
28	-	-	-0.004	-	-	-
29	-	-	-0.004	-	-	-
30	-	-	-0.004	-	-	-
31	-	-	-0.004	-	-	-
32	-	-	-0.004	-	-	-
33	-	0.016	0.010	-	-	-
34	-	0.018	-	-	-	-
36	-	0.018	-	-	-	-
37	-	0.005	-	-	-	-
39	-	0.005	-	-	-	-
40	-	0.018	-	-	-	-
42	-	0.018	-	-	-	-
43	-	0.016	-0.010	-	-	-
44	-	-	0.004	-	-	-
45	-	-	0.004	-	-	-

12 TEMPERATURE (+)25 (T_(+)25) : Node Loads Cont...

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
46	-	-	0.004	-	-	-
47	-	-	0.004	-	-	-
48	-	-	0.004	-	-	-
49	-	-	0.004	-	-	-
50	-	0.016	-0.010	-	-	-

13 TEMPERATURE (-)25 (T_(-)25) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
26	-	-0.016	0.010	-	-	-
27	-	-	0.004	-	-	-
28	-	-	0.004	-	-	-
29	-	-	0.004	-	-	-
30	-	-	0.004	-	-	-
31	-	-	0.004	-	-	-
32	-	-	0.004	-	-	-
33	-	-0.016	0.010	-	-	-
34	-	-0.018	-	-	-	-
36	-	-0.018	-	-	-	-
37	-	-0.005	-	-	-	-
39	-	-0.005	-	-	-	-
40	-	-0.018	-	-	-	-
42	-	-0.018	-	-	-	-
43	-	-0.016	0.010	-	-	-
44	-	-	-0.004	-	-	-
45	-	-	-0.004	-	-	-
46	-	-	-0.004	-	-	-
47	-	-	-0.004	-	-	-
48	-	-	-0.004	-	-	-
49	-	-	-0.004	-	-	-
50	-	-0.016	0.010	-	-	-

ATTACHMENT - 4 : STAAD INPUT

SMBS BAG SPLITTER FOUNDATION

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Software licensed to STAAD.Pro Advanced CONNECTED User: Suman Podder	Job No O19045	Sheet No 1	Rev
	Part		
Job Title MMGP	Ref		
	By SUE	Date 6/24/2020	Chd BDA
Client MAADEN	File SMBS BAG SPLITTER F	Date/Time 15-Jun-2020 23:57	

Job Information

	Engineer	Checked	Approved
Name:	SUE	BDA	AG
Date:	6/24/2020	6/25/2020	6/26/2020

Project ID	
Project Name	

Structure Type	SPACE FRAME

Number of Nodes	22	Highest Node	22
Number of Elements	11	Highest Beam	11

Number of Basic Load Cases	19
Number of Combination Load Cases	65

Included in this printout are data for:

All	The Whole Structure

Included in this printout are results for load cases:

Type	L/C	Name
Primary	1	DEAD LOAD (D)
Primary	2	ROOF LIVE LOAD (LR)
Primary	3	LIVE LOAD (L)
Primary	4	CRANE LOAD-1 (CR1)
Primary	5	CRANE LOAD-2 (CR2)
Primary	6	CRANE LOAD-3 (CR3)
Primary	7	CRANE LOAD-4 (CR4)
Primary	8	WIND LEFT-1 (W_L1)
Primary	9	WIND LEFT-2 (W_L2)
Primary	10	WIND RIGHT-1 (W_R1)
Primary	11	WIND RIGHT-2 (W_R2)
Primary	12	WIND LONG1 (W_LG1)
Primary	13	WIND LONG2 (W_LG2)
Primary	14	SEISMIC (+)X (E_(+)X)
Primary	15	SEISMIC (-)X (E_(-)X)
Primary	16	SEISMIC (+)Z (E_(+)Z)
Primary	17	SEISMIC (-)Z (E_(-)Z)
Primary	18	TEMPERATURE (+)25 (T_(+)25)
Primary	19	TEMPERATURE (-)25 (T_(-)25)
Combination	101	D+L+T_(+)25
Combination	102	D+L+T_(-)25
Combination	103	D+LR
Combination	104	D+0.75*(L+T_(+)25)+0.75*LR
Combination	105	D+0.75*(L+T_(-)25)+0.75*LR
Combination	106	D+W_L1
Combination	107	D+W_R1
Combination	108	D+W_L2
Combination	109	D+W_R2
Combination	110	D+W_LG1

Job Information Cont...

Type	L/C	Name
Combination	114	D+0.7*E_(+)Z
Combination	115	D+0.7*E_(-)Z
Combination	116	D+0.75*W_L1+0.75*L+0.75*LR
Combination	117	D+0.75*W_L2+0.75*L+0.75*LR
Combination	118	D+0.75*W_R1+0.75*L+0.75*LR
Combination	119	D+0.75*W_R2+0.75*L+0.75*LR
Combination	120	D+0.75*W_LG1+0.75*L+0.75*LR
Combination	121	D+0.75*W_LG2+0.75*L+0.75*LR
Combination	122	D+0.75*(0.7*E_(+)X)+0.75*L+0.75*LR
Combination	123	D+0.75*(0.7*E_(-)X)+0.75*L+0.75*LR
Combination	124	D+0.75*(0.7*E_(+)Z)+0.75*L+0.75*LR
Combination	125	D+0.75*(0.7*E_(-)Z)+0.75*L+0.75*LR
Combination	126	0.6*D+W_L1
Combination	127	0.6*D+W_L2
Combination	128	0.6*D+W_R1
Combination	129	0.6*D+W_R2
Combination	130	0.6*D+W_LG1
Combination	131	0.6*D+W_LG2
Combination	132	0.6*D+0.7*E_(+)X
Combination	133	0.6*D+0.7*E_(-)X
Combination	134	0.6*D+0.7*E_(+)Z
Combination	135	0.6*D+0.7*E_(-)Z
Combination	201	1.4*D
Combination	202	1.2*(D+T_(+)25)+1.6*L+0.5*LR
Combination	203	1.2*(D+T_(-)25)+1.6*L+0.5*LR
Combination	204	1.2*D+1.6*LR+0.5*L
Combination	205	1.2*D+1.6*LR+0.8*W_L1
Combination	206	1.2*D+1.6*LR+0.8*W_L2
Combination	207	1.2*D+1.6*LR+0.8*W_R1
Combination	208	1.2*D+1.6*LR+0.8*W_R2
Combination	209	1.2*D+1.6*LR+0.8*W_LG1
Combination	210	1.2*D+1.6*LR+0.8*W_LG2
Combination	211	1.2*D+1.6*W_L1+0.5*L+0.5*LR
Combination	212	1.2*D+1.6*W_L2+0.5*L+0.5*LR
Combination	213	1.2*D+1.6*W_R1+0.5*L+0.5*LR
Combination	214	1.2*D+1.6*W_R2+0.5*L+0.5*LR
Combination	215	1.2*D+1.6*W_LG1+0.5*L+0.5*LR
Combination	216	1.2*D+1.6*W_LG2+0.5*L+0.5*LR
Combination	217	1.2*D+E_(+)X+0.5*L
Combination	218	1.2*D+E_(-)X+0.5*L
Combination	219	1.2*D+E_(+)Z+0.5*L
Combination	220	1.2*D+E_(-)Z+0.5*L
Combination	221	0.9*D+1.6*W_L1
Combination	222	0.9*D+1.6*W_L2
Combination	223	0.9*D+1.6*W_R1
Combination	224	0.9*D+1.6*W_R2
Combination	225	0.9*D+1.6*W_LG1
Combination	226	0.9*D+1.6*W_LG2
Combination	227	0.9*D+E_(+)X

Job Information Cont...

Type	L/C	Name
Combination	228	0.9*D+E_(-)X
Combination	229	0.9*D+E_(+)Z
Combination	230	0.9*D+E_(-)Z

Nodes

Node	X (m)	Y (m)	Z (m)
1	-1.221	910.400	1.452
2	0	910.400	-0.000
3	0	910.400	4.300
4	4.600	910.400	-0.000
5	4.600	910.400	4.300
6	6.235	910.400	-0.000
7	6.235	910.400	4.300
8	9.993	910.400	-0.003
9	9.993	910.400	1.197
10	13.743	910.400	-0.003
11	13.743	910.400	1.197
12	9.993	911.950	-0.003
13	9.993	911.950	1.197
14	13.743	911.950	-0.003
15	13.743	911.950	1.197
16	-1.221	912.450	1.452
17	0	912.450	-0.000
18	0	912.450	4.300
19	4.600	912.450	-0.000
20	4.600	912.450	4.300
21	6.235	912.450	-0.000
22	6.235	912.450	4.300

Beams

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1	3	18	2.050	1	0
2	5	20	2.050	1	0
3	2	17	2.050	1	0
4	4	19	2.050	1	0
5	11	15	1.550	2	0
6	10	14	1.550	2	0
7	8	12	1.550	2	0
8	9	13	1.550	2	0
9	6	21	2.050	2	0
10	1	16	2.050	3	0
11	7	22	2.050	2	0

Section Properties

Prop	Section	Area (cm^2)	I_{yy} (cm^4)	I_{zz} (cm^4)	J (cm^4)	Material
1	Rect 0.60x0.50	3E+3	625E+3	900E+3	1.24E+6	CONCRETE
2	Rect 0.40x0.40	1.6E+3	213E+3	213E+3	360E+3	CONCRETE
3	Rect 0.40x0.60	2.4E+3	720E+3	320E+3	751E+3	CONCRETE

Materials

Mat	Name	E (kN/mm^2)	ν	Density (kg/m^3)	α (/°C)
1	CONCRETE	21.718	0.170	2.4E+3	10E -6
2	ALUMINUM	68.948	0.330	2.71E+3	23E -6
3	STEEL_50_KSI	199.948	0.300	7.83E+3	11.7E -6
4	STAINLESSSTEEL	197.930	0.300	7.83E+3	18E -6
5	STEEL_36_KSI	199.948	0.300	7.83E+3	11.7E -6
6	STEEL_275_NMM2	205.000	0.300	7.85E+3	12E -6
7	STEEL	205.000	0.300	7.83E+3	12E -6
8	STEEL_355_NMM2	205.000	0.300	7.85E+3	12E -6

Supports

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
1	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
2	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
3	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
4	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
5	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
6	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
7	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
8	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
9	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
10	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
11	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Releases

There is no data of this type.

Primary Load Cases

Number	Name	Type
1	DEAD LOAD (D)	None
2	ROOF LIVE LOAD (LR)	None
3	LIVE LOAD (L)	None
4	CRANE LOAD-1 (CR1)	None
5	CRANE LOAD-2 (CR2)	None
6	CRANE LOAD-3 (CR3)	None
7	CRANE LOAD-4 (CR4)	None
8	WIND LEFT-1 (W_L1)	None
9	WIND LEFT-2 (W_L2)	None
10	WIND RIGHT-1 (W_R1)	None
11	WIND RIGHT-2 (W_R2)	None
12	WIND LONG1 (W_LG1)	None
13	WIND LONG2 (W_LG2)	None
14	SEISMIC (+)X (E_(+)X)	None
15	SEISMIC (-)X (E_(-)X)	None
16	SEISMIC (+)Z (E_(+)Z)	None
17	SEISMIC (-)Z (E_(-)Z)	None
18	TEMPERATURE (+)25 (T_(+)25)	None
19	TEMPERATURE (-)25 (T_(-)25)	None

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
101	D+L+T_(+)25	1	DEAD LOAD (D)	1.00
		3	LIVE LOAD (L)	1.00
		18	TEMPERATURE (+)25 (T_(+)25)	1.00
102	D+L+T_(-)25	1	DEAD LOAD (D)	1.00
		3	LIVE LOAD (L)	1.00
		19	TEMPERATURE (-)25 (T_(-)25)	1.00
103	D+LR	1	DEAD LOAD (D)	1.00
		2	ROOF LIVE LOAD (LR)	1.00
104	D+0.75*(L+T_(+)25)+0.75*LR	1	DEAD LOAD (D)	1.00
		3	LIVE LOAD (L)	0.75
		18	TEMPERATURE (+)25 (T_(+)25)	0.75
		2	ROOF LIVE LOAD (LR)	0.75
105	D+0.75*(L+T_(-)25)+0.75*LR	1	DEAD LOAD (D)	1.00
		3	LIVE LOAD (L)	0.75
		19	TEMPERATURE (-)25 (T_(-)25)	0.75
		2	ROOF LIVE LOAD (LR)	0.75
106	D+W_L1	1	DEAD LOAD (D)	1.00
		8	WIND LEFT-1 (W_L1)	1.00
107	D+W_R1	1	DEAD LOAD (D)	1.00
		9	WIND LEFT-2 (W_L2)	1.00
108	D+W_L2	1	DEAD LOAD (D)	1.00
		10	WIND RIGHT-1 (W_R1)	1.00
109	D+W_R2	1	DEAD LOAD (D)	1.00
		11	WIND RIGHT-2 (W_R2)	1.00

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
110	D+W_LG1	1	DEAD LOAD (D)	1.00
		12	WIND LONG1 (W_LG1)	1.00
111	D+W_LG2	1	DEAD LOAD (D)	1.00
		13	WIND LONG2 (W_LG2)	1.00
112	D+0.7*E_(+)X	1	DEAD LOAD (D)	1.00
		14	SEISMIC (+)X (E_(+)X)	1.00
113	D+0.7*E_(-)X	1	DEAD LOAD (D)	1.00
		15	SEISMIC (-)X (E_(-)X)	1.00
114	D+0.7*E_(+)Z	1	DEAD LOAD (D)	1.00
		16	SEISMIC (+)Z (E_(+)Z)	1.00
115	D+0.7*E_(-)Z	1	DEAD LOAD (D)	1.00
		17	SEISMIC (-)Z (E_(-)Z)	1.00
116	D+0.75*W_L1+0.75*L+0.75*LR	1	DEAD LOAD (D)	1.00
		8	WIND LEFT-1 (W_L1)	0.75
		3	LIVE LOAD (L)	0.75
		2	ROOF LIVE LOAD (LR)	0.75
117	D+0.75*W_L2+0.75*L+0.75*LR	1	DEAD LOAD (D)	1.00
		9	WIND LEFT-2 (W_L2)	0.75
		3	LIVE LOAD (L)	0.75
		2	ROOF LIVE LOAD (LR)	0.75
118	D+0.75*W_R1+0.75*L+0.75*LR	1	DEAD LOAD (D)	1.00
		10	WIND RIGHT-1 (W_R1)	0.75
		3	LIVE LOAD (L)	0.75
		2	ROOF LIVE LOAD (LR)	0.75
119	D+0.75*W_R2+0.75*L+0.75*LR	1	DEAD LOAD (D)	1.00
		11	WIND RIGHT-2 (W_R2)	0.75
		3	LIVE LOAD (L)	0.75
		2	ROOF LIVE LOAD (LR)	0.75
120	D+0.75*W_LG1+0.75*L+0.75*LR	1	DEAD LOAD (D)	1.00
		12	WIND LONG1 (W_LG1)	0.75
		3	LIVE LOAD (L)	0.75
		2	ROOF LIVE LOAD (LR)	0.75
121	D+0.75*W_LG2+0.75*L+0.75*LR	1	DEAD LOAD (D)	1.00
		13	WIND LONG2 (W_LG2)	0.75
		3	LIVE LOAD (L)	0.75
		2	ROOF LIVE LOAD (LR)	0.75
122	D+0.75*(0.7*E_(+)X)+0.75*L+0.75*LR	1	DEAD LOAD (D)	1.00
		14	SEISMIC (+)X (E_(+)X)	0.52
		3	LIVE LOAD (L)	0.75
		2	ROOF LIVE LOAD (LR)	0.75
123	D+0.75*(0.7*E_(-)X)+0.75*L+0.75*LR	1	DEAD LOAD (D)	1.00
		15	SEISMIC (-)X (E_(-)X)	0.52
		3	LIVE LOAD (L)	0.75
		2	ROOF LIVE LOAD (LR)	0.75
124	D+0.75*(0.7*E_(+)Z)+0.75*L+0.75*LR	1	DEAD LOAD (D)	1.00
		16	SEISMIC (+)Z (E_(+)Z)	0.52
		3	LIVE LOAD (L)	0.75
		2	ROOF LIVE LOAD (LR)	0.75
125	D+0.75*(0.7*E_(-)Z)+0.75*L+0.75*LR	1	DEAD LOAD (D)	1.00

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		17	SEISMIC (-)Z (E_(-)Z)	0.52
		3	LIVE LOAD (L)	0.75
		2	ROOF LIVE LOAD (LR)	0.75
126	0.6*D+W_L1	1	DEAD LOAD (D)	0.60
		8	WIND LEFT-1 (W_L1)	1.00
127	0.6*D+W_L2	1	DEAD LOAD (D)	0.60
		9	WIND LEFT-2 (W_L2)	1.00
128	0.6*D+W_R1	1	DEAD LOAD (D)	0.60
		10	WIND RIGHT-1 (W_R1)	1.00
129	0.6*D+W_R2	1	DEAD LOAD (D)	0.60
		11	WIND RIGHT-2 (W_R2)	1.00
130	0.6*D+W_LG1	1	DEAD LOAD (D)	0.60
		12	WIND LONG1 (W_LG1)	1.00
131	0.6*D+W_LG2	1	DEAD LOAD (D)	0.60
		13	WIND LONG2 (W_LG2)	1.00
132	0.6*D+0.7*E_(+)X	1	DEAD LOAD (D)	0.60
		14	SEISMIC (+)X (E_(+)X)	0.70
133	0.6*D+0.7*E_(-)X	1	DEAD LOAD (D)	0.60
		15	SEISMIC (-)X (E_(-)X)	0.70
134	0.6*D+0.7*E_(+)Z	1	DEAD LOAD (D)	0.60
		16	SEISMIC (+)Z (E_(+)Z)	0.70
135	0.6*D+0.7*E_(-)Z	1	DEAD LOAD (D)	0.60
		17	SEISMIC (-)Z (E_(-)Z)	0.70
201	1.4*D	1	DEAD LOAD (D)	1.40
202	1.2*(D+T_(+)25)+1.6*L+0.5*LR	1	DEAD LOAD (D)	1.20
		18	TEMPERATURE (+)25 (T_(+)25)	1.20
		3	LIVE LOAD (L)	1.60
		2	ROOF LIVE LOAD (LR)	0.50
203	1.2*(D+T_(-)25)+1.6*L+0.5*LR	1	DEAD LOAD (D)	1.20
		19	TEMPERATURE (-)25 (T_(-)25)	1.20
		3	LIVE LOAD (L)	1.60
		2	ROOF LIVE LOAD (LR)	0.50
204	1.2*D+1.6*LR+0.5*L	1	DEAD LOAD (D)	1.20
		2	ROOF LIVE LOAD (LR)	1.60
		3	LIVE LOAD (L)	0.50
205	1.2*D+1.6*LR+0.8*W_L1	1	DEAD LOAD (D)	1.20
		2	ROOF LIVE LOAD (LR)	1.60
		8	WIND LEFT-1 (W_L1)	0.80
206	1.2*D+1.6*LR+0.8*W_L2	1	DEAD LOAD (D)	1.20
		2	ROOF LIVE LOAD (LR)	1.60
		9	WIND LEFT-2 (W_L2)	0.80
207	1.2*D+1.6*LR+0.8*W_R1	1	DEAD LOAD (D)	1.20
		2	ROOF LIVE LOAD (LR)	1.60
		10	WIND RIGHT-1 (W_R1)	0.80
208	1.2*D+1.6*LR+0.8*W_R2	1	DEAD LOAD (D)	1.20
		2	ROOF LIVE LOAD (LR)	1.60
		11	WIND RIGHT-2 (W_R2)	0.80
209	1.2*D+1.6*LR+0.8*W_LG1	1	DEAD LOAD (D)	1.20
		2	ROOF LIVE LOAD (LR)	1.60

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		12	WIND LONG1 (W_LG1)	0.80
210	1.2*D+1.6*LR+0.8*W_LG2	1	DEAD LOAD (D)	1.20
		2	ROOF LIVE LOAD (LR)	1.60
		13	WIND LONG2 (W_LG2)	0.80
211	1.2*D+1.6*W_L1+0.5*L+0.5*LR	1	DEAD LOAD (D)	1.20
		8	WIND LEFT-1 (W_L1)	1.60
		3	LIVE LOAD (L)	0.50
		2	ROOF LIVE LOAD (LR)	0.50
212	1.2*D+1.6*W_L2+0.5*L+0.5*LR	1	DEAD LOAD (D)	1.20
		9	WIND LEFT-2 (W_L2)	1.60
		3	LIVE LOAD (L)	0.50
		2	ROOF LIVE LOAD (LR)	0.50
213	1.2*D+1.6*W_R1+0.5*L+0.5*LR	1	DEAD LOAD (D)	1.20
		10	WIND RIGHT-1 (W_R1)	1.60
		3	LIVE LOAD (L)	0.50
		2	ROOF LIVE LOAD (LR)	0.50
214	1.2*D+1.6*W_R2+0.5*L+0.5*LR	1	DEAD LOAD (D)	1.20
		11	WIND RIGHT-2 (W_R2)	1.60
		3	LIVE LOAD (L)	0.50
		2	ROOF LIVE LOAD (LR)	0.50
215	1.2*D+1.6*W_LG1+0.5*L+0.5*LR	1	DEAD LOAD (D)	1.20
		12	WIND LONG1 (W_LG1)	1.60
		3	LIVE LOAD (L)	0.50
		2	ROOF LIVE LOAD (LR)	0.50
216	1.2*D+1.6*W_LG2+0.5*L+0.5*LR	1	DEAD LOAD (D)	1.20
		13	WIND LONG2 (W_LG2)	1.60
		3	LIVE LOAD (L)	0.50
		2	ROOF LIVE LOAD (LR)	0.50
217	1.2*D+E_(+)X+0.5*L	1	DEAD LOAD (D)	1.20
		14	SEISMIC (+)X (E_(+)X)	1.00
		3	LIVE LOAD (L)	0.50
218	1.2*D+E_(-)X+0.5*L	1	DEAD LOAD (D)	1.20
		15	SEISMIC (-)X (E_(-)X)	1.00
		3	LIVE LOAD (L)	0.50
219	1.2*D+E_(+)Z+0.5*L	1	DEAD LOAD (D)	1.20
		16	SEISMIC (+)Z (E_(+)Z)	1.00
		3	LIVE LOAD (L)	0.50
220	1.2*D+E_(-)Z+0.5*L	1	DEAD LOAD (D)	1.20
		17	SEISMIC (-)Z (E_(-)Z)	1.00
		3	LIVE LOAD (L)	0.50
221	0.9*D+1.6*W_L1	1	DEAD LOAD (D)	0.90
		8	WIND LEFT-1 (W_L1)	1.60
222	0.9*D+1.6*W_L2	1	DEAD LOAD (D)	0.90
		9	WIND LEFT-2 (W_L2)	1.60
223	0.9*D+1.6*W_R1	1	DEAD LOAD (D)	0.90
		10	WIND RIGHT-1 (W_R1)	1.60
224	0.9*D+1.6*W_R2	1	DEAD LOAD (D)	0.90
		11	WIND RIGHT-2 (W_R2)	1.60
225	0.9*D+1.6*W_LG1	1	DEAD LOAD (D)	0.90

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		12	WIND LONG1 (W_LG1)	1.60
226	0.9*D+1.6*W_LG2	1	DEAD LOAD (D)	0.90
		13	WIND LONG2 (W_LG2)	1.60
227	0.9*D+E_(+)X	1	DEAD LOAD (D)	0.90
		14	SEISMIC (+)X (E_(+)X)	1.00
228	0.9*D+E_(-)X	1	DEAD LOAD (D)	0.90
		15	SEISMIC (-)X (E_(-)X)	1.00
229	0.9*D+E_(+)Z	1	DEAD LOAD (D)	0.90
		16	SEISMIC (+)Z (E_(+)Z)	1.00
230	0.9*D+E_(-)Z	1	DEAD LOAD (D)	0.90
		17	SEISMIC (-)Z (E_(-)Z)	1.00

Load Generators

There is no data of this type.

1 DEAD LOAD (D) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	-0.539	-	-	-	-
13	-	-0.539	-	-	-	-
14	0.015	-0.234	-	-	-	-
15	0.015	-0.234	-	-	-	-
16	0.001	-1.002	0.005	-	-	-
17	0.009	-1.822	-	-	-	-
18	0.009	-1.822	-	-	-	-
19	0.009	-1.675	-0.001	-	-	-
20	0.009	-1.675	-0.001	-	-	-
21	-	-0.610	-	-	-	-
22	-	-0.610	-	-	-	-

1 DEAD LOAD (D) : Selfweight

Direction	Factor	Assigned Geometry
Y	-1.000	ALL

2 ROOF LIVE LOAD (LR) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	-	-	-	-	-
13	-	-	-	-	-	-
14	-	-	-	-	-	-
15	-	-	-	-	-	-
16	-	-0.013	-	-	-	-
17	0.018	-0.527	-	-	-	-
18	0.018	-0.527	-	-	-	-
19	-0.144	-0.381	0.001	-	-	-
20	-0.144	-0.381	0.001	-	-	-
21	-	-	-	-	-	-
22	-	-	-	-	-	-

3 LIVE LOAD (L) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	-0.672	-	-	-	-
13	-	-0.672	-	-	-	-
14	0.021	-0.339	-	-	-	-
15	0.021	-0.339	-	-	-	-
16	0.002	-3.006	0.021	-	-	-
17	-0.033	-1.556	-	-	-	-
18	-0.033	-1.556	-	-	-	-
19	0.206	0.365	-0.08	-	-	-
20	0.206	0.365	-0.08	-	-	-
21	-	-0.614	-	-	-	-
22	-	-0.614	-	-	-	-

4 CRANE LOAD-1 (CR1) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	0.007	-	-	-	-
13	-	0.007	-	-	-	-
14	0.023	-0.017	-	-	-	-
15	0.023	-0.017	-	-	-	-
16	-0.002	-0.005	-0.001	-	-	-
17	-0.082	-1.046	0.126	-	-	-
18	-0.082	-1.046	0.126	-	-	-
19	-	-0.718	0.002	-	-	-
20	-	-0.718	0.002	-	-	-
21	-	-0.415	-	-	-	-
22	-	-0.415	-	-	-	-

5 CRANE LOAD-2 (CR2) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	0.007	-	-	-	-
13	-	0.007	-	-	-	-
14	0.025	-0.018	-	-	-	-
15	0.025	-0.018	-	-	-	-
16	-0.002	0.002	-0.002	-	-	-
17	-0.046	-1.346	0.404	-	-	-
18	-0.046	-1.346	0.404	-	-	-
19	-0.057	-0.166	0.004	-	-	-
20	-0.057	-0.166	0.004	-	-	-
21	-	-0.456	-	-	-	-
22	-	-0.456	-	-	-	-

6 CRANE LOAD-3 (CR3) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	0.006	-	-	-	-
13	-	0.006	-	-	-	-
14	0.02	-0.014	-	-	-	-
15	0.02	-0.014	-	-	-	-
16	-0.002	-0.015	-	-	-	-
17	-0.115	-0.568	-	-	-	-
18	-0.115	-0.568	-	-	-	-
19	-0.201	-1.036	-0.251	-	-	-
20	-0.201	-1.036	-0.251	-	-	-
21	-	-0.307	-	-	-	-
22	-	-0.307	-	-	-	-

7 CRANE LOAD-4 (CR4) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	0.006	-	-	-	-
13	-	0.006	-	-	-	-
14	0.021	-0.015	-	-	-	-
15	0.021	-0.015	-	-	-	-
16	-0.002	-0.01	-0.001	-	-	-
17	-0.095	-0.799	-	-	-	-
18	-0.095	-0.799	-	-	-	-
19	-0.158	-0.780	-0.057	-	-	-
20	-0.158	-0.780	-0.057	-	-	-
21	-	-0.351	-	-	-	-
22	-	-0.351	-	-	-	-

8 WIND LEFT-1 (W L1) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	0.063	-	-	-	-
13	-	0.063	-	-	-	-
14	0.223	-0.158	-	-	-	-
15	0.223	-0.158	-	-	-	-
16	-0.015	0.06	-	-	-	-
17	-1.577	3.430	0.580	-	-	-
18	-1.577	3.430	0.580	-	-	-
19	-1.084	-2.319	0.023	-	-	-
20	-1.084	-2.319	0.023	-	-	-
21	0.004	-3.586	-	-	-	-
22	0.004	-3.586	-	-	-	-

9 WIND LEFT-2 (W L2) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	0.063	-	-	-	-
13	-	0.063	-	-	-	-
14	0.223	-0.158	-	-	-	-
15	0.223	-0.158	-	-	-	-
16	-0.015	0.054	-0.001	-	-	-
17	-1.655	3.071	0.579	-	-	-
18	-1.655	3.071	0.579	-	-	-
19	-1.018	-2.604	0.013	-	-	-
20	-1.018	-2.604	0.013	-	-	-
21	0.004	-3.611	-	-	-	-
22	0.004	-3.611	-	-	-	-

10 WIND RIGHT-1 (W R1) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	-0.061	-	-	-	-
13	-	-0.061	-	-	-	-
14	-0.217	0.154	-	-	-	-
15	-0.217	0.154	-	-	-	-
16	0.015	-0.059	0.002	-	-	-
17	0.505	-2.790	0.158	-	-	-
18	0.505	-2.790	0.158	-	-	-
19	1.525	2.805	-0.741	-	-	-
20	1.525	2.805	-0.741	-	-	-
21	-1.097	3.652	-	-	-	-
22	-1.097	3.652	-	-	-	-

11 WIND RIGHT-2 (W_R2) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	-0.062	-	-	-	-
13	-	-0.062	-	-	-	-
14	-0.220	0.156	-	-	-	-
15	-0.220	0.156	-	-	-	-
16	0.015	-0.067	0.002	-	-	-
17	0.441	-3.089	0.068	-	-	-
18	0.441	-3.089	0.068	-	-	-
19	1.575	2.439	-0.635	-	-	-
20	1.575	2.439	-0.635	-	-	-
21	-1.106	3.685	-	-	-	-
22	-1.106	3.685	-	-	-	-

12 WIND LONG1 (W_LG1) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	0.001	-	-	-	-
13	-	0.001	-	-	-	-
14	0.005	-0.003	-	-	-	-
15	0.005	-0.003	-	-	-	-
16	-	-0.068	0.01	-	-	-
17	-	5.896	-	-	-	-
	0.148	6.587	3.526	-	-	-
18	-	-5.896	2.478	-	-	-
	0.148	6.587	3.526	-	-	-
19	-	5.896	-	-	-	-
	-0.145	-5.987	0.043	-	-	-
20	-	-5.896	2.478	-	-	-
	-0.145	-5.987	0.043	-	-	-
21	-0.019	0.04	-	-	-	-
22	-0.019	0.04	-	-	-	-

13 WIND LONG2 (W LG2) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	0.001	-	-	-	-
13	-	0.001	-	-	-	-
14	0.005	-0.003	-	-	-	-
15	0.005	-0.003	-	-	-	-
16	-	-0.076	0.01	-	-	-
17	-	-5.896	-2.478	-	-	-
	0.083	6.256	3.526	-	-	-
18	-	5.896	-	-	-	-
	0.083	6.256	3.526	-	-	-
19	-	-5.896	-2.478	-	-	-
	-0.086	-6.311	0.044	-	-	-
20	-	5.896	-	-	-	-
	-0.086	-6.311	0.044	-	-	-
21	-0.018	0.042	-	-	-	-
22	-0.018	0.042	-	-	-	-

14 SEISMIC (+)X (E (+)X) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	0.009	-	-	-	-
13	-	0.009	-	-	-	-
14	0.032	-0.022	-	-	-	-
15	0.032	-0.022	-	-	-	-
16	-0.002	0.008	-	-	-	-
17	-0.230	0.408	0.081	-	-	-
18	-0.230	0.408	0.081	-	-	-
19	-0.144	-0.381	0.001	-	-	-
20	-0.144	-0.381	0.001	-	-	-
21	0.001	-0.510	-	-	-	-
22	0.001	-0.510	-	-	-	-

15 SEISMIC (-)X (E (-)X) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	-0.009	-	-	-	-
13	-	-0.009	-	-	-	-
14	-0.031	0.022	-	-	-	-
15	-0.031	0.022	-	-	-	-
16	0.002	-0.01	-	-	-	-
17	0.079	-0.405	-	-	-	-
18	0.079	-0.405	-	-	-	-
19	0.206	0.365	-0.08	-	-	-
20	0.206	0.365	-0.08	-	-	-
21	-0.155	0.517	-	-	-	-
22	-0.155	0.517	-	-	-	-

16 SEISMIC (+)Z (E (+)Z) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	-	-	-	-	-
13	-	-	-	-	-	-
14	-	-	-	-	-	-
15	-	-	-	-	-	-
16	-	-0.002	-0.001	-	-	-
17	-	0.370	-	-	-	-
	-	-0.722	-	-	-	-
18	-	-0.370	0.204	-	-	-
	-	-0.722	-	-	-	-
19	-	0.370	-	-	-	-
	0.001	0.713	-0.408	-	-	-
20	-	-0.370	0.204	-	-	-
	0.001	0.713	-0.408	-	-	-
21	-	-	-	-	-	-
22	-	-	-	-	-	-

17 SEISMIC (-)Z (E (-)Z) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	-	-	-	-	-
13	-	-	-	-	-	-
14	-	-	-	-	-	-
15	-	-	-	-	-	-
16	-	-0.009	0.001	-	-	-
17	-	-0.370	-0.204	-	-	-
	-0.01	0.718	0.406	-	-	-
18	-	0.370	-	-	-	-
	-0.01	0.718	0.406	-	-	-
19	-	-0.370	-0.204	-	-	-
	-	-0.718	0.002	-	-	-
20	-	0.370	-	-	-	-
	-	-0.718	0.002	-	-	-
21	-0.002	0.005	-	-	-	-
22	-0.002	0.005	-	-	-	-

18 TEMPERATURE (+)25 (T (+)25) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	0.01	-	-	-	-
13	-	0.01	-	-	-	-
14	0.036	-0.025	-	-	-	-
15	0.036	-0.025	-	-	-	-
16	0.001	-0.708	-0.007	-	-	-
17	0.024	0.597	0.081	-	-	-
18	0.024	0.597	0.081	-	-	-
19	-0.256	-0.854	0.046	-	-	-
20	-0.256	-0.854	0.046	-	-	-
21	-0.012	0.046	-	-	-	-
22	-0.012	0.046	-	-	-	-

19 TEMPERATURE (-)25 (T (-)25) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
12	-	-0.008	-	-	-	-
13	-	-0.008	-	-	-	-
14	-0.029	0.02	-	-	-	-
15	-0.029	0.02	-	-	-	-
16	-0.002	0.658	0.001	-	-	-
17	-0.099	-0.746	4.031	-	-	-
18	-0.099	-0.746	4.031	-	-	-
19	0.593	1.416	-4.230	-	-	-
20	0.593	1.416	-4.230	-	-	-
21	-	-0.208	-	-	-	-
22	-	-0.208	-	-	-	-

ATTACHMENT - 5 : STAAD OUTPUT

SMBS & PEROXIDE SHED FOUNDATION

PROPRIETARY & CONFIDENTIAL

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PAGE NO. 1

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*************************************************
*                                               *
*           STAAD.Pro CONNECT Edition           *
*           Version  22.00.00.15                *
*           Proprietary Program of              *
*           Bentley Systems, Inc.               *
*           Date=    JUN 17, 2020               *
*           Time=    14:43:20                   *
*                                               *
*  Licensed to: Larsen & Toubro Ltd.            *
*************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\MAADEN-GOLD-O19045\3890 REAGENTS SMBS & PEROXIDE\CIVIL\LATEST\SMBS_PEROXIDE FOUNDATION_.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 12-JUN-20
    4. END JOB INFORMATION
    5. INPUT WIDTH 79
    6. UNIT METER KN
    7. JOINT COORDINATES
    8. 1 0 910.4 0; 2 6.25 910.4 0; 3 12.5 910.4 0; 4 18.75 910.4 0; 5 25 910.4 0
    9. 6 31.25 910.4 0; 7 38.85 910.4 0; 8 45.1 910.4 0; 9 0 910.4 5
   10. 10 38.85 910.4 5; 11 45.1 910.4 5; 12 0 910.4 9.5; 13 38.85 910.4 9.5
   11. 14 45.1 910.4 9.5; 15 0 910.4 14.29; 16 38.85 910.4 14.29; 17 45.1 910.4 14.29
   12. 18 0 910.4 19; 19 6.25 910.4 19; 20 12.5 910.4 19; 21 18.75 910.4 19
   13. 22 25 910.4 19; 23 31.25 910.4 19; 24 38.85 910.4 19; 25 45.1 910.4 19
   14. 26 0 912.75 0; 27 6.25 912.75 0; 28 12.5 912.75 0; 29 18.75 912.75 0
   15. 30 25 912.75 0; 31 31.25 912.75 0; 32 38.85 912.75 0; 33 45.1 912.75 0
   16. 34 0 912.75 5; 35 38.85 912.75 5; 36 45.1 912.75 5; 37 0 912.75 9.5
   17. 38 38.85 912.75 9.5; 39 45.1 912.75 9.5; 40 0 912.75 14.29
   18. 41 38.85 912.75 14.29; 42 45.1 912.75 14.29; 43 0 912.75 19; 44 6.25 912.75 19
   19. 45 12.5 912.75 19; 46 18.75 912.75 19; 47 25 912.75 19; 48 31.25 912.75 19
   20. 49 38.85 912.75 19; 50 45.1 912.75 19
   21. MEMBER INCIDENCES
   22. 1 1 26; 2 2 27; 3 3 28; 4 4 29; 5 5 30; 6 6 31; 7 7 32; 8 8 33; 9 12 37
   23. 10 15 40; 11 18 43; 12 9 34; 13 19 44; 14 20 45; 15 21 46; 16 22 47; 17 23 48
   24. 18 24 49; 19 25 50; 20 14 39; 21 17 42; 22 11 36; 23 13 38; 24 16 41; 25 10 35
   25. START GROUP DEFINITION
   26. MEMBER
   27. _C1 1 8 11 19
   28. _C2 9 10 12 20 TO 22
   29. _C3 2 TO 7 13 TO 18
   30. _C4 23 TO 25
   31. JOINT
   32. _F1 1 8 18 25
   33. _F2 9 11 12 14 15 17
   34. _F3 2 TO 7 19 TO 24
   35. _F4 10 13 16
   36. END GROUP DEFINITION
   37. DEFINE MATERIAL START
   38. ISOTROPIC CONCRETE
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```
39. E 2.17185E+07
40. POISSON 0.17
41. DENSITY 23.5616
42. ALPHA 1E-05
43. DAMP 0.05
44. TYPE CONCRETE
45. STRENGTH FCU 27579
46. END DEFINE MATERIAL
47. MEMBER PROPERTY INDIAN
48. 1 TO 8 11 13 TO 19 PRIS YD 0.4 ZD 0.5
49. 9 10 12 20 TO 25 PRIS YD 0.4 ZD 0.4
50. CONSTANTS
51. MATERIAL CONCRETE ALL
52. SUPPORTS
53. 1 TO 25 FIXED
54. LOAD 1 DEAD LOAD (D)
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
          MASTER/SLAVE OR IF UNCONNECTED JOINTS.
55. SELFWEIGHT Y -1
56. JOINT LOAD
57. 43 50 FX 0 FY -0.275 FZ 0.02
58. 40 42 FX 0 FY -0.357 FZ 0
59. 37 39 FX 0 FY -0.357 FZ 0
60. 34 36 FX 0 FY -0.357 FZ 0
61. 26 33 FX 0 FY -0.286 FZ 0.02
62. 44 TO 49 FX 0 FY -1.193 FZ -0.581
63. 27 TO 32 FX 0 FY -1.193 FZ -0.581
64. LOAD 2 COLLATERAL (C)
65. JOINT LOAD
66. 43 50 FX 0 FY -0.54 FZ -0.01
67. 40 42 FX 0 FY 0.082 FZ 0
68. 37 39 FX 0 FY -0.143 FZ 0.031
69. 34 36 FX 0 FY -0.286 FZ -0.02
70. 26 33 FX 0 FY -0.367 FZ -0.031
71. 44 TO 49 FX 0 FY -0.449 FZ 0.031
72. 27 TO 32 FX 0 FY -0.02 FZ -0.031
73. LOAD 3 ROOF LIVE (LR)
74. JOINT LOAD
75. 43 50 FX 0 FY -0.54 FZ 0.071
76. 40 42 FX 0 FY -0.958 FZ 0
77. 37 39 FX 0 FY -0.958 FZ 0
78. 34 36 FX 0 FY -0.958 FZ 0
79. 26 33 FX 0 FY -0.551 FZ 0.071
80. 44 TO 49 FX 0 FY -4.007 FZ -2.264
81. 27 TO 32 FX 0 FY -4.007 FZ -2.264
82. LOAD 4 WIND LEFT-1 (W_L1)
83. JOINT LOAD
84. 43 50 FX 0 FY 0.255 FZ 1.203
85. 40 42 FX 1.662 FY 2.019 FZ 0
86. 37 39 FX 1.703 FY 2.131 FZ 0
87. 34 36 FX 1.662 FY 2.417 FZ 0
88. 26 33 FX 0 FY 2.406 FZ -1.346
89. 44 TO 49 FX 0 FY 7.994 FZ 5.211
90. 27 TO 32 FX 0 FY -5.547 FZ 1.723
91. 35 38 41 FZ -0.377
92. LOAD 5 WIND RIGHT-1 (W_R1)
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 93. JOINT LOAD
 94. 43 50 FX 0 FY 2.386 FZ -1.336
 95. 40 42 FX 1.662 FY 2.427 FZ 0
 96. 37 39 FX 1.703 FY 2.131 FZ 0
 97. 34 36 FX 1.662 FY 2.029 FZ 0
 98. 26 33 FX 0 FY 0.245 FZ 1.213
 99. 44 TO 49 FX 0 FY 5.547 FZ 1.713
100. 27 TO 32 FX 0 FY 7.994 FZ 5.211
101. 35 38 41 FZ 0.377
102. LOAD 6 WIND LEFT-2 (W_L2)
103. JOINT LOAD
104. 43 50 FX 0 FY -0.377 FZ 0.897
105. 40 42 FX -1.509 FY 1.315 FZ 0
106. 37 39 FX -1.54 FY 1.407 FZ 0
107. 34 36 FX -1.509 FY 1.723 FZ 0
108. 26 33 FX 0 FY -1.774 FZ -1.652
109. 44 TO 49 FX 0 FY 4.578 FZ 4.242
110. 27 TO 32 FX 0 FY 2.131 FZ 0.755
111. LOAD 7 WIND RIGHT-2 (W_R2)
112. JOINT LOAD
113. 43 50 FX 0 FY 1.764 FZ -1.642
114. 40 42 FX -1.509 FY 1.723 FZ 0
115. 37 39 FX -1.54 FY 1.407 FZ 0
116. 34 36 FX -1.509 FY 1.326 FZ 0
117. 26 33 FX 0 FY -0.387 FZ 0.908
118. 44 TO 49 FX 0 FY 2.131 FZ 0.744
119. 27 TO 32 FX 0 FY 4.578 FZ 4.232
120. LOAD 8 WIND LONG-1 (W_LG1)
121. JOINT LOAD
122. 43 50 FX 0 FY 2.386 FZ 0.449
123. 40 42 FX 1.662 FY 1.968 FZ 0
124. 37 39 FX 1.703 FY 1.438 FZ 0
125. 34 36 FX 1.662 FY 0.846 FZ 0
126. 26 33 FX 0 FY 2.508 FZ 0.602
127. 44 TO 49 FX 0 FY 8.657 FZ 1.968
128. 27 TO 32 FX 0 FY 7.281 FZ 2.274
129. 26 43 FY 1.386
130. 27 44 FX 1.56 FY -1.386
131. LOAD 9 WIND LONG-2 (W_LG2)
132. JOINT LOAD
133. 43 50 FX 0 FY 2.508 FZ 0.602
134. 40 42 FX -1.509 FY 0.846 FZ 0
135. 37 39 FX -1.642 FY 1.438 FZ 0
136. 34 36 FX -1.509 FY 1.968 FZ 0
137. 26 33 FX 0 FY 2.386 FZ 0.449
138. 44 TO 49 FX 0 FY 7.281 FZ 2.284
139. 27 TO 32 FX 0 FY 8.657 FZ 1.978
140. 33 50 FY 1.386
141. 32 49 FX -1.56 FY -1.386
142. LOAD 10 SEISMIC_L (E_L)
143. JOINT LOAD
144. 43 50 FX 0 FY 0.041 FZ 0.031
145. 40 42 FX 0 FY -0.041 FZ 0
146. 37 39 FX 0 FY 0.041 FZ 0
147. 34 36 FX 0 FY 0.041 FZ 0
148. 26 33 FX 0 FY -0.041 FZ -0.031
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149. 44 TO 49 FX 0 FY 0.02 FZ 0.041
150. 27 TO 32 FX 0 FY -0.02 FZ -0.041
151. 26 43 FY 0.162
152. 27 44 FX 0.183 FY -0.162
153. LOAD 11 SEISMIC_R (E_R)
154. JOINT LOAD
155. 43 50 FX 0 FY -0.041 FZ -0.031
156. 40 42 FX 0 FY 0.041 FZ 0
157. 37 39 FX 0 FY 0 FZ 0
158. 34 36 FX 0 FY -0.041 FZ 0
159. 26 33 FX 0 FY 0.041 FZ 0.031
160. 44 TO 49 FX 0 FY -0.02 FZ -0.041
161. 27 TO 32 FX 0 FY 0.02 FZ 0.041
162. 33 50 FY 0.162
163. 32 49 FX 0.183 FY -0.162
164. LOAD 12 TEMPERATURE (+)25 (T_(+)25)
165. JOINT LOAD
166. 43 50 FX 0 FY 0.153 FZ -0.102
167. 40 42 FX 0 FY 0.173 FZ 0
168. 37 39 FX 0 FY 0.051 FZ 0
169. 34 36 FX 0 FY 0.173 FZ 0
170. 26 33 FX 0 FY 0.153 FZ 0.102
171. 44 TO 49 FX 0 FY 0 FZ 0.041
172. 27 TO 32 FX 0 FY 0 FZ -0.041
173. LOAD 13 TEMPERATURE (-)25 (T_(-)25)
174. JOINT LOAD
175. 43 50 FX 0 FY -0.153 FZ 0.102
176. 40 42 FX 0 FY -0.173 FZ 0
177. 37 39 FX 0 FY -0.051 FZ 0
178. 34 36 FX 0 FY -0.173 FZ 0
179. 26 33 FX 0 FY -0.153 FZ 0.102
180. 44 TO 49 FX 0 FY 0 FZ -0.041
181. 27 TO 32 FX 0 FY 0 FZ 0.041
182. ***LOAD COMBINATION FOR STRESS CALCULATION***
183. LOAD COMB 101 D+C+T_(+)25
184. 1 1.0 2 1.0 12 1.0
185. LOAD COMB 102 D+C+T_(-)25
186. 1 1.0 2 1.0 13 1.0
187. ***
188. LOAD COMB 103 D+C+LR
189. 1 1.0 2 1.0 3 1.0
190. ***
191. LOAD COMB 104 D+C+0.75*T_(+)25+0.75*LR
192. 1 1.0 2 1.0 12 0.75 3 0.75
193. LOAD COMB 105 D+C+0.75*T_(-)25+0.75*LR
194. 1 1.0 2 1.0 13 0.75 3 0.75
195. ***
196. LOAD COMB 106 D+C+W_L1
197. 1 1.0 2 1.0 4 1.0
198. LOAD COMB 107 D+C+W_R1
199. 1 1.0 2 1.0 5 1.0
200. LOAD COMB 108 D+C+W_L2
201. 1 1.0 2 1.0 6 1.0
202. LOAD COMB 109 D+C+W_R2
203. 1 1.0 2 1.0 7 1.0
204. LOAD COMB 110 D+C+W_LG1
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205. 1 1.0 2 1.0 8 1.0
206. LOAD COMB 111 D+C+W_LG2
207. 1 1.0 2 1.0 9 1.0
208. ***
209. LOAD COMB 112 D+C+0.7*E_L
210. 1 1.0 2 1.0 10 1.0
211. LOAD COMB 113 D+C+0.7*E_R
212. 1 1.0 2 1.0 11 1.0
213. ***
214. LOAD COMB 114 D+C+0.75*W_L1+0.75*LR
215. 1 1.0 2 1.0 4 0.75 3 0.75
216. LOAD COMB 115 D+C+0.75*W_R1+0.75*LR
217. 1 1.0 2 1.0 5 0.75 3 0.75
218. LOAD COMB 116 D+C+0.75*W_L2+0.75*LR
219. 1 1.0 2 1.0 6 0.75 3 0.75
220. LOAD COMB 117 D+C+0.75*W_R2+0.75*LR
221. 1 1.0 2 1.0 7 0.75 3 0.75
222. LOAD COMB 118 D+C+0.75*W_LG1+0.75*LR
223. 1 1.0 2 1.0 8 0.75 3 0.75
224. LOAD COMB 119 D+C+0.75*W_LG2+0.75*LR
225. 1 1.0 2 1.0 9 0.75 3 0.75
226. ***
227. LOAD COMB 120 D+C+0.75*(0.7*E_L)+0.75*LR
228. 1 1.0 2 1.0 10 0.525 3 0.75
229. LOAD COMB 121 D+C+0.75*(0.7*E_R)+0.75*LR
230. 1 1.0 2 1.0 11 0.525 3 0.75
231. ***
232. LOAD COMB 122 0.6*(D+C)+W_L1
233. 1 0.6 2 0.6 4 1.0
234. LOAD COMB 123 0.6*(D+C)+W_R1
235. 1 0.6 2 0.6 5 1.0
236. LOAD COMB 124 0.6*(D+C)+W_L2
237. 1 0.6 2 0.6 6 1.0
238. LOAD COMB 125 0.6*(D+C)+W_R2
239. 1 0.6 2 0.6 7 1.0
240. LOAD COMB 126 0.6*(D+C)+W_LG1
241. 1 0.6 2 0.6 8 1.0
242. LOAD COMB 127 0.6*(D+C)+W_LG2
243. 1 0.6 2 0.6 9 1.0
244. ***
245. LOAD COMB 128 0.6*(D+C)+0.7*E_L
246. 1 0.6 2 0.6 10 0.7
247. LOAD COMB 129 0.6*(D+C)+0.7*E_R
248. 1 0.6 2 0.6 11 0.7
249. ***LOAD COMBINATION FOR STRENGTH CALCULATION***
250. LOAD COMB 201 1.4*(D+C)
251. 1 1.4 2 1.4
252. ***
253. LOAD COMB 202 1.4*(D+C+T_(+)25)+0.5*LR
254. 1 1.4 2 1.4 12 1.4 3 0.5
255. LOAD COMB 203 1.4*(D+C+T_(-)25)+0.5*LR
256. 1 1.4 2 1.4 13 1.4 3 0.5
257. ***
258. LOAD COMB 204 1.2*(D+C)+1.6*LR+0.8*W_L1
259. 1 1.2 2 1.2 3 1.6 4 0.8
260. LOAD COMB 205 1.2*(D+C)+1.6*LR+0.8*W_R1
```

```
261. 1 1.2 2 1.2 3 1.6 5 0.8
262. LOAD COMB 206 1.2*(D+C)+1.6*LR+0.8*W_L2
263. 1 1.2 2 1.2 3 1.6 6 0.8
264. LOAD COMB 207 1.2*(D+C)+1.6*LR+0.8*W_R2
265. 1 1.2 2 1.2 3 1.6 7 0.8
266. LOAD COMB 208 1.2*(D+C)+1.6*LR+0.8*W_LG1
267. 1 1.2 2 1.2 3 1.6 8 0.8
268. LOAD COMB 209 1.2*(D+C)+1.6*LR+0.8*W_LG2
269. 1 1.2 2 1.2 3 1.6 9 0.8
270. ***
271. LOAD COMB 210 1.2*(D+C)+1.6*W_L1+0.5*LR
272. 1 1.2 2 1.2 4 1.6 3 0.5
273. LOAD COMB 211 1.2*(D+C)+1.6*W_R1+0.5*LR
274. 1 1.2 2 1.2 5 1.6 3 0.5
275. LOAD COMB 212 1.2*(D+C)+1.6*W_L2+0.5*LR
276. 1 1.2 2 1.2 6 1.6 3 0.5
277. LOAD COMB 213 1.2*(D+C)+1.6*W_R2+0.5*LR
278. 1 1.2 2 1.2 7 1.6 3 0.5
279. LOAD COMB 214 1.2*(D+C)+1.6*W_LG1+0.5*LR
280. 1 1.2 2 1.2 8 1.6 3 0.5
281. LOAD COMB 215 1.2*(D+C)+1.6*W_LG2+0.5*LR
282. 1 1.2 2 1.2 9 1.6 3 0.5
283. ***
284. LOAD COMB 216 1.2*(D+C)+E_L
285. 1 1.2 2 1.2 10 1.0
286. LOAD COMB 217 1.2*(D+C)+E_R
287. 1 1.2 2 1.2 11 1.0
288. ***
289. LOAD COMB 218 0.9*(D+C)+1.6*W_L1
290. 1 0.9 2 0.9 4 1.6
291. LOAD COMB 219 0.9*(D+C)+1.6*W_R1
292. 1 0.9 2 0.9 5 1.6
293. LOAD COMB 220 0.9*(D+C)+1.6*W_L2
294. 1 0.9 2 0.9 6 1.6
295. LOAD COMB 221 0.9*(D+C)+1.6*W_R2
296. 1 0.9 2 0.9 7 1.6
297. LOAD COMB 222 0.9*(D+C)+1.6*W_LG1
298. 1 0.9 2 0.9 8 1.6
299. LOAD COMB 223 0.9*(D+C)+1.6*W_LG2
300. 1 0.9 2 0.9 9 1.6
301. ***
302. LOAD COMB 224 0.9*(D+C)+E_L
303. 1 0.9 2 0.9 10 1.0
304. LOAD COMB 225 0.9*(D+C)+E_R
305. 1 0.9 2 0.9 11 1.0
306. ***
307. PERFORM ANALYSIS
```

STAAD SPACE -- PAGE NO. 7

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 50 NUMBER OF MEMBERS 25
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 25

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 150
TOTAL LOAD COMBINATION CASES = 54 SO FAR.

308. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUN 17,2020 TIME= 14:43:23 ****

STAAD SPACE -- PAGE NO. 8

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://www.bentley.com/en/support/                      *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2017 Bentley Systems, Inc.        *
*                http://www.bentley.com                     *
*************************************************************
```

ATTACHMENT - 6 : STAAD OUTPUT

SMBS BAG SPLITTER FOUNDATION

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PAGE NO. 1

```
*************************************************
*                                               *
*           STAAD.Pro CONNECT Edition           *
*           Version  22.00.00.15                *
*           Proprietary Program of              *
*           Bentley Systems, Inc.               *
*           Date=    JUN 15, 2020               *
*           Time=    23:58:11                   *
*                                               *
*  Licensed to: Larsen & Toubro Ltd.            *
*************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\MAADEN-GOLD-O19045\3890 REAGENTS SMBS & PEROXIDE\CIVIL\LATEST\SMBS BAG SPLITTER FOUNDAT.. .STD
    2. START JOB INFORMATION
    3. ENGINEER NAME SUE
    4. ENGINEER DATE 6/24/2020
    5. CHECKER NAME BDA
    6. APPROVED NAME AG
    7. JOB NAME MMGP
    8. JOB NO O19045
    9. JOB CLIENT MAADEN
   10. CHECKER DATE 6/25/2020
   11. APPROVED DATE 6/26/2020
   12. END JOB INFORMATION
   13. INPUT WIDTH 79
   14. UNIT METER MTON
   15. JOINT COORDINATES
   16. 1 -1.221 910.4 1.452; 2 0 910.4 -1.00136E-05; 3 0 910.4 4.3
   17. 4 4.6001 910.4 -1.00136E-05; 5 4.6001 910.4 4.3; 6 6.23501 910.4 -1.00136E-05
   18. 7 6.23501 910.4 4.3; 8 9.99302 910.4 -0.00300026; 9 9.99302 910.4 1.197
   19. 10 13.7431 910.4 -0.00300026; 11 13.7431 910.4 1.197
   20. 12 9.99302 911.95 -0.00300026; 13 9.99302 911.95 1.197
   21. 14 13.7431 911.95 -0.00300026; 15 13.7431 911.95 1.197; 16 -1.221 912.45 1.452
   22. 17 0 912.45 -1.00136E-05; 18 0 912.45 4.3; 19 4.6001 912.45 -1.00136E-05
   23. 20 4.6001 912.45 4.3; 21 6.23501 912.45 -1.00136E-05; 22 6.23501 912.45 4.3
   24. MEMBER INCIDENCES
   25. 1 3 18; 2 5 20; 3 2 17; 4 4 19; 5 11 15; 6 10 14; 7 8 12; 8 9 13; 9 6 21
   26. 10 1 16; 11 7 22
   27. START GROUP DEFINITION
   28. MEMBER
   29. _C5 1 TO 4
   30. _C6 10
   31. _C7_C7A 7 TO 9 11
   32. _C8 5 6
   33. JOINT
   34. _F5 2 3
   35. _F6 4 5
   36. _F7 6 7
   37. _F8 8 9
   38. _F9 10 11
```

STAAD SPACE -- PAGE NO. 2

```
 39. END GROUP DEFINITION
 40. DEFINE MATERIAL START
 41. ISOTROPIC CONCRETE
 42. E 2.21467E+06
 43. POISSON 0.17
 44. DENSITY 2.40261
 45. ALPHA 1E-05
 46. DAMP 0.05
 47. TYPE CONCRETE
 48. STRENGTH FCU 2812.28
 49. END DEFINE MATERIAL
 50. MEMBER PROPERTY INDIAN
 51. 1 TO 4 PRIS YD 0.6 ZD 0.5
 52. 5 TO 9 11 PRIS YD 0.4 ZD 0.4
 53. 10 PRIS YD 0.4 ZD 0.6
 54. CONSTANTS
 55. MATERIAL CONCRETE ALL
 56. SUPPORTS
 57. 1 TO 11 FIXED
 58. LOAD 1 DEAD LOAD (D)
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
           MASTER/SLAVE OR IF UNCONNECTED JOINTS.
 59. SELFWEIGHT Y -1
 60. JOINT LOAD
 61. 17 18 FX 0.009 FY -1.822 FZ 0
 62. 19 20 FX 0.009 FY -1.675 FZ -0.001
 63. 16 FX 0.001 FY -1.002 FZ 0.005
 64. 21 22 FX 0 FY -0.61 FZ 0
 65. 12 13 FX 0 FY -0.539 FZ 0
 66. 14 15 FX 0.015 FY -0.234 FZ 0
 67. LOAD 2 ROOF LIVE LOAD (LR)
 68. JOINT LOAD
 69. 17 18 FX 0.018 FY -0.527 FZ 0
 70. 19 20 FX -0.144 FY -0.381 FZ 0.001
 71. 16 FX 0 FY -0.013 FZ 0
 72. 21 22 FX 0 FY 0 FZ 0
 73. 12 13 FX 0 FY 0 FZ 0
 74. 14 15 FX 0 FY 0 FZ 0
 75. LOAD 3 LIVE LOAD (L)
 76. JOINT LOAD
 77. 17 18 FX -0.033 FY -1.556 FZ 0
 78. 19 20 FX 0.206 FY 0.365 FZ -0.08
 79. 16 FX 0.002 FY -3.006 FZ 0.021
 80. 21 22 FX 0 FY -0.614 FZ 0
 81. 12 13 FX 0 FY -0.672 FZ 0
 82. 14 15 FX 0.021 FY -0.339 FZ 0
 83. LOAD 4 CRANE LOAD-1 (CR1)
 84. JOINT LOAD
 85. 17 18 FX -0.082 FY -1.046 FZ 0.126
 86. 19 20 FX 0 FY -0.718 FZ 0.002
 87. 16 FX -0.002 FY -0.005 FZ -0.001
 88. 21 22 FX 0 FY -0.415 FZ 0
 89. 12 13 FX 0 FY 0.007 FZ 0
 90. 14 15 FX 0.023 FY -0.017 FZ 0
 91. LOAD 5 CRANE LOAD-2 (CR2)
 92. JOINT LOAD
```

STAAD SPACE -- PAGE NO. 3

```
 93. 17 18 FX -0.046 FY -1.346 FZ 0.404
 94. 19 20 FX -0.057 FY -0.166 FZ 0.004
 95. 16 FX -0.002 FY 0.002 FZ -0.002
 96. 21 22 FX 0 FY -0.456 FZ 0
 97. 12 13 FX 0 FY 0.007 FZ 0
 98. 14 15 FX 0.025 FY -0.018 FZ 0
 99. LOAD 6 CRANE LOAD-3 (CR3)
100. JOINT LOAD
101. 17 18 FX -0.115 FY -0.568 FZ 0
102. 19 20 FX -0.201 FY -1.036 FZ -0.251
103. 16 FX -0.002 FY -0.015 FZ 0
104. 21 22 FX 0 FY -0.307 FZ 0
105. 12 13 FX 0 FY 0.006 FZ 0
106. 14 15 FX 0.02 FY -0.014 FZ 0
107. LOAD 7 CRANE LOAD-4 (CR4)
108. JOINT LOAD
109. 17 18 FX -0.095 FY -0.799 FZ 0
110. 19 20 FX -0.158 FY -0.78 FZ -0.057
111. 16 FX -0.002 FY -0.01 FZ -0.001
112. 21 22 FX 0 FY -0.351 FZ 0
113. 12 13 FX 0 FY 0.006 FZ 0
114. 14 15 FX 0.021 FY -0.015 FZ 0
115. LOAD 8 WIND LEFT-1 (W_L1)
116. JOINT LOAD
117. 17 18 FX -1.577 FY 3.43 FZ 0.58
118. 19 20 FX -1.084 FY -2.319 FZ 0.023
119. 16 FX -0.015 FY 0.06 FZ 0
120. 21 22 FX 0.004 FY -3.586 FZ 0
121. 12 13 FX 0 FY 0.063 FZ 0
122. 14 15 FX 0.223 FY -0.158 FZ 0
123. LOAD 9 WIND LEFT-2 (W_L2)
124. JOINT LOAD
125. 17 18 FX -1.655 FY 3.071 FZ 0.579
126. 19 20 FX -1.018 FY -2.604 FZ 0.013
127. 16 FX -0.015 FY 0.054 FZ -0.001
128. 21 22 FX 0.004 FY -3.611 FZ 0
129. 12 13 FX 0 FY 0.063 FZ 0
130. 14 15 FX 0.223 FY -0.158 FZ 0
131. LOAD 10 WIND RIGHT-1 (W_R1)
132. JOINT LOAD
133. 17 18 FX 0.505 FY -2.79 FZ 0.158
134. 19 20 FX 1.525 FY 2.805 FZ -0.741
135. 16 FX 0.015 FY -0.059 FZ 0.002
136. 21 22 FX -1.097 FY 3.652 FZ 0
137. 12 13 FX 0 FY -0.061 FZ 0
138. 14 15 FX -0.217 FY 0.154 FZ 0
139. LOAD 11 WIND RIGHT-2 (W_R2)
140. JOINT LOAD
141. 17 18 FX 0.441 FY -3.089 FZ 0.068
142. 19 20 FX 1.575 FY 2.439 FZ -0.635
143. 16 FX 0.015 FY -0.067 FZ 0.002
144. 21 22 FX -1.106 FY 3.685 FZ 0
145. 12 13 FX 0 FY -0.062 FZ 0
146. 14 15 FX -0.22 FY 0.156 FZ 0
147. LOAD 12 WIND LONG1 (W_LG1)
148. JOINT LOAD
```

STAAD SPACE -- PAGE NO. 4

```
149. 17 18 FX 0.148 FY 6.587 FZ 3.526
150. 19 20 FX -0.145 FY -5.987 FZ 0.043
151. 16 FX 0 FY -0.068 FZ 0.01
152. 21 22 FX -0.019 FY 0.04 FZ 0
153. 12 13 FX 0 FY 0.001 FZ 0
154. 14 15 FX 0.005 FY -0.003 FZ 0
155. 18 20 FY -5.896 FZ 2.478
156. 17 19 FY 5.896
157. LOAD 13 WIND LONG2 (W_LG2)
158. JOINT LOAD
159. 17 18 FX 0.083 FY 6.256 FZ 3.526
160. 19 20 FX -0.086 FY -6.311 FZ 0.044
161. 16 FX 0 FY -0.076 FZ 0.01
162. 21 22 FX -0.018 FY 0.042 FZ 0
163. 12 13 FX 0 FY 0.001 FZ 0
164. 14 15 FX 0.005 FY -0.003 FZ 0
165. 17 19 FY -5.896 FZ -2.478
166. 18 20 FY 5.896
167. LOAD 14 SEISMIC (+)X (E_(+)X)
168. JOINT LOAD
169. 17 18 FX -0.23 FY 0.408 FZ 0.081
170. 19 20 FX -0.144 FY -0.381 FZ 0.001
171. 16 FX -0.002 FY 0.008 FZ 0
172. 21 22 FX 0.001 FY -0.51 FZ 0
173. 12 13 FX 0 FY 0.009 FZ 0
174. 14 15 FX 0.032 FY -0.022 FZ 0
175. LOAD 15 SEISMIC (-)X (E_(-)X)
176. JOINT LOAD
177. 17 18 FX 0.079 FY -0.405 FZ 0
178. 19 20 FX 0.206 FY 0.365 FZ -0.08
179. 16 FX 0.002 FY -0.01 FZ 0
180. 21 22 FX -0.155 FY 0.517 FZ 0
181. 12 13 FX 0 FY -0.009 FZ 0
182. 14 15 FX -0.031 FY 0.022 FZ 0
183. LOAD 16 SEISMIC (+)Z (E_(+)Z)
184. JOINT LOAD
185. 17 18 FX 0 FY -0.722 FZ 0
186. 19 20 FX 0.001 FY 0.713 FZ -0.408
187. 16 FX 0 FY -0.002 FZ -0.001
188. 21 22 FX 0 FY 0 FZ 0
189. 12 13 FX 0 FY 0 FZ 0
190. 14 15 FX 0 FY 0 FZ 0
191. 18 20 FY -0.37 FZ 0.204
192. 17 19 FY 0.37
193. LOAD 17 SEISMIC (-)Z (E_(-)Z)
194. JOINT LOAD
195. 17 18 FX -0.01 FY 0.718 FZ 0.406
196. 19 20 FX 0 FY -0.718 FZ 0.002
197. 16 FX 0 FY -0.009 FZ 0.001
198. 21 22 FX -0.002 FY 0.005 FZ 0
199. 12 13 FX 0 FY 0 FZ 0
200. 14 15 FX 0 FY 0 FZ 0
201. 17 19 FY -0.37 FZ -0.204
202. 18 20 FY 0.37
203. LOAD 18 TEMPERATURE (+)25 (T_(+)25)
204. JOINT LOAD
```

STAAD SPACE -- PAGE NO. 5

```
205. 17 18 FX 0.024 FY 0.597 FZ 0.081
206. 19 20 FX -0.256 FY -0.854 FZ 0.046
207. 16 FX 0.001 FY -0.708 FZ -0.007
208. 21 22 FX -0.012 FY 0.046 FZ 0
209. 12 13 FX 0 FY 0.01 FZ 0
210. 14 15 FX 0.036 FY -0.025 FZ 0
211. LOAD 19 TEMPERATURE (-)25 (T_(-)25)
212. JOINT LOAD
213. 17 18 FX -0.099 FY -0.746 FZ 4.031
214. 19 20 FX 0.593 FY 1.416 FZ -4.23
215. 16 FX -0.002 FY 0.658 FZ 0.001
216. 21 22 FX 0 FY -0.208 FZ 0
217. 12 13 FX 0 FY -0.008 FZ 0
218. 14 15 FX -0.029 FY 0.02 FZ 0
219. ***LOAD COMBINATION FOR STRESS CALCULATION***
220. LOAD COMB 101 D+L+T_(+)25
221. 1 1.0 3 1.0 18 1.0
222. LOAD COMB 102 D+L+T_(-)25
223. 1 1.0 3 1.0 19 1.0
224. ***
225. LOAD COMB 103 D+LR
226. 1 1.0 2 1.0
227. ***
228. LOAD COMB 104 D+0.75*(L+T_(+)25)+0.75*LR
229. 1 1.0 3 0.75 18 0.75 2 0.75
230. LOAD COMB 105 D+0.75*(L+T_(-)25)+0.75*LR
231. 1 1.0 3 0.75 19 0.75 2 0.75
232. ***
233. LOAD COMB 106 D+W_L1
234. 1 1.0 8 1.0
235. LOAD COMB 107 D+W_R1
236. 1 1.0 9 1.0
237. LOAD COMB 108 D+W_L2
238. 1 1.0 10 1.0
239. LOAD COMB 109 D+W_R2
240. 1 1.0 11 1.0
241. LOAD COMB 110 D+W_LG1
242. 1 1.0 12 1.0
243. LOAD COMB 111 D+W_LG2
244. 1 1.0 13 1.0
245. ***
246. LOAD COMB 112 D+0.7*E_(+)X
247. 1 1.0 14 1.0
248. LOAD COMB 113 D+0.7*E_(-)X
249. 1 1.0 15 1.0
250. LOAD COMB 114 D+0.7*E_(+)Z
251. 1 1.0 16 1.0
252. LOAD COMB 115 D+0.7*E_(-)Z
253. 1 1.0 17 1.0
254. ***
255. LOAD COMB 116 D+0.75*W_L1+0.75*L+0.75*LR
256. 1 1.0 8 0.75 3 0.75 2 0.75
257. LOAD COMB 117 D+0.75*W_L2+0.75*L+0.75*LR
258. 1 1.0 9 0.75 3 0.75 2 0.75
259. LOAD COMB 118 D+0.75*W_R1+0.75*L+0.75*LR
260. 1 1.0 10 0.75 3 0.75 2 0.75
```

STAAD SPACE -- PAGE NO. 6

```
261. LOAD COMB 119 D+0.75*W_R2+0.75*L+0.75*LR
262. 1 1.0 11 0.75 3 0.75 2 0.75
263. LOAD COMB 120 D+0.75*W_LG1+0.75*L+0.75*LR
264. 1 1.0 12 0.75 3 0.75 2 0.75
265. LOAD COMB 121 D+0.75*W_LG2+0.75*L+0.75*LR
266. 1 1.0 13 0.75 3 0.75 2 0.75
267. ***
268. LOAD COMB 122 D+0.75*(0.7*E_(+)X)+0.75*L+0.75*LR
269. 1 1.0 14 0.525 3 0.75 2 0.75
270. LOAD COMB 123 D+0.75*(0.7*E_(-)X)+0.75*L+0.75*LR
271. 1 1.0 15 0.525 3 0.75 2 0.75
272. LOAD COMB 124 D+0.75*(0.7*E_(+)Z)+0.75*L+0.75*LR
273. 1 1.0 16 0.525 3 0.75 2 0.75
274. LOAD COMB 125 D+0.75*(0.7*E_(-)Z)+0.75*L+0.75*LR
275. 1 1.0 17 0.525 3 0.75 2 0.75
276. ***
277. LOAD COMB 126 0.6*D+W_L1
278. 1 0.6 8 1.0
279. LOAD COMB 127 0.6*D+W_L2
280. 1 0.6 9 1.0
281. LOAD COMB 128 0.6*D+W_R1
282. 1 0.6 10 1.0
283. LOAD COMB 129 0.6*D+W_R2
284. 1 0.6 11 1.0
285. LOAD COMB 130 0.6*D+W_LG1
286. 1 0.6 12 1.0
287. LOAD COMB 131 0.6*D+W_LG2
288. 1 0.6 13 1.0
289. ***
290. LOAD COMB 132 0.6*D+0.7*E_(+)X
291. 1 0.6 14 0.7
292. LOAD COMB 133 0.6*D+0.7*E_(-)X
293. 1 0.6 15 0.7
294. LOAD COMB 134 0.6*D+0.7*E_(+)Z
295. 1 0.6 16 0.7
296. LOAD COMB 135 0.6*D+0.7*E_(-)Z
297. 1 0.6 17 0.7
298. ***LOAD COMBINATION FOR STRENGTH CALCULATION***
299. LOAD COMB 201 1.4*D
300. 1 1.4
301. ***
302. LOAD COMB 202 1.2*(D+T_(+)25)+1.6*L+0.5*LR
303. 1 1.2 18 1.2 3 1.6 2 0.5
304. LOAD COMB 203 1.2*(D+T_(-)25)+1.6*L+0.5*LR
305. 1 1.2 19 1.2 3 1.6 2 0.5
306. ***
307. LOAD COMB 204 1.2*D+1.6*LR+0.5*L
308. 1 1.2 2 1.6 3 0.5
309. ***
310. LOAD COMB 205 1.2*D+1.6*LR+0.8*W_L1
311. 1 1.2 2 1.6 8 0.8
312. LOAD COMB 206 1.2*D+1.6*LR+0.8*W_L2
313. 1 1.2 2 1.6 9 0.8
314. LOAD COMB 207 1.2*D+1.6*LR+0.8*W_R1
315. 1 1.2 2 1.6 10 0.8
316. LOAD COMB 208 1.2*D+1.6*LR+0.8*W_R2
```

STAAD SPACE -- PAGE NO. 7

```
317. 1 1.2 2 1.6 11 0.8
318. LOAD COMB 209 1.2*D+1.6*LR+0.8*W_LG1
319. 1 1.2 2 1.6 12 0.8
320. LOAD COMB 210 1.2*D+1.6*LR+0.8*W_LG2
321. 1 1.2 2 1.6 13 0.8
322. ***
323. LOAD COMB 211 1.2*D+1.6*W_L1+0.5*L+0.5*LR
324. 1 1.2 8 1.6 3 0.5 2 0.5
325. LOAD COMB 212 1.2*D+1.6*W_L2+0.5*L+0.5*LR
326. 1 1.2 9 1.6 3 0.5 2 0.5
327. LOAD COMB 213 1.2*D+1.6*W_R1+0.5*L+0.5*LR
328. 1 1.2 10 1.6 3 0.5 2 0.5
329. LOAD COMB 214 1.2*D+1.6*W_R2+0.5*L+0.5*LR
330. 1 1.2 11 1.6 3 0.5 2 0.5
331. LOAD COMB 215 1.2*D+1.6*W_LG1+0.5*L+0.5*LR
332. 1 1.2 12 1.6 3 0.5 2 0.5
333. LOAD COMB 216 1.2*D+1.6*W_LG2+0.5*L+0.5*LR
334. 1 1.2 13 1.6 3 0.5 2 0.5
335. ***
336. LOAD COMB 217 1.2*D+E_(+)X+0.5*L
337. 1 1.2 14 1.0 3 0.5
338. LOAD COMB 218 1.2*D+E_(-)X+0.5*L
339. 1 1.2 15 1.0 3 0.5
340. LOAD COMB 219 1.2*D+E_(+)Z+0.5*L
341. 1 1.2 16 1.0 3 0.5
342. LOAD COMB 220 1.2*D+E_(-)Z+0.5*L
343. 1 1.2 17 1.0 3 0.5
344. ***
345. LOAD COMB 221 0.9*D+1.6*W_L1
346. 1 0.9 8 1.6
347. LOAD COMB 222 0.9*D+1.6*W_L2
348. 1 0.9 9 1.6
349. LOAD COMB 223 0.9*D+1.6*W_R1
350. 1 0.9 10 1.6
351. LOAD COMB 224 0.9*D+1.6*W_R2
352. 1 0.9 11 1.6
353. LOAD COMB 225 0.9*D+1.6*W_LG1
354. 1 0.9 12 1.6
355. LOAD COMB 226 0.9*D+1.6*W_LG2
356. 1 0.9 13 1.6
357. ***
358. LOAD COMB 227 0.9*D+E_(+)X
359. 1 0.9 14 1.0
360. LOAD COMB 228 0.9*D+E_(-)X
361. 1 0.9 15 1.0
362. LOAD COMB 229 0.9*D+E_(+)Z
363. 1 0.9 16 1.0
364. LOAD COMB 230 0.9*D+E_(-)Z
365. 1 0.9 17 1.0
366. ***
367. PERFORM ANALYSIS
```

STAAD SPACE -- PAGE NO. 8

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	22	NUMBER OF MEMBERS	11
NUMBER OF PLATES	0	NUMBER OF SOLIDS	0
NUMBER OF SURFACES	0	NUMBER OF SUPPORTS	11

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 19, TOTAL DEGREES OF FREEDOM = 66
TOTAL LOAD COMBINATION CASES = 65 SO FAR.

368. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUN 15,2020 TIME= 23:58:15 ****

STAAD SPACE -- PAGE NO. 9

```
     ************************************************************
     *   For technical assistance on STAAD.Pro, please visit    *
     *   http://www.bentley.com/en/support/                     *
     *                                                          *
     *   Details about additional assistance from               *
     *   Bentley and Partners can be found at program menu      *
     *   Help->Technical Support                                *
     *                                                          *
     *      Copyright (c) 1997-2017 Bentley Systems, Inc.       *
     *                http://www.bentley.com                    *
     ************************************************************
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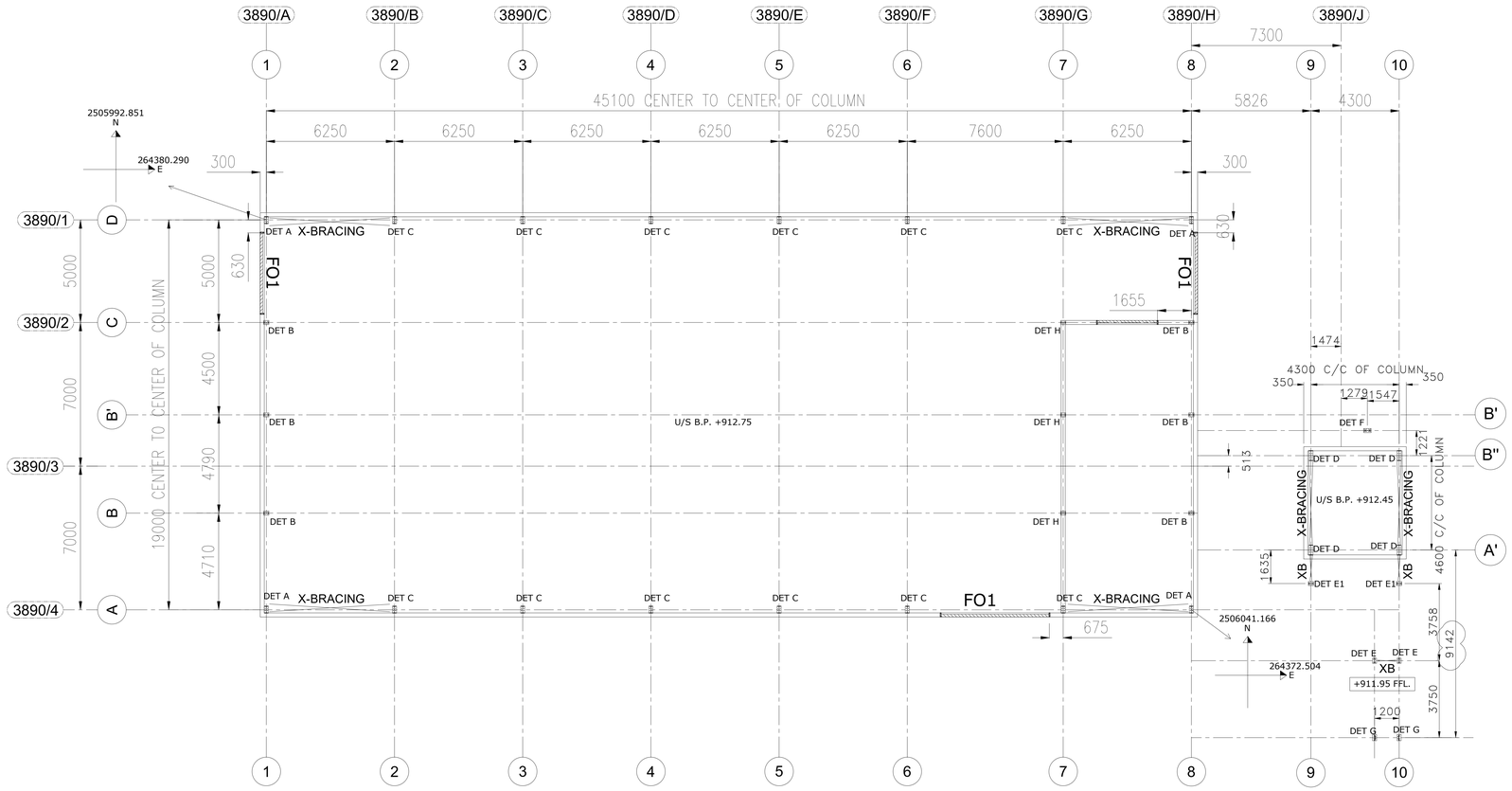
ATTACHMENT - 7 : INPUT DRAWING.

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DRAWING NUMBER MD-216-3890-EG-ST-GAD-2364 REV A01



ANCHOR BOLT PLAN AT BASE PLATE RL 912.75/912.45 (U/S BASE PLATE)

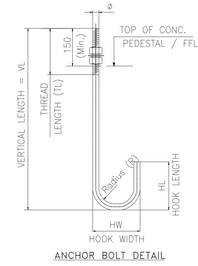
F01 – DENOTES FRAMED OPENING FOR RUD (4000WX5000H)– 3 NO'S.
F02 – DENOTES FRAMED OPENING FOR RUD (3000WX3000H)– 1 NO'S.
XB – DENOTES DIAGONAL BRACING.

NOTE: ANCHOR BOLT SPECIFICATION (AS PER ASTM 1554, GR.36,50,105)

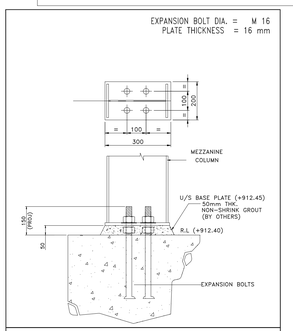
QTY.	DIA. OF BOLT = ø	DIA. OF HOLE	THREAD LENGTH	HOOK LENGTH	HOOK WIDTH	RADIUS	VERTICAL LENGTH (VL)	ANC.ROD LENGTH
--	ø	D	TL	HL	HW	R	VL	AL
8	M16	22	150	110	200	100	575	710
124	M20	26	150	110	200	100	575	710
--	M24	30	200	140	256	128	675	851
--	M30	36	250	170	320	160	975	1189
--	M36	42	250	210	384	192	1075	1338

REFERENCE NOTES :–

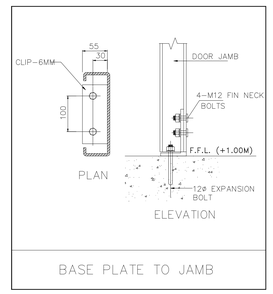
1. BRACING REACTIONS ARE TO BE CONSIDERED WITH THE MAIN FRAME REACTIONS.
2. STANDARD AIC ANCHOR BOLTS CONSIST OF 2 NUTS + ONE WASHER.
3. BOTTOM OF BASE PLATES ARE AT THE SAME ELEVATION (UNLESS NOTED OTHERWISE).
4. AIC MAY USE A DIFFERENT SHAPE OF ANCHOR BOLT BASED ON AVAILABILITY OF STOCK.
5. 50mm GROUT (BY OTHERS) IS REQUIRED AT ALL BASE OF COLUMN.
6. 40 x 40 CONCRETE NOTCH IS FOR SINGLE SKIN PANEL ONLY. MAY VARY DEPENDING ON THE ACTUAL WALL PANEL CONDITION.
7. ALL DIMENSIONS ARE IN MILLIMETERS.



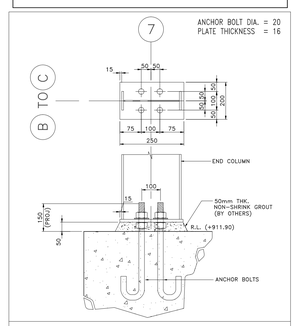
ANCHOR BOLT DETAIL



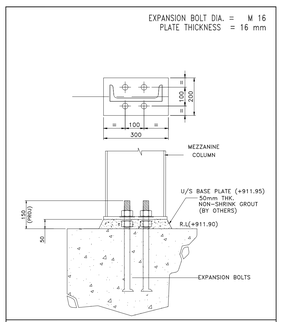
DETAIL–E1



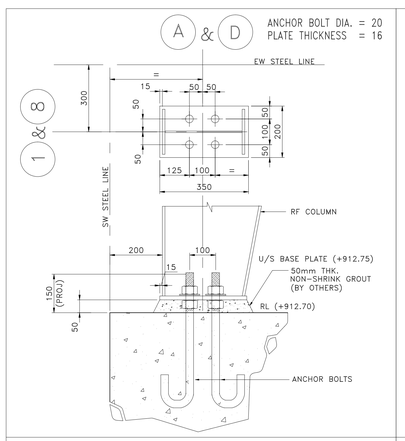
BASE PLATE TO JAMB



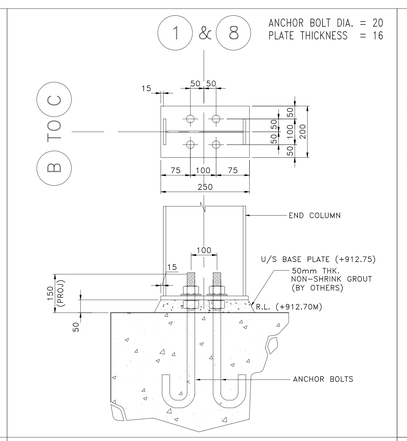
DETAIL–H



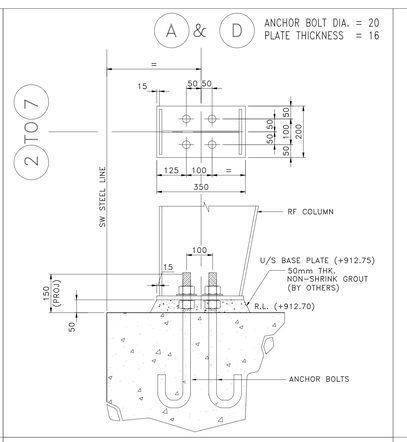
DETAIL–G



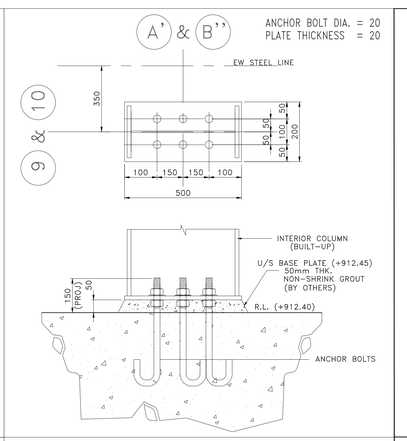
DETAIL–A



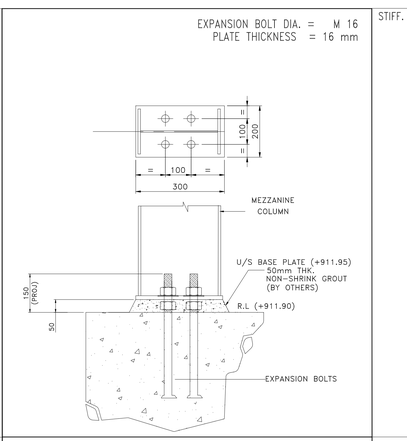
DETAIL–B



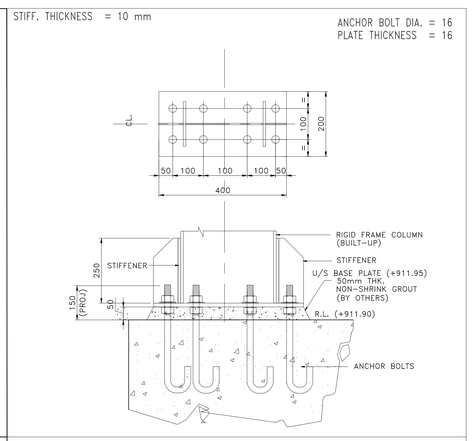
DETAIL–C



DETAIL–D



DETAIL–E



DETAIL–F

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3890-EG-ME-GAD-1007-A01	LOAD PLAN REAGENTS – SMBS & PEROXIDE PLAN VIEW	A01	03/01/2020	ISSUED FOR REFERENCE					
MD-216-3890-EG-ST-GAD-1001-A01	STEEL STRUCTURE GUIDE DRAWING SMBS AND PEROXIDE 3D VIEW	A01	17/01/2020	ISSUED FOR REFERENCE					
MD-216-3890-EG-ST-GAD-1002	ELE inputs	A01	17/01/2020	ISSUED FOR REFERENCE					
MD-216-3890-EG-ST-GAD-1002-A01	STEEL STR GUIDE DRAWING SMBS & PEROXIDE PLAN & SECTIONS	A01	17/01/2020	ISSUED FOR REFERENCE					

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01		ISSUED FOR REFERENCE	PRI	MNP	PRI	QRS	

aic arabian international company
P.O.Box : 41017 Jeddah 21521,K.S.A
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Fax : +966 2 691 0099
E-mail : aic@aicsteel.com
Site : www.aicsteel.com

DRAWING NUMBER						REVISION
MD-P20131-B07-A001						00
DATE	14/06/2020	DATE	14/06/2020	DATE	14/06/2020	
DRAWN	7015	CHECKED	7013/7028	APPROVED	7001	

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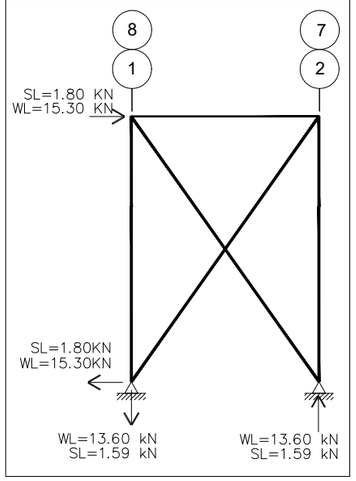
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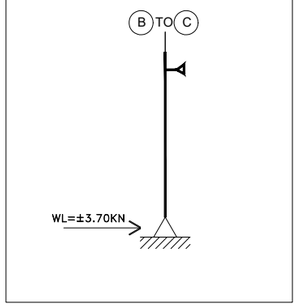
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019045_450_S_G_26083_001						0
OUTOTEC DRAWING NUMBER						DOCUMENT ID / REVISION
900707 _ GAU01 _ AA04 _ MTL01_ 39801 _ 0U100676487						0U100676487 / 0
DRAWN	--	--	MANSOURAH – MASSARAH GOLD PROJECT			
CHECKED	MNP	14/06/2020	GENERAL ARRANGEMENT DRAWING			
ENGINEER	--	--	REAGENTS – SMBS & PEROXIDE STORAGE AND MIXING STATION			
APPROVED	--	14/06/2020	COLUMN ANCHOR PLAN AND DETAILS			
EMPLOYER						
SCALE	1:100	A0	DRAWING NUMBER			REV
			MD-216-3890-EG-ST-GAD-2364			A01

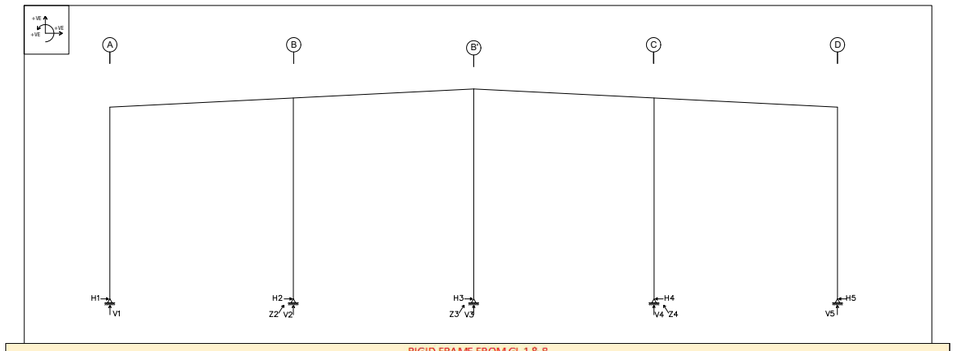
DRAWING NUMBER MD-216-3890-EG-ST-GAD-2365 REV A01

SIDEWALL (XB) BRACING REACTION AT GL A & D

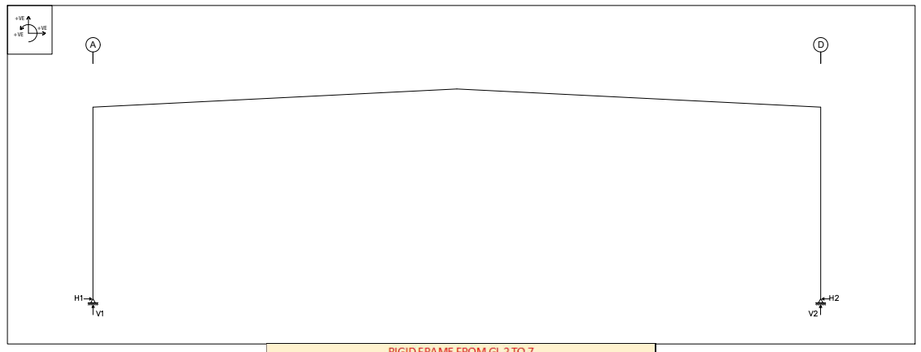


PARTITION COLUMN REACTION AT GL. 7

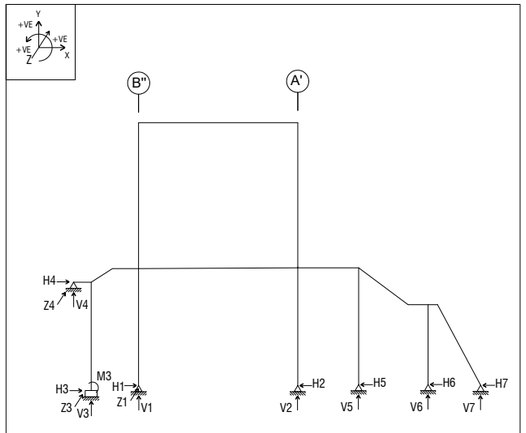




RIGID FRAME FROM GL 1 & 8															
CATEGORY	LOAD CASE	LEFT COL. @GL A		LEFT COL. @GL B			COL. @GL B'			COL. @GL C			RIGHT COL. @GL D	
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)	Z2 (KN)	H3 (KN)	V3 (KN)	Z3 (KN)	H4 (KN)	V4 (KN)	Z4 (KN)	H5 (KN)	V5 (KN)
Dead	Dead	-0.20	-2.70	0.00	-3.50	0.00	0.00	-3.50	0.00	0.00	-3.50	0.00	0.20	-2.80
Collateral	Collateral	0.10	-5.30	0.60	0.80	0.00	-0.30	-1.40	0.00	-0.20	-2.80	0.00	-0.30	-3.60
Roof Live	Live	-0.70	-5.30	0.00	-9.40	0.00	0.00	-9.40	0.00	0.00	-9.40	0.00	0.70	-5.40
Wind	Wind_Left1	-11.80	2.50	0.00	19.80	-16.30	0.00	20.90	-16.70	0.00	23.70	-16.30	-13.20	23.60
	Wind_Right1	13.10	23.40	0.00	23.80	-16.30	0.00	20.90	-16.70	0.00	19.90	-16.30	11.90	2.40
	Wind_Left2	-8.80	-3.70	0.00	12.90	14.80	0.00	13.80	15.10	0.00	16.90	14.80	-16.20	-17.40
	Wind_Right2	16.10	17.30	0.00	16.90	14.80	0.00	13.80	15.10	0.00	13.00	14.80	8.90	-3.80
	Wind_Long1	-4.40	23.40	0.00	19.30	-16.30	0.00	14.10	-16.70	0.00	8.30	-16.30	5.90	24.60
	Wind_Long2	-5.90	24.60	0.00	8.30	14.80	0.00	14.10	16.10	0.00	19.30	14.80	4.40	23.40
Seismic	Seismic_L	-0.30	0.40	0.00	-0.40	0.00	0.00	0.40	0.00	0.00	0.40	0.00	-0.30	-0.40
	Seismic_R	0.30	-0.40	0.00	0.40	0.00	0.00	0.00	0.00	0.00	-0.40	0.00	0.30	0.40
Temperature	25Degree	1.00	1.50	0.00	1.70	0.00	0.00	0.50	0.00	0.00	1.70	0.00	1.00	1.50
	25Degree/-VE	-1.00	-1.50	0.00	-1.70	0.00	0.00	-0.50	0.00	0.00	-1.70	0.00	1.00	-1.50



RIGID FRAME FROM GL 2 TO 7					
CATEGORY	LOAD CASE	LEFT COL. @GL A		MEZZ COL. @GL D	
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)
Dead	Dead	5.70	-11.70	-5.70	-11.70
Collateral	Collateral	-0.30	-4.40	-0.30	-0.20
Roof Live	Live	22.20	-39.30	-22.20	-39.30
Wind	Wind_Left1	-51.10	78.40	16.90	-54.40
	Wind_Right1	-16.80	54.40	51.10	78.40
	Wind_Left2	-41.60	44.90	7.40	20.90
	Wind_Right2	-7.30	20.90	41.50	44.90
	Wind_Long1	-19.30	84.90	22.30	71.40
	Wind_Long2	-22.40	71.40	19.40	84.90
Seismic	Seismic_L	-0.40	0.20	-0.40	-0.20
	Seismic_R	0.40	-0.20	0.40	0.20
Temperature	25Degree	-0.40	0.00	-0.40	0.00
	25Degree/-VE	0.40	0.00	0.40	0.00



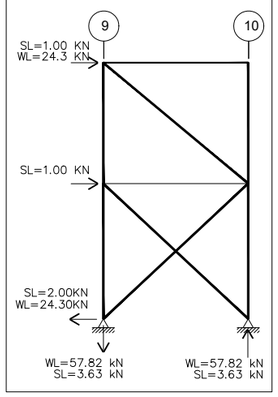
LOADS LEGEND	
Wind_Left1	Denotes wind load from left to right with internal pressure
Wind_Right1	Denotes wind load from right to left with internal pressure
Wind_Left2	Denotes wind load from left to right with internal suction
Wind_Right2	Denotes wind load from right to left with internal suction
Wind_Long1	Denotes wind load parallel to ridge from end wall with internal pressure
Wind_Long2	Denotes wind load parallel to ridge from end wall with internal suction
Seismic_L	Denotes seismic load from left
Seismic_R	Denotes seismic load from right

IMPORTANT NOTES :

(1) BEARING CAPACITY OF CONCRETE UNDER BASE PLATE SHOULD NOT BE LESS THAN 1.05 KN/CM2
(2) SAFETY OF CONCRETE PEDESTAL & FOOTINGS IS NOT THE RESPONSIBILITY OF AIC STEEL , CUSTOMER TO DESIGN THE CONCRETE FOOTING
(3) CUSTOMER TO DESIGN THE CONCRETE COLUMN FOR THE CONCRETE BREAKOUT & EDGE DISTANCE IS CALCULATED USING ACI PROVISIONS.
(4) GIVEN REACTIONS ARE FOR UNFACTORED LOADS.
(5) BRACING REACTIONS TO BE ADDED WITH MAIN FRAME COLUMN REACTIONS FOR THE DESIGN OF FOUNDATION.
(6) LOADS IN THE SAME CATEGORY , NOT TO BE COMBINED IN THE SAME LOAD COMBINATION .

RIGID FRAME FROM GL TO A' & B"																		
CATEGORY	LOAD CASE	LEFT COL. @GL B"			MEZZ COL. @GL A'			STAIR COL.				STAIR COL.		STAIR COL.		STAIR COL.		STAIR COL.	
		H1 (KN)	V1 (KN)	Z1 (KN)	H2 (KN)	V2 (KN)	Z2 (KN)	H3 (KN)	V3 (KN)	Z3 (KN)	M3(KN)	H4 (KN)	V4 (KN)	Z4 (KN)	H5 (KN)	V5 (KN)	H6 (KN)	V6 (KN)	H7 (KN)	V7 (KN)
Dead	Dead	0.09	-17.87	0.00	-0.09	-16.43	-0.01	0.01	-9.83	0.05	-0.01	0.00	1.64	-0.02	0.00	-5.98	0.00	-5.29	-0.15	-2.29
Roof Live	Live	0.18	-5.17	0.00	1.41	-3.74	0.01	0.00	-0.13	0.00	0.01	0.00	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00
LIVE LOAD	LL	-0.32	-15.26	0.00	-2.02	3.58	-0.78	0.02	-29.48	0.21	-0.05	0.00	7.44	-0.06	0.00	-6.02	0.00	-6.59	-0.21	-3.32
CRANE	CR1	-0.80	-10.26	1.24	0.00	-7.04	0.02	-0.02	-0.05	-0.01	0.05	0.00	0.02	0.00	0.00	-4.07	0.00	0.07	-0.23	-0.17
	CR2	-0.45	-13.20	3.96	0.56	-1.63	0.04	-0.02	0.02	-0.02	0.05	0.00	-0.01	0.00	0.00	-4.47	0.00	0.07	-0.25	-0.18
	CR3	-1.13	-5.57	0.00	1.97	-10.16	-2.46	-0.02	-0.15	0.00	0.05	0.00	0.06	0.00	0.00	-3.01	0.00	0.06	-0.20	-0.14
	CR4	-0.93	-7.84	0.00	1.55	-7.65	-0.56	-0.02	-0.10	-0.01	0.05	0.00	0.04	0.00	0.00	-3.44	0.00	0.06	-0.21	-0.15
Wind	Wind Left1	-15.47	33.63	5.69	10.63	-22.74	0.23	-0.15	0.59	0.00	0.47	0.00	-0.10	0.00	-0.04	-35.17	0.00	0.62	-2.19	-1.55
	Wind Left2	-16.23	30.12	5.68	9.98	-25.54	0.13	-0.15	0.53	-0.01	0.47	0.00	-0.08	0.00	-0.04	-35.41	0.00	0.62	-2.19	-1.55
	Wind Right1	4.95	-27.36	1.55	-14.96	27.51	-7.27	0.15	-0.58	0.02	-0.47	0.00	0.11	-0.01	10.76	35.82	0.00	-0.60	2.13	1.51
	Wind Right2	4.32	-30.29	0.67	-15.45	23.92	-6.23	0.15	-0.66	0.02	-0.47	0.00	0.14	-0.01	10.85	36.14	0.00	-0.61	2.16	1.53
	Wind+Long1	1.45	64.60	34.58	1.42	-58.71	0.42	0.00	-0.67	0.10	0.01	0.00	0.33	-0.03	0.19	0.39	0.00	0.01	-0.05	-0.03
	Wind+Long2	0.81	61.35	34.58	0.84	-61.89	0.43	0.00	-0.75	0.10	0.00	0.00	0.36	-0.03	0.18	0.41	0.00	0.01	-0.05	-0.03
Seismic	S+X	-2.26	4.00	0.79	1.41	-3.74	0.01	-0.02	0.08	0.00	0.07	0.00	-0.01	0.00	-0.01	-5.00	0.00	0.09	-0.31	-0.22
	S-X	0.77	-3.97	0.00	-2.02	3.58	-0.78	0.02	-0.10	0.00	-0.07	0.00	0.02	0.00	1.52	5.07	0.00	-0.09	0.30	0.22
	S-Z	0.00	-7.08	0.00	-0.01	6.99	-4.00	0.00	-0.02	-0.01	0.00	0.00	-0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	S+Z	-0.01	7.04	3.98	0.00	-7.04	0.02	0.00	-0.09	0.01	0.00	0.00	0.04	0.00	0.02	0.05	0.00	0.00	0.00	0.00
Temperature	25Degree	0.24	5.85	0.79	2.51	-8.38	0.45	0.01	-6.94	-0.07	-0.07	0.00	4.19	0.00	0.12	0.45	0.00	0.10	-0.35	-0.25
	25Degree/-VE	-0.97	-7.32	39.53	-5.82	13.89	-41.48	-0.02	6.45	0.01	0.08	0.00	-4.09	0.02	0.00	-2.04	0.00	-0.08	0.28	0.20

SIDEWALL (PB) BRACING REACTION AT GL 9 & 10



Building reactions are based on the following building data:

ROOF LOADS			
DEAD LOAD (kN/m2)		=	0.10
ROOF LIVE LOAD (kN/m2)		=	1.00 REDUCIBLE AS PER SBC
BASIC LOADS			
WIND LOAD	Wind Speed (3 sec)	=	155 (Km/Hr)
	Wind Code	=	ASCE (7-16)
	EXPOSURE	=	C
	CLASSIFICATION	=	ENCLOSED
SEIS LOAD IBC 2006	Ss	=	10%g
	S1	=	2%g
	SITE CLASS	=	B
TEMPERATURE LOAD	TEMPERATURE LOAD	=	± 25 C
PLATFORM LOADS			
LIVE LOAD (kN/m2)		=	3.00
DEAD LOAD (kN/m2)		=	0.50

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION
MD-216-3890-EG-ME-GAD-1007-A01	LOAD PLAN REAGENTS - SMBS & PEROXIDE PLAN VIEW	A01	03/01/2020	ISSUED FOR REFERENCE
MD-216-3890-EG-ST-GAD-1001-A01	STEEL STRUCTURE GUIDE DRAWING SMBS AND PEROXIDE 3D VIEW	A01	17/01/2020	ISSUED FOR REFERENCE
MD-216-3890-EG-ST-GAD-1002	ELE inputs	A01	17/01/2020	ISSUED FOR REFERENCE
MD-216-3890-EG-ST-GAD-1002-A01	STEEL STR GUIDE DRAWING SMBS & PEROXIDE PLAN & SECTIONS	A01	17/01/2020	ISSUED FOR REFERENCE

DRN CHK ENG APP EMP: QRS | A01 | ISSUED FOR REFERENCE | PRI MNP PRI QRS

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Site: www.aicsteel.com

DRAWING NUMBER	REVISION
MD-P20131-B07-A002	00

	DATE		DATE		DATE
DRAWN	7015	CHECKED	7013/7028	APPROVED	7001
	14/06/2020		14/06/2020		14/06/2020

LARSEN & TOUBRO DRAWING NUMBER: 019045_450_S_G_26083_002 — REVISION 0

OUTOTEC DRAWING NUMBER: 900707 _ GAU01 _ AA04 _ MTL01 _ 39802 _ 0U100676488 — DOCUMENT ID: 0U100676488 — REVISION 0

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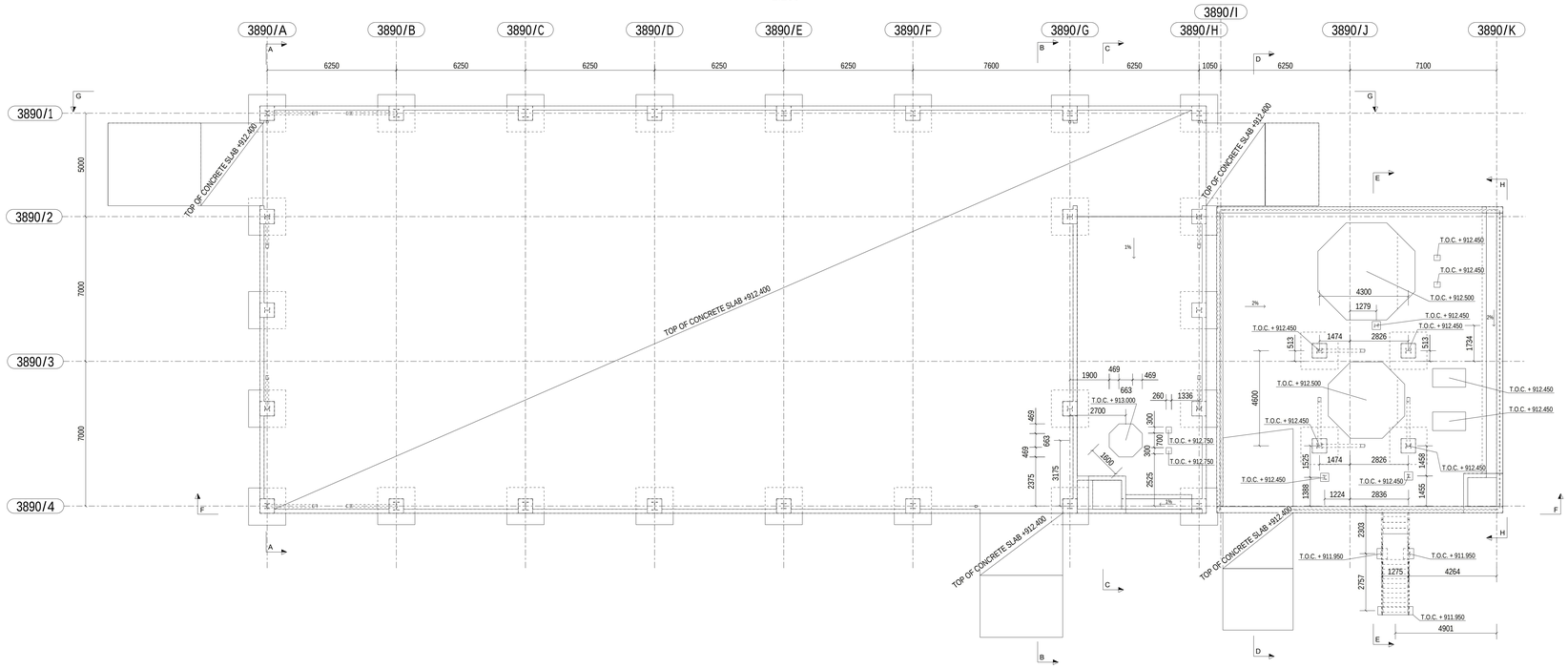
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CHECKED	MNP		14/06/2020
ENGINEER	--	--	
APPROVED	--		14/06/2020
EMPLOYER			
SCALE	N.T.S	A0	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
REAGENTS - SMBS & PEROXIDE STORAGE AND MIXING STATION
COLUMN REACTIONS

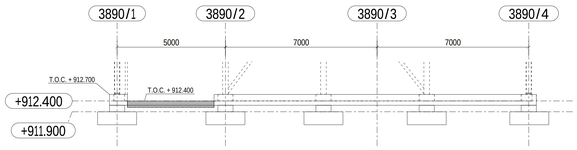
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DRAWING NUMBER: MD-216-3890-EG-CV-GAD-1001
REV: A01

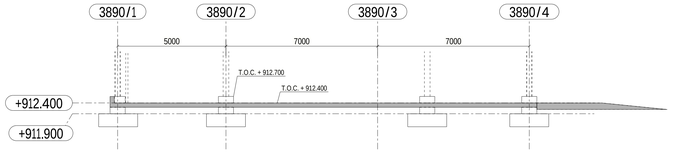
LEVEL +912.400
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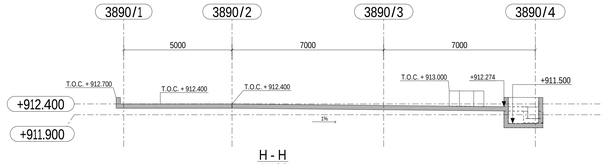
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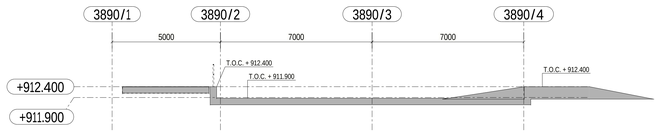
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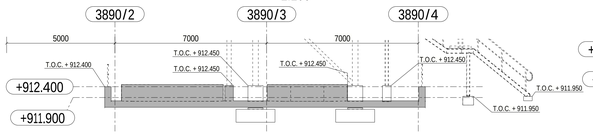
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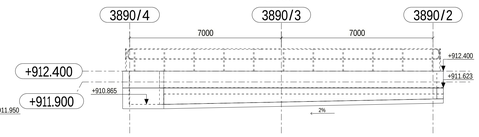
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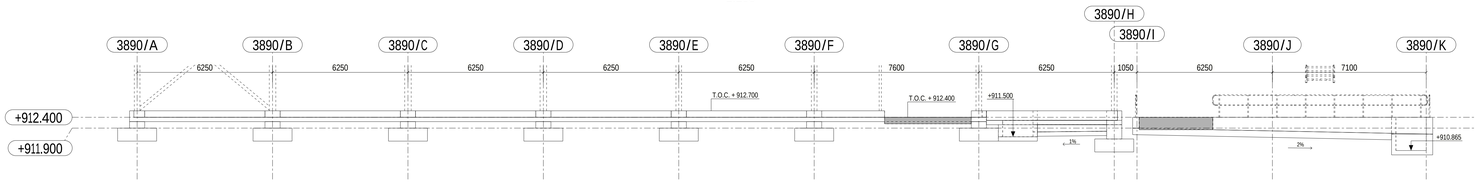
E - E
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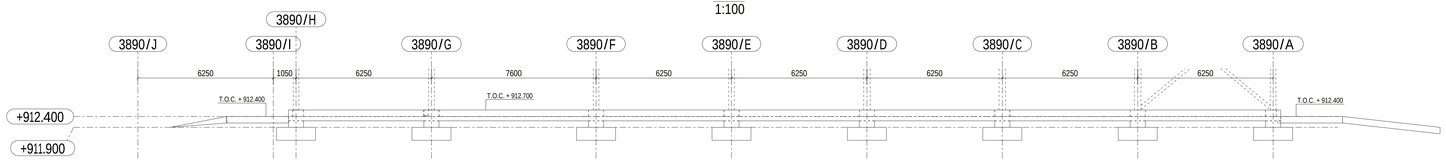
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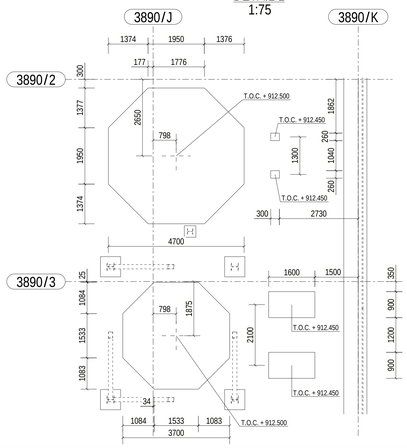
F - F
1:100



G - G
1:100



DETAIL 1
1:75



THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

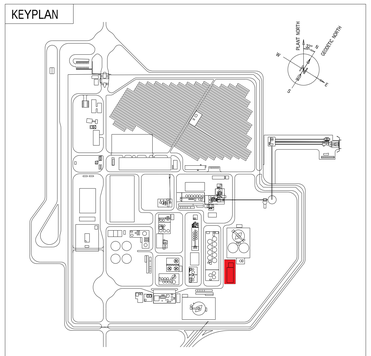
FINISHED TOP GROUND LEVEL (F.G.L) FOR OUTSIDE AREA 3890 = RL(+)911.900 m

FINISHED TOP GROUND LEVEL (F.G.L) FOR INSIDE AREA 3890 = RL(+)912.400 m

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY, THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3890-EG-ST-GAD-1002	PLAN AND SECTIONS		
MD-216-1000-EG-ST-CRT-0001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3890-EG-ST-GAD-1001	3D VIEW
MD-216-3890-EG-ME-GAD-1001	PLAN VIEW	MD-216-3890-EG-ME-GAD-1002	SECTION A-A, B-B & C-C

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	17.01.2020	ISSUED FOR APPROVAL	Kir.Jok	Mik.Lai	Lui.Piz	Suo.Cha	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER | REVISION 0

OUTOTEC DRAWING NUMBER: 900707_GAU01_AA04_CTA04_38901 | DOCUMENT ID: OU100668170 | REVISION 0

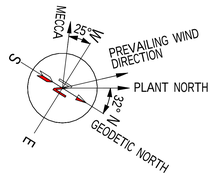
DRAWN	Kir.Jok	17.01.2020	
CHECKED	Mik.Lai	17.01.2020	
ENGINEER	Lui.Piz	17.01.2020	
APPROVED	Suo.Cha	17.01.2020	
EMPLOYER			
SCALE	1:100	A0	

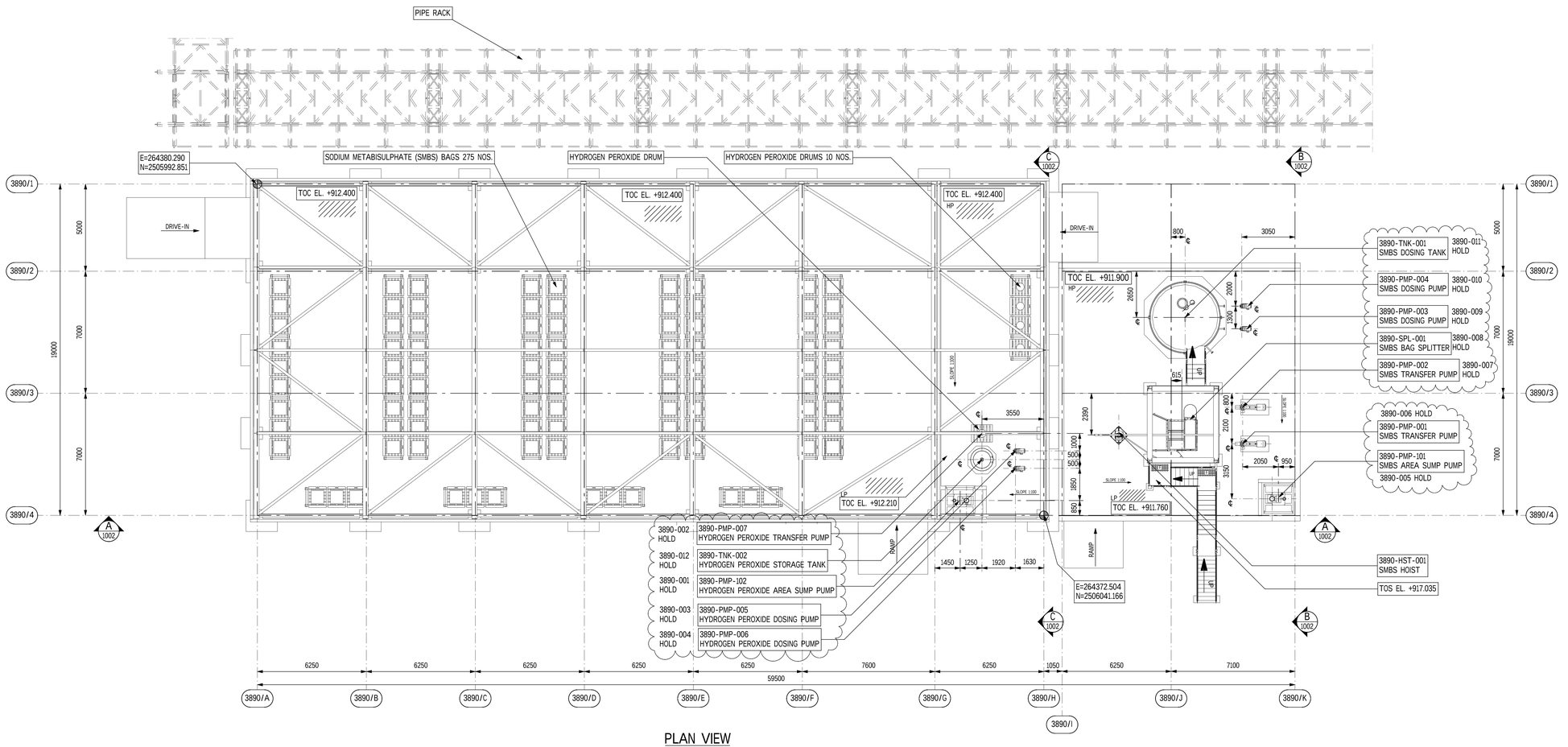
MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
SMBS AND PEROXIDE
PLAN AT EL. +912.400 AND +911.900

DRAWING NUMBER: MD-216-3890-EG-CV-GAD-1001 | REV: A01

Tekla Structures

DRAWING NUMBER MD-216-3890-EG-ME-GAD-1001 REV A02



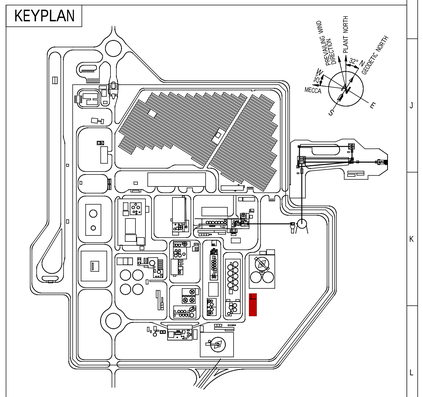


PLAN VIEW

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METERS UNLESS OTHERWISE STATED.
2. FINISHED GROUND LEVEL (F.G.L) FOR PROCESS AREA 3890 = RL(+)911.900 m
3. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE. T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE. T.O.G. = TOP OF GRATING. HP = HIGHT POINT. LP = LOW POINT.
4. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3890-EG-ME-GAD-1002	SECTION A-A, B-B & C-C

DOCUMENT NUMBER	REFERENCE DOCUMENTS

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	19-12-2019	ISSUED FOR DESIGN	SSS	TV	SC	RL	
A01	15-10-2019	ISSUED FOR APPROVAL	MA	TV	SC	RL	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_450_M_G_03257_001		1
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707 _GAU01 _AG22_MTL01 _38901	OU100664257	1

DRAWN	AM	15-10-2019
CHECKED	TV	15-10-2019
ENGINEER	SC	15-10-2019
APPROVED	RL	15-10-2019
EMPLOYER		
SCALE	1:100	A0

MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
REAGENTS - SMBS & PEROXIDE
PLAN VIEW

DRAWING NUMBER MD-216-3890-EG-ME-GAD-1001 REV A02